

UM2xA646

Datasheet

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Revision Record

Revision	Date	Updated or modified Sections
1	2022/11/7	Initial issue
2	2023/2/13	Add the operating temperature (page 14)
3	2025/5/13	Fix typo in ***Table 0-3***, ***Table 0-125***, *and* ***Table 0-131*** Update IP version in ***Table 0-2***, ***Table 0-3*** and ***Table 0-4*** Update strap pin in ***Table 0-4*** and ***Table 0-8***. Update pin-mux table in ***Figure 0-3***, ***Table 0-87***, ***Table 0-90***, ***Table 0-91***, and ***Table 0-94*** Remove the unused register in ***Table 0-79***. Add CPU reset control register in ***Table 0-84***. ADD PCIE iddq control register in ***Table 0-86***. Fix typo in ***Table 0-32***

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Architecture

This chapter contains the following sections:

1. Feature
2. SoC Architecture Overview
3. Semiconductor Foundry and Library
4. IP Block List

1. Feature

The primary features of Unimap-TX5 design are as follows:

- ARM Cortex-A53 Quad Core Processor
  - CPU core: 1200MHz
  - L1 cache : Instruction cache: 32KB, Data cache: 32KB
  - L2 cache : 1024KB
  - NEON
  - CRYPTO
  - CoreSight, ETB: 16KB
- Embedded Memory
  - ROM : 128KB
  - DATA RAM : 512KB
- Bus Structure
  - ARM CoreLink NIC-400 network interconnect
  - AMBA-APB bus segment (APB) for lower speed peripheral blocks (FTX2P030)
  - AMBA-APB bus segment (APB) for lower speed peripheral blocks (FTX2P030_u1)
- Clocks / PLLs
  - One dedicated generic PLL is required is used to synthesize the CPU clock and GPU clock.
  - One dedicated generic PLL is required is used to synthesize the pixel clock.
  - One dedicated generic PLL is required is used to synthesize the bus clock. The output of this PLL will be divided down to create the AXI, AHB, APB and peripheral clocks.
  - One dedicated generic PLL is required is used to synthesize the MAC clock. The output of this PLL will also be divided down to create the MAC and USB clocks.
  - One dedicated SSCG PLL is required for the DDR_TOP.
- Operating Frequencies
  - CPU Clock : 1200 MHz Maximum Frequency
  - GPU Clock : 600 MHz Maximum Frequency
  - DDR4 Clock : 600 MHz Maximum Frequency
  - AMBA-AXI bus : 400 MHz Maximum Frequency
  - AMBA-AHB Bus : 100 MHz Maximum Frequency
  - AMBA-APB bus : 50 MHz Maximum Frequency
  - LCDC Clock : 148.5 MHz Maximum Frequency
- Embedded function

  - DMA Controller
    - 8 channel *2
  - 3D Graphic Engine
    - 1920x1080P @60Hz
  - Soteria Engine
    - AES / RSA / CRC / HMAC / TRNG
    - eMemory OTP : 4K word x 16bit
  - RTC block
    - 1.8V alive domain
- External memory I/F
  - QSPI flash * 1 set [1]
    - CS: 1 pins
    - Clock freq.: 100 MHz
  - DDR3/DDR4 combo * 1 set
    - CS: 1 pins
    - DDR4 support feature
      - Clock freq.: DDR4-2400
      - Data width: DDR4 -32bits
      - Storage size: support up to 8GB
    - DDR3 support feature
      - Clock freq.: DDR3-1866
      - Data width: DDR3-16bits
      - Storage size: support up to 1Gb
  - SD card IF * 1 set
    - Support 3.3V I/O
    - Clock freq.: Max. 50MHz (High speed mode)
    - Data width: 4bits
    - Data rate: Max. 25MB/s
  - EMMC IF * 1 set
    - Support 1.8V I/O
    - Clock freq.: Max. 200MHz (HS200)
    - Data width: 4bits
    - Data rate: Max. 100MB/s
- Display IF
  - LVDS IF *2 set

        - 1920x1080P @ 60Hz
        - Support Even/Odd frame type of panel
- High Speed IF
    - USB 3.2 Gen1 OTG/Dual role *1set
    - USB 2.0 OTG/Dual role *2set
    - PCIe Gen3 4 lanes
- Communication IF
    - Gigabit Ethernet *2ports
    - CAN FD bus * 3 sets
- Low speed communication IF
    - UART * 4 sets
        - Baud rate : 115200 bps
    - USART * 2 sets
        - Baud rate : 115200 bps
    - I2C * 4 sets
        - Fast-mode
    - SPI * 4 sets
        - Slave mode of SCLK: Max. 33.3 MHz
        - Master mode of SCLK: Max. 50 MHz
- ADC/DAC IF
    - Temperature Converter *1 set
        - 10 bits, 1 KSPS
    - ADC *1 set
        - 12 bits, 1 MSPS, 8 to 1 mux
    - DAC *4 sets
        - 12 bits, 1 MSPS
- Motor Control IF
    - eQEP *12 sets
    - ePWM *9 sets
    - Cordic
        - Sin/Cos Math Hardware
- Customized IF
    - XMC interface with TX2
    - Customized SPI interface for external ADC
- GPIO

- 32 GPIOs

[1] If system boot up with Winbond NAND flash, TX5 only support the flash device with “default BUF=1”.

For example: W25N01GVxx**IG** / W25M02GVxx**IG**.

2. SoC Architecture Overview

Figure 0-1 is the SoC architecture, including all related modules needed for platform integration. The architectural block diagram of Unimap-TX5 is shown in Figure 0-1.

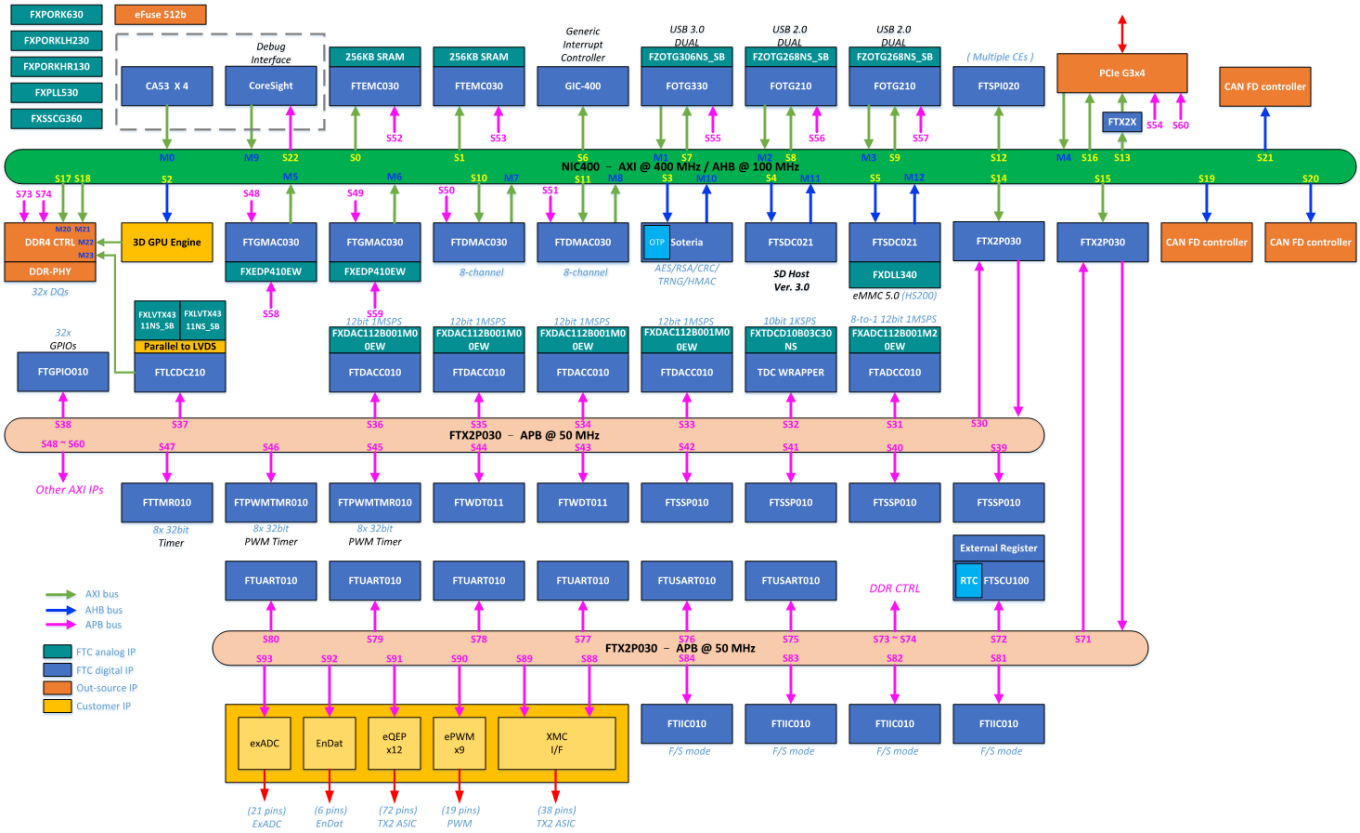


Figure 0-1.SoC Architecture

3. Semiconductor Foundry and Library

Unimap-TX5 will be implemented using the UMC 28 nm HPC+ process. The process libraries for synthesis, floor planning, place and route, and power analysis will be provided by Unimap Technology Corporation.

A summary of the semiconductor information for this project can be found in *Table 0-1*.

Table 0-1. Semiconductor Foundry and Library Information

Semiconductor Manufacturing Parameter	Value	Description
Foundry	UMC	United Microelectronics Corporation
Process node	28nm HPC+	UMC 28nm HPC+ logic process
Core library name(s)	FSJ0P_CLD	UMC 28nm HPC+ logic process 9-track cell LVT C30 library
	FSJ0P_CHS	UMC 28nm HPC+ logic process 9-track cell HVT C35 library
	FSJ0P_CRS	UMC 28nm HPC+ logic process 9-track cell RVT C35 library
	FSJ0P_CLS	UMC 28nm HPC+ logic process 9-track cell LVT C35 library
	FSJ0P_CHU	UMC 28nm HPC+ logic process 9-track cell HVT C40 library
	FSJ0P_CRU	UMC 28nm HPC+ logic process 9-track cell RVT C40 library
	FSJ0P_CLU	UMC 28nm HPC+ logic process 9-track cell LVT C40 library
I/O library name	FOJ0P_QRS18	UMC 28nm HPC+ logic process 1.8V/3.3V IO cell library
RTC library name	FSJ0P_RRS18	UMC 28nm HPC+ logic process 1.8V RTC core library
RTC I/O library name	FOJ0P_VRS18	UMC 28nm HPC+ logic process 1.8V RTC I/O cell library

4. IP Block List

This SoC is constructed using multiple blocks of Intellectual Property. The list of the required IPs as well as vendor information can be found in the following sections and tables.

4.1 Digital IP

This section describes the necessary digital IP used in Unimap-TX5. A list of the digital IP blocks is shown in Table 0-2.

Table 0-2. Digital IP

IP Name	IP Part Number	IP Vendor	IP Version	Total Instances Required	Description / Note
USB 3.2 Gen 1 Dual Role Controller	FOTG330	Unimap	1.10.0	1	-
USB 2.0 OTG Controller	FOTG210	Unimap	1.31.0	2	
10/100/1000 Ethernet Controller	FTGMAC030	Unimap	1.9.0	2	
Security Engine	Soteria	Unimap		1	
Embedded Memory Controller	FTEMC030	Unimap	1.3.0	2	
Direct Memory Access Controller	FTDMAC030	Unimap	1.5.0	2	
LCD Controller	FTLCDC210_UP	Unimap	1.0.0	1	
AXI to APB Bridge	FTX2P030	Unimap	1.4.1	2	
AXI to AXI Bridge	FTX2X030	Unimap	1.8.1	1	
SPI Flash Controller	FTSPI020	Unimap	1.9.0	1	
SD Card Host Controller	FTSDC021	Unimap	1.13.0	2	
System controller Unit	FTSCU100	Unimap	3.1.0	1	
Timer and PWM function	FTPWMTMR010	Unimap	1.3.0	2	
UART and IrDA controller	FTUART010	Unimap	1.12.1	4	

IP Name	IP Part Number	IP Vendor	IP Version	Total Instances Required	Description / Note
Universal Synchronous Receiver/ Transmitter	FTUSART010	Unimap	1.0.0	2	
Independent Watchdog timer	FTWDT011	Unimap	1.1.0	2	
Timer controller	FTTMR010	Unimap	1.10.0	1	
General purpose Input / Output Controller	FTGPIO010	Unimap	1.9.0	1	
Analog to Digital Converter Controller	FTADCC010	Unimap	1.1.0	1	
I2C Bus interface Controller	FTIIC010	Unimap	1.22.0	4	
Digital to Analog Converter Controller	FTDACC010	Unimap	0.1.0	4	
Synchronous Serial Port Controller	FTSSP010	Unimap	1.28.0	4	
Temperature sensor controller	FTTDCC010	Unimap	1.3.0	1	
DDR BIST controller	FTDDRXBIST100	Unimap	1.1.1	1	
CA53 Quad Core	CA53Q_subsystem	ARM	0.3.0	1	
NIC400	nic400_UniMap	ARM	0.5.0	1	
GIC400	GIC400	ARM	0.1.0	1	
DDR4 Controller	DWC_DDR_UMCTL2	Synopsys	0.9.0	1	
PCI-E Controller	dwc_pcie_3.0_standard_ AMBA_controller	Synopsys	0.10.0	1	
CAN bus Controller	FlexCAN_controller	Silvaco	0.2.0	3	
GPU		VIVANTE		1	

4.2 Analog and Mixed Signal IP

This section describes the necessary analog IP and Mixed signal IP used in this project. A list of the analog and mixed signal IP required is shown in Table 0-3.

Table 0-3. Analog and Mixed Signal IP

IP Name	IP Part Number	IP Version	Total Instances Required	Description
POR	FXPORK630HJ0P	0.1.0	1	Core 0.9V
POR	FXPORKLH230HJ0P	0.1.0	1	Core 1.8V
POR	FXPORKHR130HJ0P	0.2.0	1	RTC 1.8V
PLL	FXPLL530HJ0P	0.4.0	4	-
PLL	FXSSCG360HJ0P	0.2.0	1	-
DLL	FXDLL340HJ0P	0.2.0	1	
USB3 PHY	FZOTG306NSHJ0P_SB	0.4.2	1	
USB2 PHY	FZOTG268NSHJ0P_SB	0.2.1	2	
ADC	FXADC112B001M20EWHJ0P	0.3.1	1	
TDC	FXTDCD10B03C30NSHJ0P	0.2.0	1	
DAC	FXDAC112B001M00EWHJ0P	0.2.0	4	
LVDSTX	FXLVTX4311NSHJ0P_SB	0.1.0	2	
1 Port GPHY	FXEDP410EWHJ0P	0.7.1	2	
DDR4 PHY	dwc_lpddr4_multiphy	0.6.0	1	
PCI-E PHY	dwc_pcie_3.0_standard_AMBA_controller	0.10.0	1	
eFuse	u028efucp01603218400	0.1.0	1	-
OTP	EG004K16TI028CW07	0.2.0	1	

4.3 Special Cells, IP, and I/O Cells

This section describes the special cells, IP, and I/O cells required.

Table 0-4. Special Cells, IP, and I/O Cells

IP Name	IP Vendor	IP Version	Total Instances Required	Description
Motor_Top	Unimap	Motor_Top_6.eco_2	1	
P2LVDS	Unimap	-	1	LCD to LVDS block

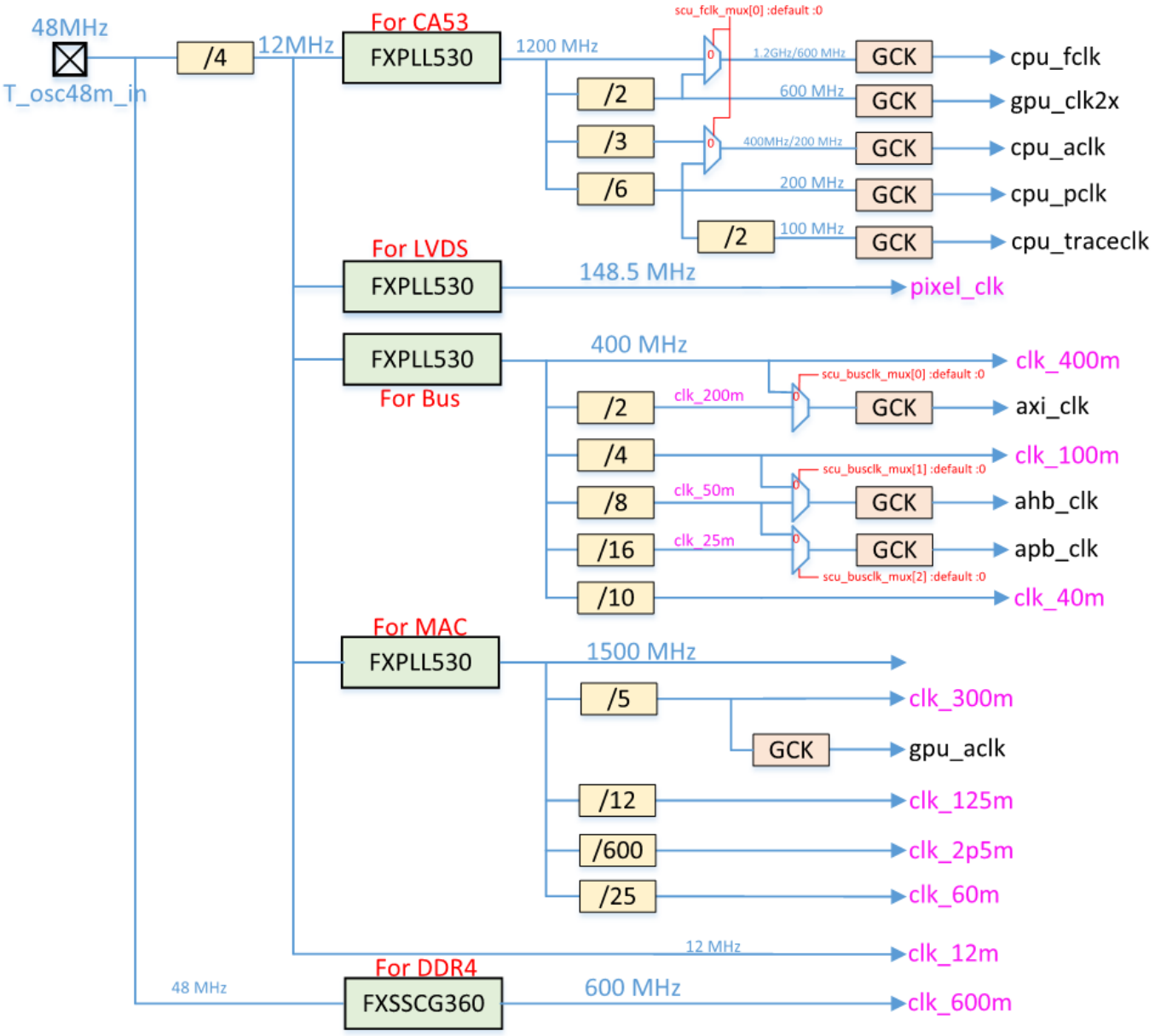
Clock and Reset

This chapter contains the following sections:

5. System Clock Diagram
6. External Clock Sources
7. PLL Programming Values
8. Summary Clock Information
9. Reset sheme

5. System Clock Diagram

The clock diagram is shown in *Figure 0-1* and *Figure 0-2*. The main crystal input is 48 MHz to provide the clock source for PLLs. PLLs are used to generate clock for CPU, AHB bus, APB bus, MAC and peripheral IPs.



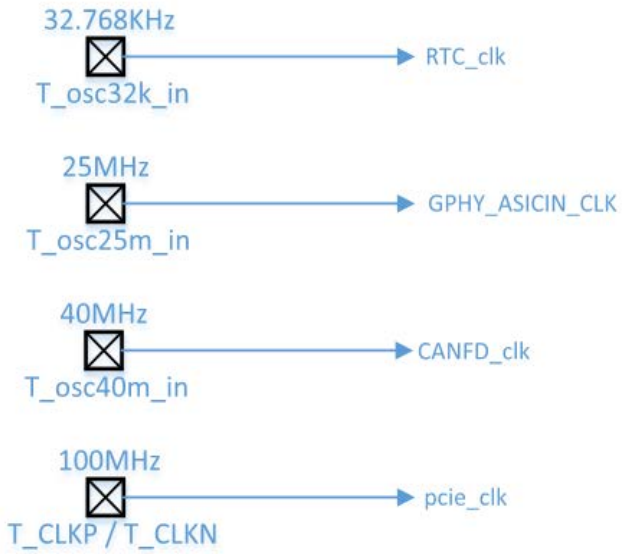


Figure 0-1. PLL Clock Distribution Diagram

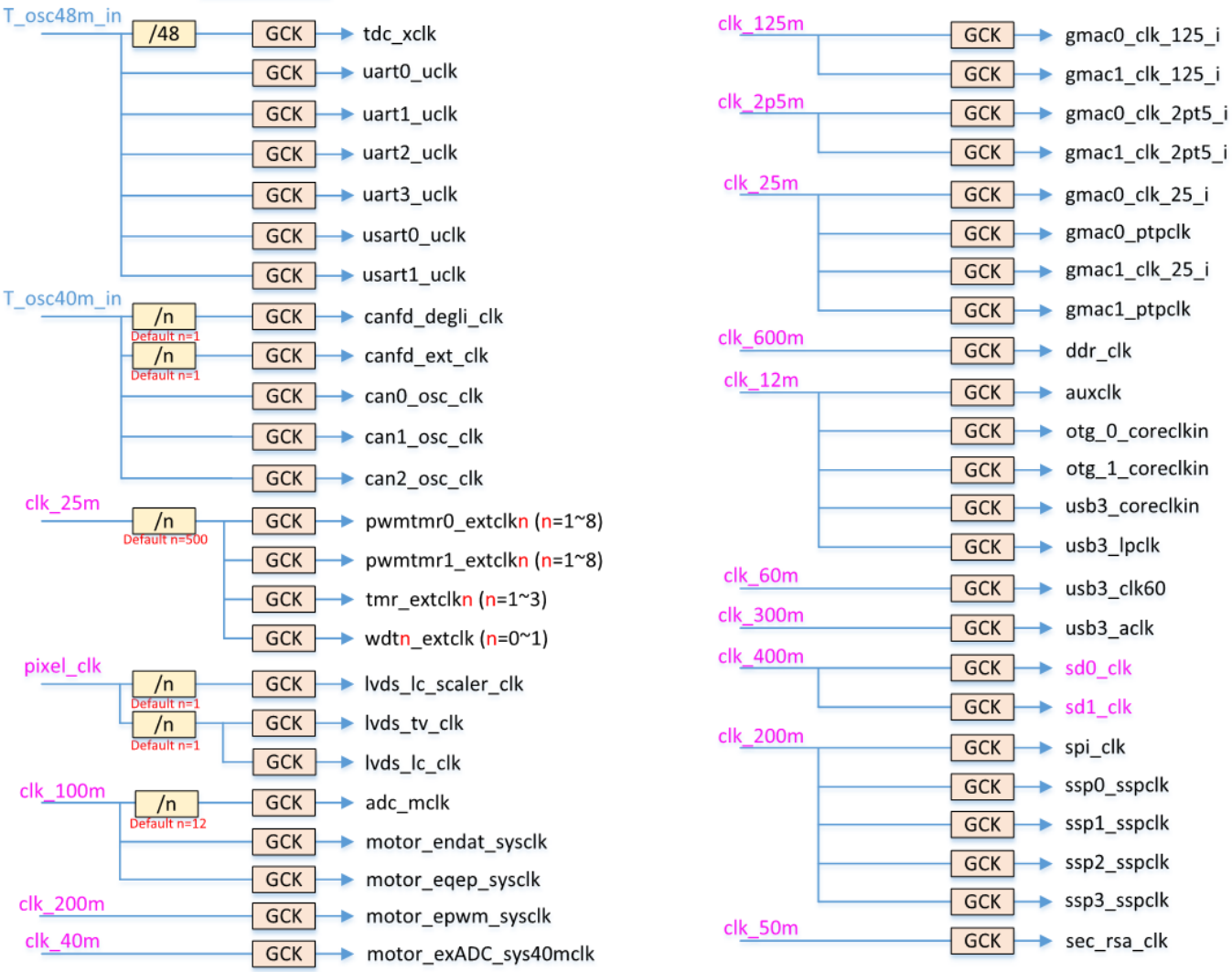


Figure 0-2. Peripheral IP Clock Distribution Diagram

6. External Clock Sources

There is five external clock source connected to the device. The clock source is shown in Table 0-1. For typical case, crystal pad power up stable time is 10ms. Recommend that power on stable time of crystal oscillator component has to be less than 10ms.

Table 0-1. External Clock Source

Clock Name	Clock Frequency	Detailed Information
X_OSC25M_IN X_OSC25M_IO	25 MHz	Used to drive GEPHY. Default enabled
X_OSC40M_IN X_OSC40M_IO	40 MHz	Used to drive CAN-BUS. Default enabled
X_OSC48M_IN X_OSC48M_IO	48MHz	Used to drive PLL and UART. Default enabled
X_PCIE_CLKN X_PCIE_CLKP	100MHz	Used to drive PCIE block Default enabled
X_OSC32K_IN X_OSC32K_IO	32.768KHz	Used to drive RTC block. Default enabled

7. PLL Programming Values

Table 0-2 defines the PLL used in this device as well as the default output frequencies and the corresponding default programming values. All input registers of PLLs could be changed through FCS flow.

Table 0-2. PLL Default Frequencies and Programming Values

PLL Instance	Programming MS	Programming NS	Programming PS	PLL Clock Out (MHz)
u_PLL_CPU	1	100	0	1200
u_PLL_LVDS	1	99	4	148.5
u_PLL_BUS	3	400	2	400
u_PLL_GMAC	1	125	0	1500

PLL Instance	SSEN	NF[23:0]	NS[11:0]	NV[23:0]	PLL Clock Out (MHz)
u_PLL_DDR4	0	819200	0	0	600

8. Summary Clock Information

This section summarizes the clock of PLL names, frequencies, and source.

Table 0-3. Clock Summary

IP Block	Clock Name	Frequency (MHz)	Clock Source from	Clock Group
CA53Q_subsystem	CA53 FCLK	1200 / 600	u_PLL_CPU/CKOUT u_PLL_CPU/CKOUT div 2	pll0_ckout
CA53Q_subsystem	CA53 AXI clock	400 / 200	u_PLL_CPU/CKOUT div 3 u_PLL_CPU/CKOUT div 6	pll0_ckout
CA53Q_subsystem	CA53 pclk	200	u_PLL_CPU/CKOUT div 6	pll0_ckout
CA53Q_subsystem	CA53 trace clock	100	u_PLL_CPU/CKOUT div 12	pll0_ckout
VIVANTE_GPU	GPU clock	600	u_PLL_CPU/CKOUT div 2	pll0_ckout
	pixel_clk	148.5	u_PLL_LVDS/CKOUT	pll1_ckout
FTLCDC210_UP	lvds_lc_scaler_clk	pixel_clk div n	u_PLL_LVDS/CKOUT div n	pll1_ckout
FTLCDC210_UP	lvds_tv_clk	pixel_clk div n	u_PLL_LVDS/CKOUT div n	pll1_ckout
FTLCDC210_UP	lvds_lc_clk	pixel_clk div n	u_PLL_LVDS/CKOUT div n	pll1_ckout
	axi_clk	400 / 200	u_PLL_BUS/CKOUT u_PLL_BUS/CKOUT div 2	pll2_ckout
	ahb_clk	100 / 50	u_PLL_BUS/CKOUT div 4 u_PLL_BUS/CKOUT div 8	pll2_ckout
	apb_clk	50 / 25	u_PLL_BUS/CKOUT div 8 u_PLL_BUS/CKOUT div 16	pll2_ckout
FTSDC021	SD card 2x clock	400	u_PLL_BUS/CKOUT	pll2_ckout
FTSDC021	SD card 1x clock	200	u_PLL_BUS/CKOUT div 2	pll2_ckout
FTSPI020	spi_clk	200	u_PLL_BUS/CKOUT div 2	pll2_ckout
FTSSP010	sspX_sspclk (X=0~3)	200	u_PLL_BUS/CKOUT div 2	pll2_ckout
FTADCC010	adc_mclk	100 div n	u_PLL_BUS/CKOUT div 4	pll2_ckout
Motor_Top	motor_endat_sysclk	100	u_PLL_BUS/CKOUT div 4	pll2_ckout
Motor_Top	motor_epwm_sysclk	200	u_PLL_BUS/CKOUT div 4	pll2_ckout
Motor_Top	motor_eqep_sysclk	100	u_PLL_BUS/CKOUT div 4	pll2_ckout

IP Block	Clock Name	Frequency (MHz)	Clock Source from	Clock Group
Motor_Top	motor_exADC_sys40mclk	40	u_PLL_BUS/CKOUT div 10	pll2_ckout
Soteria	sec_rsa_clk	50	u_PLL_BUS/CKOUT div 8	pll2_ckout
FTPWMTMR010	pwmtmr0_extclkX (X=1~8)	25 div n	u_PLL_BUS/CKOUT div 16	pll2_ckout
FTPWMTMR010	pwmtmr1_extclkX (X=1~8)	25 div n	u_PLL_BUS/CKOUT div 16	pll2_ckout
FTTMR010	tmr_extclkX (X=1~3)	25 div n	u_PLL_BUS/CKOUT div 16	pll2_ckout
FTWDT011	wdtX_extclk (X=0~1)	25 div n	u_PLL_BUS/CKOUT div 16	pll2_ckout
FTGMAC030	gmacX_clk_25_i (X=0~1)	25	u_PLL_BUS/CKOUT div 16	pll2_ckout
FTGMAC030	gmacX_ptpclk (X=0~1)	25	u_PLL_BUS/CKOUT div 16	pll2_ckout
VIVANTE_GPU	gpu_aclk	300	u_PLL_GMAC/CKOUT div 5	pll3_ckout
FOTG330	usb3_aclk	300	u_PLL_GMAC/CKOUT div 5	pll3_ckout
FOTG330	usb3_clk60	60	u_PLL_GMAC/CKOUT div 25	pll3_ckout
FTGMAC030	gmacX_clk_125_i (X=0~1)	125	u_PLL_GMAC/CKOUT div 12	pll3_ckout
FTGMAC030	gmacX_clk_2pt5_i (X=0~1)	2.5	u_PLL_GMAC/CKOUT div 600	pll3_ckout
SBS_DDR_TOP	ddr_clk	600	u_PLL_DDR4/CKOUT1	pll4_ckout
FTUART010	uartX_uclk (X=0~3)	48	T_osc48m_in	OSC 48M
FTUSART010	usartX_uclk (X=0~1)	48	T_osc48m_in	OSC 48M
SBS_PCIE	auxclk	12	T_osc48m_in div 4	OSC 48M
FZOTG268NSHJ0P	otg_0_coreclkin	12	T_osc48m_in div 4	OSC 48M
FZOTG268NSHJ0P	otg_1_coreclkin	12	T_osc48m_in div 4	OSC 48M
FZOTG306NSHJ0P	usb3_coreclkin	12	T_osc48m_in div 4	OSC 48M
FOTG330	usb3_lpclk	12	T_osc48m_in div 4	OSC 48M
FTTDCC010	tdc_xclk	1	T_osc48m_in div 48	OSC 48M
FXEDP412EWHJ0P	I_ASICIN	25	25MHz OSC	OSC 25M

IP Block	Clock Name	Frequency (MHz)	Clock Source from	Clock Group
d_ip_flexcan3_syn	canfd_degli_clk	40 div n	T_osc40m_in div n	OSC 40M
d_ip_flexcan3_syn	canfd_ext_clk	40 div n	T_osc40m_in div n	OSC 40M
d_ip_flexcan3_syn	canX_osc_clk (X=0~2)	40	T_osc40m_in	OSC 40M
SBS_PCIE	refclk_p	100	T_CLKP	PCB 100M OSC
SBS_PCIE	refclk_n	100	T_CLKN	PCB 100M OSC

9. Reset sheme

Unimap-TX5 supports one external reset event. Table 0-4 defines the reset events which cause the SoC system to return to the initial power-up state.

Table 0-4. Global Reset Events

Reset Event	Pin Name/ Signal Name	IP Blocks NOT Reset by this Event	Initialization State Machine Invoked by this Event	Description
Power On Reset	POR	None	Yes	From internal Power-On Reset circuit
Hardware Reset	X_RSTN	None	Yes	The reset is provided by external pin.
Software Reset	scu_sw_reset[109:0]	All blocks except the target of the reset are left in their alive, non-reset, operational state.	No	This reset allows software to reset an individual IP block. Reset is assumed to be asynchronous and active low. Reset will be synchronized to the clock domain in which it is used.

Memory Map

This chapter contains the following section:

10. Bus Architecture
11 CA53 CoreSight address mapping

10. Bus Architecture

The bus architecture in SoC is constructed by AXI, AHB and APB bus. *Figure 0-1* describes all base addresses of the AMBA slaves.

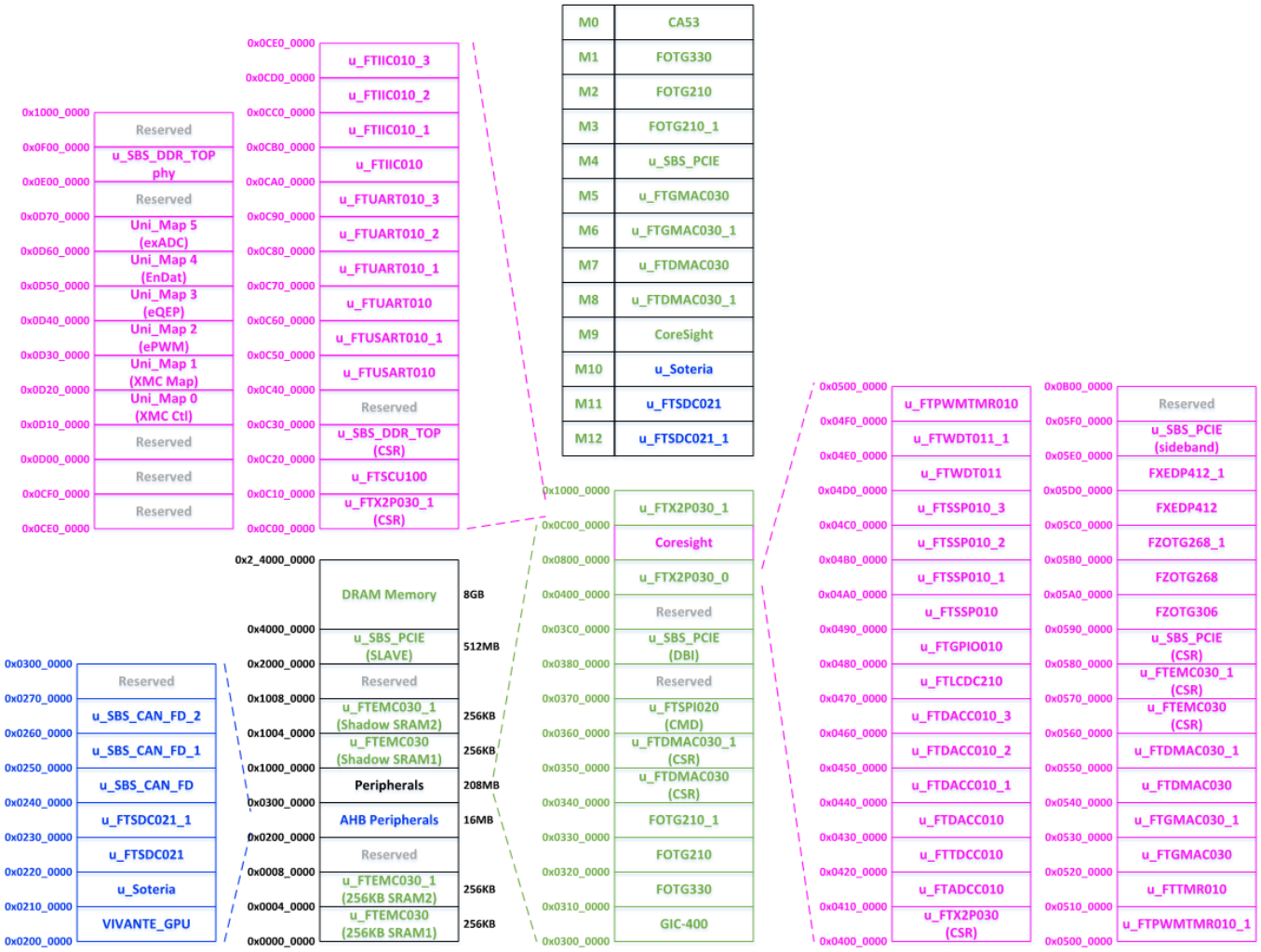


Figure 0-1. Memory Mapping Diagram

Table 0-1 and *Table 0-2* show all master and slave channels in NIC interconnect.
Table 0-3 and *Table 0-4* show all APB slave information.

Table 0-1. NIC400 Master Channel

	Instance Name	Description	Bus Type	Bus Width (bits)
M0	CA53	CA53 CPU	AXI	128
M1	u_SBS_USB3_TOP	USB3 controller	AXI	64
M2	u_SBS_OTG2_TOP	USB2 controller 0	AXI	64
M3	u_SBS_OTG2_TOP_1	USB2 controller 1	AXI	64
M4	u_SBS_PCIE	PCIE	AXI	128
M5	u_SBS_GMAC	GMAC controller 0	AXI	64
M6	u_SBS_GMAC_1	GMAC controller 1	AXI	64
M7	u_FTDMAC030	DMA controller 0	AXI	64
M8	u_FTDMAC030_1	DMA controller 1	AXI	64
M9	CA53 CoreSight	CA53 CoreSight	AXI	64
M10	u_Soteria	Security Engine	AHB	32
M11	u_FTSDC021	SD Card controller 0 registers	AHB	32
M12	u_FTSDC021_1	SD Card controller 1 registers	AHB	32

Table 0-2. NIC400 Slave Memory Space Address Map

	Instance Name	Description	Base Definition	Size (bytes)
S0	u_FTEMC030_0	SRAM1	0x0000_0000	256K
S1	u_FTEMC030_1	SRAM2	0x0004_0000	256K
S2	u_VIVANTE_GPU	GPU register	0x0200_0000	1M
S3	u_Soteria	Security Engine	0x0210_0000	1M
S4	u_FTSDC021_0	SD Card controller 0 registers	0x0220_0000	1M
S5	u_FTSDC021_1	SD Card controller 1 registers	0x0230_0000	1M
S6	u_GIC400	Interrupt Controller registers	0x0300_0000	1M
S7	u_SBS_USB3_TOP	USB3 controller registers	0x0310_0000	1M
S8	u_SBS_OTG2_TOP	USB2 controller 0 registers	0x0320_0000	1M
S9	u_SBS_OTG2_TOP_1	USB2 controller 1 registers	0x0330_0000	1M
S10	u_FTDMAC	DMA controller 0	0x0340_0000	1M
S11	u_FTDMAC_1	DMA controller 1	0x0350_0000	1M
S12	u_FTSPI020	SPI-Flash controller command port	0x0360_0000	1M
S13	u_SBS_PCIE	PCIE DBI bus	0x0380_0000	4M
S14	u_FTX2P030	AXI to APB bridge 0	0x0400_0000	64M
S15	u_FTX2P030_1	AXI to APB bridge 1	0x0C00_0000	64M
S16	u_SBS_PCIE	PCIE slave bus	0x2000_0000	512M
S17	u_SBS_DDR_TOP	DDR channel 0	0x4000_0000	8G
S18	u_SBS_DDR_TOP	DDR channel 1	0x4000_0000	8G
S19	u_SBS_CAN_FD	CAN-bus controller 0 registers	0x0240_0000	1M
S20	u_SBS_CAN_FD_1	CAN-bus controller 1 registers	0x0250_0000	1M
S21	u_SBS_CAN_FD_2	CAN-bus controller 2 registers	0x0260_0000	1M

	Instance Name	Description	Base Definition	Size (bytes)
S22	CoreSight	CA53 CoreSight APB registers	0x0800_0000	64M
S0’	u_FTEMC030_0 (shadow of 0x0000_0000)	SRAM1	0x1000_0000	256K
S1’	u_FTEMC030_1 (shadow of 0x0004_0000)	SRAM2	0x1004_0000	256K

Table 0-3. AXI to APB Bridge 0 Memory Space Address Map

	Instance Name	Description	Base Definition	Size (byte)
S30	u_FTX2P030	AXI to APB bridge 0 registers	0x0400_0000	1M
S31	u_FTADCC010	ADC controller registers	0x0410_0000	1M
S32	u_FTTDCC010	TDC controller registers	0x0420_0000	1M
S33	u_FTDACC010	DAC controller 0 registers	0x0430_0000	1M
S34	u_FTDACC010_1	DAC controller 1 registers	0x0440_0000	1M
S35	u_FTDACC010_2	DAC controller 2 registers	0x0450_0000	1M
S36	u_FTDACC010_3	DAC controller 3 registers	0x0460_0000	1M
S37	u_FTLCDC210_UP	LCDC controller registers	0x0470_0000	1M
S38	u_FTGPIO010	GPIO controller registers	0x0480_0000	1M
S39	u_FTSSP010	SSP controller 0 registers	0x0490_0000	1M
S40	u_FTSSP010_1	SSP controller 1 registers	0x04A0_0000	1M
S41	u_FTSSP010_2	SSP controller 2 registers	0x04B0_0000	1M
S42	u_FTSSP010_3	SSP controller 3 registers	0x04C0_0000	1M
S43	u_FTWDT011	Watchdog controller 0 registers	0x04D0_0000	1M
S44	u_FTWDT011_1	Watchdog controller 1 registers	0x04E0_0000	1M
S45	u_FTPWMTMR010	PWM timer controller 0 registers	0x04F0_0000	1M
S46	u_FTPWMTMR010_1	PWM timer controller 1 registers	0x0500_0000	1M

	Instance Name	Description	Base Definition	Size (byte)
S47	u_FTTMR010	Timer controller registers	0x0510_0000	1M
S48	u_SBS_GMAC	GMAC controller 0 registers	0x0520_0000	1M
S49	u_SBS_GMAC_1	GMAC controller 1 registers	0x0530_0000	1M
S50	u_FTDMAC030	DMAC controller 0 registers	0x0540_0000	1M
S51	u_FTDMAC030_1	DMAC controller 1 registers	0x0550_0000	1M
S52	u_FTEMC030	Embedded memory controller 0 registers	0x0560_0000	1M
S53	u_FTEMC030_1	Embedded memory controller 1 registers	0x0570_0000	1M
S54	u_SBS_PCIE_TOP	PCIE CR interface registers	0x0580_0000	1M
S55	u_SBS_USB3_TOP	USB3 PHY registers	0x0590_0000	1M
S56	u_SBS_OTG2_TOP	USB2 PHY 0 registers	0x05A0_0000	1M
S57	u_SBS_OTG2_TOP_1	USB2 PHY 1 registers	0x05B0_0000	1M
S58	u_SBS_GMAC	Ethernet PHY 0 registers	0x05C0_0000	1M
S59	u_SBS_GMAC_1	Ethernet PHY 1 registers	0x05D0_0000	1M
S60	u_SBS_PCIE	PCIE sideband registers	0x05E0_0000	1M

Table 0-4. AXI to APB Bridge 1 Memory Space Address Map

	Instance Name	Description	Base Definition	Size (byte)
S71	u_FTX2P030_1	AXI to APB Bridge 1 registers	0x0C00_0000	1M
S72	u_FTSCU100	FTSCU100 registers	0x0C10_0000	1M
S73	u_SBS_DDR_TOP	DDR controller registers	0x0C20_0000	1M
S74	u_SBS_DDR_TOP	DDR PHY registers	0x0E00_0000	16M
S75	u_FTUSART010_0	USART controller 0 registers	0x0C40_0000	1M

	Instance Name	Description	Base Definition	Size (byte)
S76	u_FTUSART010_1	USART controller 1 registers	0x0C50_0000	1M
S77	u_FTUART010_0	UART controller 0 registers	0x0C60_0000	1M
S78	u_FTUART010_1	UART controller 1 registers	0x0C70_0000	1M
S79	u_FTUART010_2	UART controller 2 registers	0x0C80_0000	1M
S80	u_FTUART010_3	UART controller 3 registers	0x0C90_0000	1M
S81	u_FTIIC010_0	IIC controller 0 registers	0x0CA0_0000	1M
S82	u_FTIIC010_1	IIC controller 1 registers	0x0CB0_0000	1M
S83	u_FTIIC010_2	IIC controller 2 registers	0x0CC0_0000	1M
S84	u_FTIIC010_3	IIC controller 3 registers	0x0CD0_0000	1M
S85	reserved	-	0x0CE0_0000	1M
S86	reserved	-	0x0CF0_0000	1M
S87	reserved	-	0x0D00_0000	1M
S88	u_Motor_Top	XMC I/F CTL registers	0x0D10_0000	1M
S89	u_Motor_Top	XMC I/F MAP registers	0x0D20_0000	1M
S90	u_Motor_Top	ePWM registers	0x0D30_0000	1M
S91	u_Motor_Top	eQEP registers	0x0D40_0000	1M
S92	u_Motor_Top	EnDAT registers	0x0D50_0000	1M
S93	u_Motor_Top	exADC registers	0x0D60_0000	1M

Figure 0-2 shows the mutual access permissions between all AMBA masters and slaves on the NIC400 bus matrix.

		M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Remark
		CA53	USB3_SBS	USB2_SBS_0	USB2_SBS_1	PCI-E SBS	GMAC_0	GMAC_1	DMAC_0	DMAC_1	CoreSight	Soteria	SDC021_0	SDC021_1	
Name	Module														
S0	SRAM1 (EMC_1)	0	0	0	0	0	0	0	0	0	0	0	0	0	
S1	SRAM2 (EMC_2)	0	0	0	0	0	0	0	0	0	0	0	0	0	
S2	VIVANTE_GPU	0									0	0			
S3	Soteria	0									0	0			
S4	SDC021_0 reg	0							0	0	0	0			
S5	SDC021_1 reg	0							0	0	0	0			
S6	GIC-400	0									0	0			
S7	USB3_subsys	0				0					0	0			
S8	USB2_subsys_0	0									0	0			
S9	USB2_subsys_1	0									0	0			
S10	DMAC_0	0									0	0			
S11	DMAC_1	0									0	0			
S12	SPI020 cmd	0		0					0	0	0	0			
S13	PCI-E subsys DBI	0							0	0	0	0			
S14	FTX2P030_0	0							0	0	0	0			
S15	FTX2P030_1	0							0	0	0	0			
S16	PCI-E subsys	0							0	0	0	0			
S17	DDR4 Memory	0		0		0		0	0		0	0	0		
S18	DDR4 Memory		0		0		0			0				0	
S19	CAN-FD_0	0							0	0	0	0			
S20	CAN-FD_1	0							0	0	0	0			
S21	CAN-FD_2	0							0	0	0	0			
S22	Coresight	0							0	0	0				

Figure 0-2. NIC400 Interconnect matrix

11. CA53 CoreSight address mapping table

CoreSight SoC-400 is a solution for debug and trace of complex SoCs. *Table 0-5* shows the address mapping table of CoreSight in Unimap-TX5.

Table 0-5. ARM CoreSight SoC-400 address mapping table (base-address : 0x0800_0000)

Address Offset	Description
0x0000_0000	CS ROM Table
0x0000_1000	CoreSight Cross Trigger Interface (CTI)
0x0000_2000	CoreSight Embedded Trace Buffer (ETB)
0x0000_3000	CoreSight ATB Funnel
0x0000_4000	TimeStamp
0x0000_5000	TimeStamp_read
0x0000_6000	Timer
0x0000_7000	Timer_read
0x0000_8000	CoreSight ATB Replicator
0x0000_9000	CoreSight Trace Port Interface Unit (TPIU)
0x0100_0000	Cortex-A53 ROM Table
0x0101_0000	Cortex-A53 CPU0 DBG
0x0102_0000	Cortex-A53 CPU0 CTI
0x0103_0000	Cortex-A53 CPU0 PMU
0x0104_0000	Cortex-A53 CPU0 ETM
0x0111_0000	Cortex-A53 CPU1 DBG
0x0112_0000	Cortex-A53 CPU1 CTI
0x0113_0000	Cortex-A53 CPU1 PMU

Address Offset	Description
0x0114_0000	Cortex-A53 CPU1 ETM
0x0121_0000	Cortex-A53 CPU2 DBG
0x0122_0000	Cortex-A53 CPU2 CTI
0x0123_0000	Cortex-A53 CPU2 PMU
0x0124_0000	Cortex-A53 CPU2 ETM
0x0131_0000	Cortex-A53 CPU3 DBG
0x0132_0000	Cortex-A53 CPU3 CTI
0x0133_0000	Cortex-A53 CPU3 PMU
0x0134_0000	Cortex-A53 CPU3 ETM

Interrupts and DMA Connection

This chapter contains the following section:

12. Interrupt Source
13. DMA Request and Ack connection

12. Interrupt Source

Interrupts within Unimap-TX5 as listed in *Table 0-1*.

Table 0-1. IRQ Interrupt Sources

	Instance	Pin Name	Trigger Mode (LEVEL, EDGE)	Trigger Level LEVEL: HIGH, LOW EDGE: RISING, FALLING
IRQ 0	u_FTSCU100	sys_int	LEVEL	HIGH
IRQ 1	u_SBS_USB3_TOP	fotg330_intr	LEVEL	HIGH
IRQ 2	u_SBS_OTG2_TOP	sys_int_n [1]	LEVEL	HIGH
IRQ 3	u_SBS_OTG2_TOP_1	sys_int_n [1]	LEVEL	HIGH
IRQ 4	u_SBS_PCIE	PCIE_INTR	LEVEL	HIGH
IRQ 5	u_SBS_GMAC/ u_FTGMAC030	mac_int	LEVEL	HIGH
IRQ 6	u_SBS_GMAC_1/ u_FTGMAC030	mac_int	LEVEL	HIGH
IRQ 7	u_FTDMAC030	DMACIntr	LEVEL	HIGH
IRQ 8	u_FTDMAC030	DMACIntr_tc	LEVEL	HIGH
IRQ 9	u_FTDMAC030	DMACIntr_err	LEVEL	HIGH
IRQ 10	u_FTDMAC030_1	DMACIntr	LEVEL	HIGH
IRQ 11	u_FTDMAC030_1	DMACIntr_tc	LEVEL	HIGH
IRQ 12	u_FTDMAC030_1	DMACIntr_err	LEVEL	HIGH
IRQ 13	u_SOTERIA2	sec_subsys_intr	LEVEL	HIGH
IRQ 14	u_FTSDC021	sdc_int	LEVEL	HIGH
IRQ 15	u_FTSDC021_1	sdc_int	LEVEL	HIGH
IRQ 16	u_VIVANTE_GPU	xaq2_intr	LEVEL	HIGH
IRQ 17	u_FTLCDC210_UP	LC_INTR	LEVEL	HIGH

	Instance	Pin Name	Trigger Mode (LEVEL, EDGE)	Trigger Level LEVEL: HIGH, LOW EDGE: RISING, FALLING
IRQ 18	u_FTLCDC210_UP	LC_VSTATUSINTR	LEVEL	HIGH
IRQ 19	u_FTLCDC210_UP	LC_BAUPDINTR	LEVEL	HIGH
IRQ 20	u_FTLCDC210_UP	LC_FURINTR	LEVEL	HIGH
IRQ 21	u_FTLCDC210_UP	MERRINTR	LEVEL	HIGH
IRQ 22	-	-	-	-
IRQ 23	-	-	-	-
IRQ 24	u_FTSPI020	intr	LEVEL	HIGH
IRQ 25	u_SBS_DDR_TOP	dfi_alert_err_intr	LEVEL	HIGH
IRQ 26	u_FTX2P030	DecErrIntr	LEVEL	HIGH
IRQ 27	u_FTX2P030_1	DecErrIntr	LEVEL	HIGH
IRQ 28	u_FTADCC010	adc_int	LEVEL	HIGH
IRQ 29	U_FTTDCC010	tdc_int	LEVEL	HIGH
IRQ 30	u_FTDACC010	dacc_intr	LEVEL	HIGH
IRQ 31	u_FTDACC010_1	dacc_intr	LEVEL	HIGH
IRQ 32	u_FTDACC010_2	dacc_intr	LEVEL	HIGH
IRQ 33	u_FTDACC010_3	dacc_intr	LEVEL	HIGH
IRQ 34	u_FTGPIO010	gpio_intr	LEVEL	HIGH
IRQ 35	u_FTSSP010	ssp_intr	LEVEL	HIGH
IRQ 36	u_FTSSP010_1	ssp_intr	LEVEL	HIGH
IRQ 37	u_FTSSP010_2	ssp_intr	LEVEL	HIGH
IRQ 38	u_FTSSP010_3	ssp_intr	LEVEL	HIGH
IRQ 39	u_FTWDT011	wdt_intr	LEVEL	HIGH

	Instance	Pin Name	Trigger Mode (LEVEL, EDGE)	Trigger Level LEVEL: HIGH, LOW EDGE: RISING, FALLING
IRQ 40	u_FTWDT011	wdt_ovf_intr	LEVEL	HIGH
IRQ 41	u_FTWDT011	wdt_udf_intr	LEVEL	HIGH
IRQ 42	u_FTWDT011_1	wdt_intr	LEVEL	HIGH
IRQ 43	u_FTWDT011_1	wdt_ovf_intr	LEVEL	HIGH
IRQ 44	u_FTWDT011_1	wdt_udf_intr	LEVEL	HIGH
IRQ 45	u_FTPWMTMR010	tmr_intr	LEVEL	HIGH
IRQ 46	u_FTPWMTMR010	tmr1_intr	LEVEL	HIGH
IRQ 47	u_FTPWMTMR010	tmr2_intr	LEVEL	HIGH
IRQ 48	u_FTPWMTMR010	tmr3_intr	LEVEL	HIGH
IRQ 49	u_FTPWMTMR010	tmr4_intr	LEVEL	HIGH
IRQ 50	u_FTPWMTMR010	tmr5_intr	LEVEL	HIGH
IRQ 51	u_FTPWMTMR010	tmr6_intr	LEVEL	HIGH
IRQ 52	u_FTPWMTMR010	tmr7_intr	LEVEL	HIGH
IRQ 53	u_FTPWMTMR010	tmr8_intr	LEVEL	HIGH
IRQ 54	u_FTPWMTMR010_1	tmr_intr	LEVEL	HIGH
IRQ 55	u_FTPWMTMR010_1	tmr1_intr	LEVEL	HIGH
IRQ 56	u_FTPWMTMR010_1	tmr2_intr	LEVEL	HIGH
IRQ 57	u_FTPWMTMR010_1	tmr3_intr	LEVEL	HIGH
IRQ 58	u_FTPWMTMR010_1	tmr4_intr	LEVEL	HIGH
IRQ 59	u_FTPWMTMR010_1	tmr5_intr	LEVEL	HIGH
IRQ 60	u_FTPWMTMR010_1	tmr6_intr	LEVEL	HIGH
IRQ 61	u_FTPWMTMR010_1	tmr7_intr	LEVEL	HIGH

	Instance	Pin Name	Trigger Mode (LEVEL, EDGE)	Trigger Level LEVEL: HIGH, LOW EDGE: RISING, FALLING
IRQ 62	u_FTPWMTMR010_1	tmr8_intr	LEVEL	HIGH
IRQ 63	u_FTTMR010	tmr_intr	LEVEL	HIGH
IRQ 64	u_FTTMR010	tmr1_intr	EDGE	RISING
IRQ 65	u_FTTMR010	tmr2_intr	EDGE	RISING
IRQ 66	u_FTTMR010	tmr3_intr	EDGE	RISING
IRQ 67	u_FTUSART010	usart_int	LEVEL	HIGH
IRQ 68	u_FTUSART010_1	usart_int	LEVEL	HIGH
IRQ 69	u_FTUART010	irda_intr1	LEVEL	HIGH
IRQ 70	u_FTUART010_1	irda_intr1	LEVEL	HIGH
IRQ 71	u_FTUART010_2	irda_intr1	LEVEL	HIGH
IRQ 72	u_FTUART010_3	irda_intr1	LEVEL	HIGH
IRQ 73	u_FTIIC010	isi2c	LEVEL	HIGH
IRQ 74	u_FTIIC010_1	isi2c	LEVEL	HIGH
IRQ 75	u_FTIIC010_2	isi2c	LEVEL	HIGH
IRQ 76	u_FTIIC010_3	isi2c	LEVEL	HIGH
IRQ 77	u_CAN_FD	canfd_irq	LEVEL	HIGH
IRQ 78	u_CAN_FD_1	canfd_irq	LEVEL	HIGH
IRQ 79	u_CAN_FD_2	canfd_irq	LEVEL	HIGH
IRQ 80	u_Motor_Top	ePWM_INT[1]	LEVEL	HIGH
IRQ 81	u_Motor_Top	ePWM_INT[2]	LEVEL	HIGH
IRQ 82	u_Motor_Top	ePWM_INT[3]	LEVEL	HIGH
IRQ 83	u_Motor_Top	ePWM_INT[4]	LEVEL	HIGH

	Instance	Pin Name	Trigger Mode (LEVEL, EDGE)	Trigger Level LEVEL: HIGH, LOW EDGE: RISING, FALLING
IRQ 84	u_Motor_Top	ePWM_INT[5]	LEVEL	HIGH
IRQ 85	u_Motor_Top	ePWM_INT[6]	LEVEL	HIGH
IRQ 86	u_Motor_Top	ePWM_INT[7]	LEVEL	HIGH
IRQ 87	u_Motor_Top	ePWM_INT[8]	LEVEL	HIGH
IRQ 88	u_Motor_Top	ePWM_INT[9]	LEVEL	HIGH
IRQ 89	u_Motor_Top	eQep_int[0]	LEVEL	HIGH
IRQ 90	u_Motor_Top	eQep_int[1]	LEVEL	HIGH
IRQ 91	u_Motor_Top	eQep_int[2]	LEVEL	HIGH
IRQ 92	u_Motor_Top	eQep_int[3]	LEVEL	HIGH
IRQ 93	u_Motor_Top	eQep_int[4]	LEVEL	HIGH
IRQ 94	u_Motor_Top	eQep_int[5]	LEVEL	HIGH
IRQ 95	u_Motor_Top	eQep_int[6]	LEVEL	HIGH
IRQ 96	u_Motor_Top	eQep_int[7]	LEVEL	HIGH
IRQ 97	u_Motor_Top	eQep_int[8]	LEVEL	HIGH
IRQ 98	u_Motor_Top	eQep_int[9]	LEVEL	HIGH
IRQ 99	u_Motor_Top	eQep_int[10]	LEVEL	HIGH
IRQ 100	u_Motor_Top	eQep_int[11]	LEVEL	HIGH
IRQ 101	u_Motor_Top	EnDat_int[0]	LEVEL	HIGH
IRQ 102	u_Motor_Top	EnDat_int[1]	LEVEL	HIGH
IRQ 103	u_Motor_Top	EnDat_int[2]	LEVEL	HIGH
IRQ 104	u_SBS_DDR_TOP	dwc_ddrphy_int_n [1]	LEVEL	HIGH
IRQ 105	u_CAN_FD	camfd_wake_irq	LEVEL	HIGH

	Instance	Pin Name	Trigger Mode (LEVEL, EDGE)	Trigger Level LEVEL: HIGH, LOW EDGE: RISING, FALLING
IRQ 106	u_CAN_FD_1	camfd_wake_irq	LEVEL	HIGH
IRQ 107	u_CAN_FD_2	camfd_wake_irq	LEVEL	HIGH
IRQ 108	u_SBS_GMAC/ u_FXEDP410EWHJ0P	po_int_n [1]	LEVEL	HIGH
IRQ 109	u_SBS_GMAC_1/ u_FXEDP410EWHJ0P	po_int_n [1]	LEVEL	HIGH
IRQ 110	CA53Q_subsystem	nPMUIRQ[0]	LEVEL	LOW
IRQ 111	CA53Q_subsystem	nPMUIRQ[1]	LEVEL	LOW
IRQ 112	CA53Q_subsystem	nPMUIRQ[2]	LEVEL	LOW
IRQ 113	CA53Q_subsystem	nPMUIRQ[3]	LEVEL	LOW
IRQ 114	CA53Q_subsystem	nCOMMIRQ[0]	LEVEL	LOW
IRQ 115	CA53Q_subsystem	nCOMMIRQ[1]	LEVEL	LOW
IRQ 116	CA53Q_subsystem	nCOMMIRQ[2]	LEVEL	LOW
IRQ 117	CA53Q_subsystem	nCOMMIRQ[3]	LEVEL	LOW
IRQ 118	CA53Q_subsystem	nVCPUMNTIRQ[0]	LEVEL	LOW
IRQ 119	CA53Q_subsystem	nVCPUMNTIRQ[1]	LEVEL	LOW
IRQ 120	CA53Q_subsystem	nVCPUMNTIRQ[2]	LEVEL	LOW
IRQ 121	CA53Q_subsystem	nVCPUMNTIRQ[3]	LEVEL	LOW

[1] The original interrupt signal polarity is low active. But for the consistency of the SoC integration, we inserted an inverter at the output side of the signals.

13. DMA Request and Ack connection

The DMA hardware handshaking mode request/ack connections to the Unimap DMA controller (FTDMAC030) are listed below in *Table 0-2*.

Table 0-2. DMA REQuest and ACK connection

Control Register	Source Instance 1	Source Pin Name 1	Source Instance 0	Source Pin Name 0	Destination DMAC	DMA channel Pin Name
-	-	-	u_FTSPI020	dma_req	u_FTDMAC030	dma_req[0]
-	-	-	u_FTUART010	irda_ntxrdy	u_FTDMAC030	dma_req[1]
-	-	-	u_FTUART010	irda_nrxrdy	u_FTDMAC030	dma_req[2]
-	-	-	u_FTUART010_1	irda_ntxrdy	u_FTDMAC030	dma_req[3]
-	-	-	u_FTUART010_1	irda_nrxrdy	u_FTDMAC030	dma_req[4]
0x0c10_087C[5]	u_FTSDC021	sd_dma_req	u_FTUSART010	dma_tx_req	u_FTDMAC030	dma_req[5]
0x0c10_087C[6]	u_FTSDC021_1	sd_dma_req	u_FTUSART010	dma_rx_req	u_FTDMAC030	dma_req[6]
-	-	-	u_FTSSP010	ssp_txdmareq_r	u_FTDMAC030	dma_req[7]
-	-	-	u_FTSSP010	ssp_rxdmareq_r	u_FTDMAC030	dma_req[8]
-	-	-	u_FTSSP010_1	ssp_txdmareq_r	u_FTDMAC030	dma_req[9]
-	-	-	u_FTSSP010_1	ssp_rxdmareq_r	u_FTDMAC030	dma_req[10]
0x0c10_087C[11]	u_FTPWMTMR010_1	tmr1_dma_req	u_ADC_TOP	adc_dma_req[0]	u_FTDMAC030	dma_req[11]
0x0c10_087C[12]	u_FTPWMTMR010_1	tmr2_dma_req	u_ADC_TOP	adc_dma_req[1]	u_FTDMAC030	dma_req[12]
0x0c10_087C[13]	u_FTPWMTMR010_1	tmr3_dma_req	u_ADC_TOP	adc_dma_req[2]	u_FTDMAC030	dma_req[13]
0x0c10_087C[14]	u_FTPWMTMR010_1	tmr4_dma_req	u_ADC_TOP	adc_dma_req[3]	u_FTDMAC030	dma_req[14]
-	-	-	u_CAN_FD_1	canfd_req	u_FTDMAC030	dma_req[15]
0x0c10_087C[16]	u_FTPWMTMR010_1	tmr5_dma_req	u_DAC_TOP	dac_dma_req	u_FTDMAC030_1	dma_req[0]
0x0c10_087C[17]	u_FTPWMTMR010_1	tmr6_dma_req	u_DAC_TOP_1	dac_dma_req	u_FTDMAC030_1	dma_req[1]

Control Register	Source Instance 1	Source Pin Name 1	Source Instance 0	Source Pin Name 0	Destination DMAC	DMA channel Pin Name
0x0c10_087C[18]	u_FTPWMTMR010_1	tmr7_dma_req	u_DAC_TOP_2	dac_dma_req	u_FTDMAC030_1	dma_req[2]
0x0c10_087C[19]	u_FTPWMTMR010_1	tmr8_dma_req	u_DAC_TOP_3	dac_dma_req	u_FTDMAC030_1	dma_req[3]
-	-	-	u_FTUART010_2	irda_ntxrdy	u_FTDMAC030_1	dma_req[4]
-	-	-	u_FTUART010_2	irda_nrxrdy	u_FTDMAC030_1	dma_req[5]
-	-	-	u_FTUART010_3	irda_ntxrdy	u_FTDMAC030_1	dma_req[6]
-	-	-	u_FTUART010_3	irda_nrxrdy	u_FTDMAC030_1	dma_req[7]
-	-	-	u_FTUSART010_1	dma_tx_req	u_FTDMAC030_1	dma_req[8]
-	-	-	u_FTUSART010_1	dma_rx_req	u_FTDMAC030_1	dma_req[9]
-	-	-	u_FTSSP010_2	ssp_txdmareq_r	u_FTDMAC030_1	dma_req[10]
0x0c10_087C[27]	u_FTSDC021	sd_dma_req	u_FTSSP010_2	ssp_rxdmareq_r	u_FTDMAC030_1	dma_req[11]
-	-	-	u_FTSSP010_3	ssp_txdmareq_r	u_FTDMAC030_1	dma_req[12]
0x0c10_087C[29]	u_FTSDC021_1	sd_dma_req	u_FTSSP010_3	ssp_rxdmareq_r	u_FTDMAC030_1	dma_req[13]
-	-	-	u_CAN_FD_2	canfd_req	u_FTDMAC030_1	dma_req[14]
-	-	-	u_CAN_FD_3	canfd_req	u_FTDMAC030_1	dma_req[15]

Figure 0-1 shows the transmitted waveform of FTDACC010 DMA access. When the user wants to trigger the DMA function of FTDACC010, DMA needs to check the last transferred data to ensure the time when the data transfer is complete. For this purpose, *Table 0-3* shows the control registers to determine which DMA channel of FTDMAC030_1 is used.

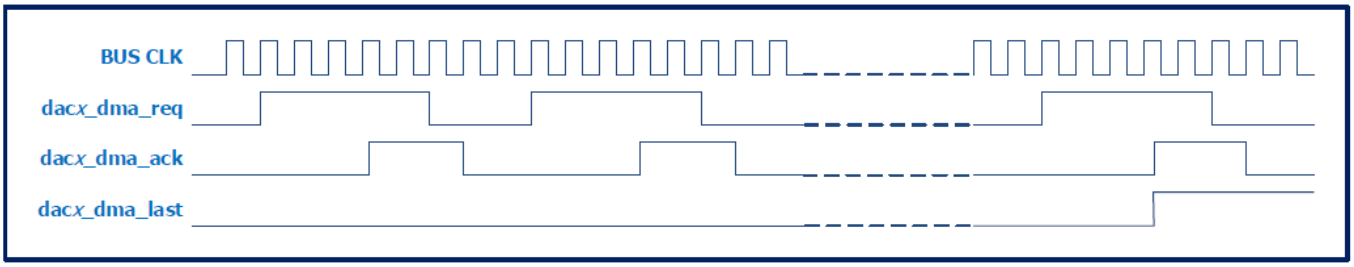


Figure 0-1. FTDACC010 DMA Request/ACK timing chart

Table 0-3. FTDACC010's dma_last Connetion Selection Register

Control Register	Register Name	Description
0x0c10_0878[19:16]	GLUE_reg_DAC_LAST_SEL[3:0]	FTDACC010_0 dma_last selection. Determine which DMA channel of FTDMAC030_1 is triggered by FTDACC010_0 DMA
0x0c10_0878[23:20]	GLUE_reg_DAC_LAST_SEL[7:4]	FTDACC010_1 dma_last selection. Determine which DMA channel of FTDMAC030_1 is triggered by FTDACC010_1 DMA
0x0c10_0878[27:24]	GLUE_reg_DAC_LAST_SEL[11:8]	FTDACC010_2 dma_last selection. Determine which DMA channel of FTDMAC030_1 is triggered by FTDACC010_2 DMA
0x0c10_0878[31:28]	GLUE_reg_DAC_LAST_SEL[15:12]	FTDACC010_3 dma_last selection. Determine which DMA channel of FTDMAC030_1 is triggered by FTDACC010_3 DMA

I/O and Package

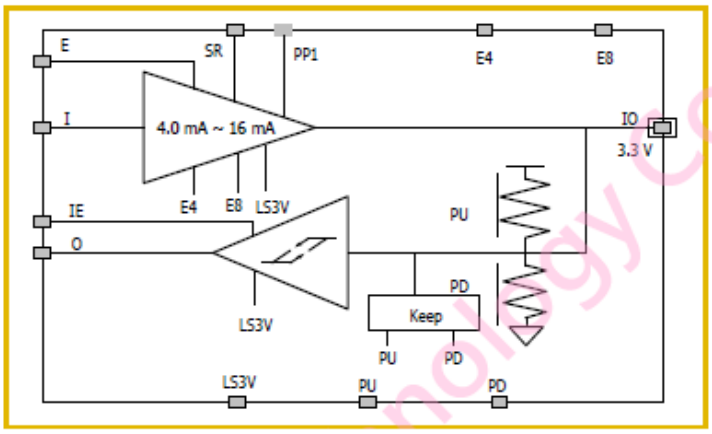
This chapter contains the following sections:

14 Signal Pad Descriptions
15 Power and Ground Pin Descriptions
16 Dedicate Selection Pin
17 Hardware Strap Pin
18 Package Information

14. Signal Pad Descriptions

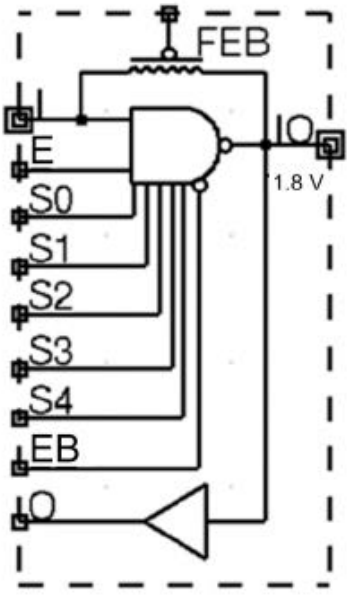
14.1 IO specification

Figure 0-1 and **Figure 0-2** shows the IO pad symbol diagram and truth table which used in Unimap-TX5.



I	E	IO	IE	O
0	1	0	1	0
1	1	1	1	1
0	1	0	0	0
1	1	1	0	0
X	0	Z	1	X
X	0	1	1	1
X	0	0	1	0
X	0	Pull-down	1	0
X	0	Pull-up	1	1
X	0	None/Keeper	1	IO
X	0	X	0	0

Figure 0-1. Bi-direction IO pad Symbol diagram and truth table



Truth Table

I	E	EB	IO	O
0	1	0	1	1
1	1	0	0	0

Oscillation Mode

I	E	EB	IO	O
X	1	1	Z	X
X	1	1	0	0
X	1	1	1	1

Tri-state Mode

I	E	EB	IO	O
X	0	0	1	1

Power-down (Stop) Mode

I	E	EB	IO	O
X	0	1	X	0

Parallel Mode

Figure 0-2. OSC IO pad Symbol diagram and truth table

14.2 Signal Pads

Table 0-1 shows all signal pads in Unimap-TX5 SoC system.

Table 0-1. Signal Pads

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
1	X_OSC32K_IO	1	I/O	1.8		RTC domain, 32KHz OSC PAD
2	X_OSC32K_IN	1	I	1.8		
3	X_RSTN	1	I	1.8	PU/SMT	RTC domain, hardware reset, active low
4	X_PWRBTN	1	I	1.8	PU	RTC domain, power button control, active low
5	X_WAKE[3:0]	4	I	1.8	PU	RTC domain, wake up control, active low
6	X_PWRCTRL[3:0]	4	O	1.8		RTC domain, power control output
7	X_DVCTL[1:0]	2	I	1.8	PD	RTC domain, dual voltage group control 0: 1.8V pad, 1: 3.3V pad
8	X_OM	1	I	1.8	PD	RTC domain, DFT test mode
9	X_OTP_PENVDD2	1	I	1.8		OTP VDD2 power enable
10	X_OSC25M_IO	1	I/O	1.8		25MHz OSC PAD
11	X_OSC25M_IN	1	I/O	1.8		
12	X_OSC40M_IO	1	I/O	1.8		40MHz OSC PAD
13	X_OSC40M_IN	1	I/O	1.8		
14	X_OSC48M_IO	1	I/O	1.8		48MHz OSC PAD
15	X_OSC48M_IN	1	I/O	1.8		
16	X_BOOT_FAIL	1	I/O	3.3	16mA	Normal is high, fail is low
17	X_CA53_TRSTn	1	I/O	3.3	PU	CA53 JTAG/TRACE Port
18	X_CA53_TCK	1	I/O	3.3	PU	
19	X_CA53_TMS	1	I/O	3.3	PU	
20	X_CA53_TDI	1	I/O	3.3	PU	

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
21	X_CA53_TDO	1	I/O	3.3		
22	X_CA53_TRACECLK	1	I/O	3.3		
23	X_CA53_TRACECTL	1	I/O	3.3		
24	X_CA53_TRACED[3:0]	4	I/O	3.3		
25	X_SEC_CPU_TRSTn	1	I/O	3.3	PU	
26	X_SEC_CPU_TCK	1	I/O	3.3	PU	
27	X_SEC_CPU_TMS	1	I/O	3.3	PU	Soteria ibex CPU JTAG Port
28	X_SEC_CPU_TDI	1	I/O	3.3	PU	
29	X_SEC_CPU_TDO	1	I/O	3.3		
30	X_SPIF_SCK	1	I/O	3.3		
31	X_SPIF_CS_N	1	I/O	3.3		
32	X_SPIF_TX	1	I/O	3.3		SPI Flash
33	X_SPIF_RX	1	I/O	3.3		
34	X_SPIF_WP_N	1	I/O	3.3		
35	X_SPIF_HOLD_N	1	I/O	3.3		
36	X_SPI0_SCLK	1	I/O	Dual grp1		
37	X_SPI0_FS	1	I/O	Dual grp1		
38	X_SPI0_FS1	1	I/O	Dual grp1		SSP 0
39	X_SPI0_TX	1	I/O	Dual grp1		
40	X_SPI0_RX	1	I/O	Dual grp1		
41	X_SPI1_SCLK	1	I/O	3.3		
42	X_SPI1_FS	1	I/O	3.3		SSP 1
43	X_SPI1_FS1	1	I/O	3.3		

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
44	X_SPI1_TX	1	I/O	3.3		
45	X_SPI1_RX	1	I/O	3.3		
46	X_I2C0_SCL	1	I/O	3.3		I2C 0
47	X_I2C0_SDA	1	I/O	3.3		
48	X_I2C1_SCL	1	I/O	3.3		I2C 1
49	X_I2C1_SDA	1	I/O	3.3		
50	X_I2C2_SCL	1	I/O	Dual grp0		I2C 2
51	X_I2C2_SDA	1	I/O	Dual grp0		
52	X_I2C3_SCL	1	I/O	Dual grp0		I2C 3
53	X_I2C3_SDA	1	I/O	Dual grp0		
54	X_GPIO[18:0]	19	I/O	3.3		GPIO
55	X_SD0_CLK	1	I/O	3.3		
56	X_SD0_CMD_RSP	1	I/O	3.3		
57	X_SD0_DAT0	1	I/O	3.3		
58	X_SD0_DAT1	1	I/O	3.3		
59	X_SD0_DAT2	1	I/O	3.3		SD CARD Interface
60	X_SD0_DAT3	1	I/O	3.3		
61	X_SD0_WP	1	I/O	3.3	PU	
62	X_SD0_CD	1	I/O	3.3	PU	
63	X_SD0_RSTN	1	I/O	3.3		
64	X_SD1_CLK	1	I/O	1.8		
65	X_SD1_CMD_RSP	1	I/O	1.8		eMMC Interface
66	X_SD1_DAT0	1	I/O	1.8		

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
67	X_SD1_DAT1	1	I/O	1.8		
68	X_SD1_DAT2	1	I/O	1.8		
69	X_SD1_DAT3	1	I/O	1.8		
70	X_SD1_WP	1	I/O	1.8	PU	
71	X_SD1_CD	1	I/O	1.8	PU	
72	X_SD1_RSTN	1	I/O	1.8		
73	X_UART0_TX	1	I/O	3.3		
74	X_UART0_RX	1	I/O	3.3	PU	UART 0
75	X_UART0_CTSN	1	I/O	3.3	PU	
76	X_UART0_RTSN	1	I/O	3.3		
77	X_UART1_TX	1	I/O	3.3		
78	X_UART1_RX	1	I/O	3.3	PU	UART 1
79	X_UART1_CTSN	1	I/O	3.3	PU	
80	X_UART1_RTSN	1	I/O	3.3		
81	X_UART2_TX	1	I/O	Dual grp0		
82	X_UART2_RX	1	I/O	Dual grp0	PU	UART 2
83	X_UART2_CTSN	1	I/O	Dual grp0	PU	
84	X_UART2_RTSN	1	I/O	Dual grp0		
85	X_USART0_CLK	1	I/O	3.3		
86	X_USART0_TX	1	I/O	3.3		
87	X_USART0_RX	1	I/O	3.3	PU	USART 0
88	X_USART0_CTSN	1	I/O	3.3	PU	
89	X_USART0_RTSN	1	I/O	3.3		

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
90	X_CANFD0_TX	1	I/O	3.3		CANFD 0
91	X_CANFD0_RX	1	I/O	3.3		
92	X_CANFD1_TX	1	I/O	3.3		CANFD 1
93	X_CANFD1_RX	1	I/O	3.3		
94	X_TA0_OUTP	1	O			LVDS_TX_0 FXLVTX4311NSHJ0P_SB Even side
95	X_TA0_OUTN	1	O			
96	X_TB0_OUTP	1	O			
97	X_TB0_OUTN	1	O			
98	X_TC0_OUTP	1	O			
99	X_TC0_OUTN	1	O			
100	X_TD0_OUTP	1	O			
101	X_TD0_OUTN	1	O			
102	X_TCLK0_OUTP	1	O			
103	X_TCLK0_OUTN	1	O			
104	X_LVTX_RPI_E	1	I			
105	X_TA1_OUTP	1	O			LVDS_TX_1 FXLVTX4311NSHJ0P_SB Odd side
106	X_TA1_OUTN	1	O			
107	X_TB1_OUTP	1	O			
108	X_TB1_OUTN	1	O			
109	X_TC1_OUTP	1	O			
110	X_TC1_OUTN	1	O			
111	X_TD1_OUTP	1	O			
112	X_TD1_OUTN	1	O			

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
113	X_TCLK1_OUTP	1	O			
114	X_TCLK1_OUTN	1	O			
115	X_LVTX_RPI_O	1	I			
116	X_ADC_XVRB	1	I			
117	X_ ADC_XVRT	1	I			ADC FXADC112B001M20EWHJ0P
118	X_ ADC_XAIN	8	I			
119	X_DAC0_VREFT	1	I			
120	X_DAC0_VREFB	1	I			DAC 0 FXDAC112B001M00EWHJ0P
121	X_DAC0_AOUT	1	O			
122	X_DAC1_VREFT	1	I			
123	X_DAC1_VREFB	1	I			DAC 1 FXDAC112B001M00EWHJ0P
124	X_DAC1_AOUT	1	O			
125	X_DAC2_VREFT	1	I			
126	X_DAC2_VREFB	1	I			DAC 2 FXDAC112B001M00EWHJ0P
127	X_DAC2_AOUT	1	O			
128	X_DAC3_VREFT	1	I			
129	X_DAC3_VREFB	1	I			DAC 3 FXDAC112B001M00EWHJ0P
130	X_DAC3_AOUT	1	O			
131	X_DDR_CLK	1	O			
132	X_DDR_CLKB	1	O			
133	X_DDR_CKE	1	O			DDR3/DDR4 combo
134	X_DDR_RSTn	1	O			
135	X_DDR_ACTn	1	O			

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
136	X_DDR_CSn	1	O			
137	X_DDR_ODT	1	O			
138	X_DDR_BG	2	O			
139	X_DDR_BA	2	O			
140	X_DDR_ADDR	17	O			
141	X_DDR_DQ	32	I/O			
142	X_DDR_DQS	4	I/O			
143	X_DDR_DQSB	4	I/O			
144	X_DDR_DM	4	I/O			
145	X_DDR_VREF	1	I			
146	X_DDR_ERRn	1	I/O			
147	X_DDR_PAR	1	I/O			
148	X_DDR_ZN	1	O			
149	X_DDR_ZN_SENSE	1	I			
150	X_P0MDIAP	1	I/O			
151	X_P0MDIAN	1	I/O			
152	X_P0MDIBP	1	I/O			
153	X_P0MDIBN	1	I/O			
154	X_P0MDICP	1	I/O			GPHY 0 FXEDP410EWHJ0P
155	X_P0MDICN	1	I/O			
156	X_P0MDIDP	1	I/O			
157	X_P0MDIDN	1	I/O			
158	X_P0_SD	1	I			

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
159	X_P0_RSET_BG	1	O			
160	X_P0_LED0	1	I/O	3.3		LED0 output of GPHY 0
161	X_P0_LED1	1	I/O	3.3		LED1 output of GPHY 0
162	X_P1MDIAP	1	I/O			GPHY 1 FXEDP410EWHJ0P
163	X_P1MDIAN	1	I/O			
164	X_P1MDIBP	1	I/O			
165	X_P1MDIBN	1	I/O			
166	X_P1MDICP	1	I/O			
167	X_P1MDICN	1	I/O			
168	X_P1MDIDP	1	I/O			
169	X_P1MDIDN	1	I/O			
170	X_P1_SD	1	I			
171	X_P1_RSET_BG	1	O			
172	X_P1_LED0	1	I/O	3.3		LED0 output of GPHY 1
173	X_P1_LED1	1	I/O	3.3		LED1 output of GPHY 1
174	X_U3_SSRXA	1	I			USB3 FZOTG306NSHJ0P_SB
175	X_U3_SSRXB	1	I			
176	X_U3_SSTXA	1	O			
177	X_U3_SSTXB	1	O			
178	X_U3_DM	1	I/O			
179	X_U3_DP	1	I/O			
180	X_U3_RREF	1	I/O			
181	X_U3_VBUS	1	I	3.3	PD	USB3 VBUS, active high

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
182	X_U3_PORT_PWR_EN	1	O	3.3		Power switch enable for USB3
183	X_U3_ID_PAD	1	I	3.3		ID pad input for USB3
184	X_U20_DM	1	I/O			USB2_0 FZOTG268NSHJ0P_SB
185	X_U20_DP	1	I/O			
186	X_U20_RREF	1	I/O			
187	X_U20_DRVBUS	1	O	3.3		USB2_0 drive VBUS
188	X_U20_VBUS	1	I	3.3	PD	USB2_0 VBUS, active high
189	X_U21_DM	1	I/O			USB2_1 FZOTG268NSHJ0P_SB
190	X_U21_DP	1	I/O			
191	X_U21_RREF	1	I/O			
192	X_U21_DRVBUS	1	O	3.3		USB2_1 drive VBUS
193	X_U21_VBUS	1	I	3.3	PD	USB2_1 VBUS, active high
194	X_PCIE_CLKP	1	I			PCIE Gen3 4 lanes
195	X_PCIE_CLKN	1	I			
196	X_PCIE_RREF	1	I/O			
197	X_PCIE_TXP_L0	1	O			
198	X_PCIE_TXN_L0	1	O			
199	X_PCIE_RXP_L0	1	I			
200	X_PCIE_RXN_L0	1	I			
201	X_PCIE_TXP_L1	1	O			
202	X_PCIE_TXN_L1	1	O			
203	X_PCIE_RXP_L1	1	I			
204	X_PCIE_RXN_L1	1	I			

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
205	X_PCIE_TXP_L2	1	O			
206	X_PCIE_TXN_L2	1	O			
207	X_PCIE_RXP_L2	1	I			
208	X_PCIE_RXN_L2	1	I			
209	X_PCIE_TXP_L3	1	O			
210	X_PCIE_TXN_L3	1	O			
211	X_PCIE_RXP_L3	1	I			
212	X_PCIE_RXN_L3	1	I			
213	X_PCIE_PERSTN	1	O	3.3		PCIE PERSTN out
214	X_PCIE_CLKREQN	1	I/O	3.3	PU	PCIE CLKREQN
215	X_PCIE_WAKEN	1	I/O	3.3	PU	PCIE WAKEN
216	X_PWM1A	1	I/O	3.3		
217	X_PWM2A	1	I/O	3.3		
218	X_PWM3A	1	I/O	3.3		
219	X_PWM4A	1	I/O	3.3		
220	X_PWM5A	1	I/O	3.3		
221	X_PWM6A	1	I/O	3.3		
222	X_PWM1B	1	I/O	3.3		Motor_Top ePWM
223	X_PWM2B	1	I/O	3.3		
224	X_PWM3B	1	I/O	3.3		
225	X_PWM4B	1	I/O	3.3		
226	X_PWM5B	1	I/O	3.3		
227	X_PWM6B	1	I/O	3.3		

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
228	X_PWM_OE	1	I/O	3.3		
229	X_EQEP_AIN	12	I/O	3.3		Motor_Top EQEP
230	X_EQEP_BIN	12	I/O	3.3		
231	X_EQEP_I	12	I/O	3.3		
232	X_EQEP_S	12	I/O	3.3		
233	X_EQEP_AOUT	12	I/O	3.3		
234	X_EQEP_BOUT	12	I/O	3.3		
235	X_ENDAT_TCLK	3	I/O	3.3		Motor_Top ENDAT
236	X_ENDAT_OE	3	I/O	3.3		
237	X_ENDAT_DIN	3	I/O	3.3		
238	X_ENDAT_DOUT	3	I/O	3.3		
239	X_EXADC1_SCLK	1	I/O	3.3		Motor_Top EXADC
240	X_EXADC1_CSN	1	I/O	3.3		
241	X_EXADC1_A	3	I/O	3.3		
242	X_EXADC1_DOUTA	1	I/O	3.3		
243	X_EXADC1_DOUTB	1	I/O	3.3		
244	X_EXADC2_SCLK	1	I/O	3.3		
245	X_EXADC2_CSN	1	I/O	3.3		
246	X_EXADC2_A	3	I/O	3.3		
247	X_EXADC2_DOUTA	1	I/O	3.3		
248	X_EXADC2_DOUTB	1	I/O	3.3		
249	X_EXADC3_SCLK	1	I/O	3.3		
250	X_EXADC3_CSN	1	I/O	3.3		

No.	Name	Width	Dir.	Voltage (source)	Default Setting	Pin Description
251	X_EXADC3_A	3	I/O	3.3		
252	X_EXADC3_DOUTA	1	I/O	3.3		
253	X_EXADC3_DOUTB	1	I/O	3.3		
254	X_XMC_CSN	4	I/O	3.3		
255	X_XMC_OEN	1	I/O	3.3		
256	X_XMC_WEN	1	I/O	3.3		Motor_Top XMC
257	X_XMC_ADDR	16	I/O	3.3		
258	X_XMC_DATA	16	I/O	3.3		

14.3 Pin Mux Table

Figure 0-3 shows the control registers and how to configure pin mux function.

PAD name	SCU register setting	function 0	SCU register setting	function 1	SCU register setting	function 2
X_GPIO[0]	0x0c10_0898[1:0]=2'b00	u_FTGPIO010/GPIO[0]	0x0c10_0898[1:0]=2'b01	u_FTSSP010_2/SPI2_SCLK	0x0c10_0898[1:0]=2'b10	u_GLUE_top/dbg_pin_out[0]
X_GPIO[1]	0x0c10_0898[3:2]=2'b00	u_FTGPIO010/GPIO[1]	0x0c10_0898[3:2]=2'b01	u_FTSSP010_2/SPI2_FS	0x0c10_0898[3:2]=2'b10	u_GLUE_top/dbg_pin_out[1]
X_GPIO[2]	0x0c10_0898[5:4]=2'b00	u_FTGPIO010/GPIO[2]	0x0c10_0898[5:4]=2'b01	u_FTSSP010_2/SPI2_TX	0x0c10_0898[5:4]=2'b10	u_GLUE_top/dbg_pin_out[2]
X_GPIO[3]	0x0c10_0898[7:6]=2'b00	u_FTGPIO010/GPIO[3]	0x0c10_0898[7:6]=2'b01	u_FTSSP010_2/SPI2_RX	0x0c10_0898[7:6]=2'b10	u_GLUE_top/dbg_pin_out[3]
X_GPIO[4]	0x0c10_0898[9:8]=2'b00	u_FTGPIO010/GPIO[4]	0x0c10_0898[9:8]=2'b01	u_FTPWMTMR010/tmr1_out	0x0c10_0898[9:8]=2'b10	u_GLUE_top/dbg_pin_out[4]
X_GPIO[5]	0x0c10_0898[11:10]=2'b00	u_FTGPIO010/GPIO[5]	0x0c10_0898[11:10]=2'b01	u_FTPWMTMR010/tmr1_out_n	0x0c10_0898[11:10]=2'b10	u_GLUE_top/dbg_pin_out[5]
X_GPIO[6]	0x0c10_0898[13:12]=2'b00	u_FTGPIO010/GPIO[6]	0x0c10_0898[13:12]=2'b01	u_FTPWMTMR010/tmr2_out	0x0c10_0898[13:12]=2'b10	u_GLUE_top/dbg_pin_out[6]
X_GPIO[7]	0x0c10_0898[15:14]=2'b00	u_FTGPIO010/GPIO[7]	0x0c10_0898[15:14]=2'b01	u_FTPWMTMR010/tmr2_out_n	0x0c10_0898[15:14]=2'b10	u_GLUE_top/dbg_pin_out[7]
X_GPIO[8]	0x0c10_0898[17:16]=2'b00	u_FTGPIO010/GPIO[8]	0x0c10_0898[17:16]=2'b01	u_FTPWMTMR010/tmr3_out	0x0c10_0898[17:16]=2'b10	u_GLUE_top/dbg_pin_out[8]
X_GPIO[9]	0x0c10_0898[19:18]=2'b00	u_FTGPIO010/GPIO[9]	0x0c10_0898[19:18]=2'b01	u_FTPWMTMR010/tmr3_out_n	0x0c10_0898[19:18]=2'b10	u_GLUE_top/dbg_pin_out[9]
X_GPIO[10]	0x0c10_0898[21:20]=2'b00	u_FTGPIO010/GPIO[10]	0x0c10_0898[21:20]=2'b01	u_Motor_Top/DS_DIN[0]	0x0c10_0898[21:20]=2'b10	u_GLUE_top/dbg_pin_out[10]
X_GPIO[11]	0x0c10_0898[23:22]=2'b00	u_FTGPIO010/GPIO[11]	0x0c10_0898[23:22]=2'b01	u_Motor_Top/DS_DIN[1]	0x0c10_0898[23:22]=2'b10	u_GLUE_top/dbg_pin_out[11]
X_GPIO[12]	0x0c10_0898[25:24]=2'b00	u_FTGPIO010/GPIO[12]	0x0c10_0898[25:24]=2'b01	u_Motor_Top/DS_DIN[2]	0x0c10_0898[25:24]=2'b10	u_GLUE_top/dbg_pin_out[12]
X_GPIO[13]	0x0c10_0898[27:26]=2'b00	u_FTGPIO010/GPIO[13]	0x0c10_0898[27:26]=2'b01	u_Motor_Top/DS_DIN[3]	0x0c10_0898[27:26]=2'b10	u_GLUE_top/dbg_pin_out[13]
X_GPIO[14]	0x0c10_0898[29:28]=2'b00	u_FTGPIO010/GPIO[14]	0x0c10_0898[29:28]=2'b01	u_Motor_Top/DS_DIN[4]	0x0c10_0898[29:28]=2'b10	u_GLUE_top/dbg_pin_out[14]
X_GPIO[15]	0x0c10_0898[31:30]=2'b00	u_FTGPIO010/GPIO[15]	0x0c10_0898[31:30]=2'b01	u_Motor_Top/DS_DIN[5]	0x0c10_0898[31:30]=2'b10	u_GLUE_top/dbg_pin_out[15]
X_GPIO[16]	0x0c10_089c[1:0]=2'b00	u_FTGPIO010/GPIO[16]	0x0c10_089c[1:0]=2'b01	u_DAC_TOP/DAC0_EXTRG	0x0c10_089c[1:0]=2'b10	u_GLUE_top/dbg_pin_out[16]
X_GPIO[17]	0x0c10_089c[3:2]=2'b00	u_FTGPIO010/GPIO[17]	0x0c10_089c[3:2]=2'b01	u_SBS_CAN_FD_2/CANFD2_TX	0x0c10_089c[3:2]=2'b10	u_GLUE_top/dbg_pin_out[17]
X_GPIO[18]	0x0c10_089c[5:4]=2'b00	u_FTGPIO010/GPIO[18]	0x0c10_089c[5:4]=2'b01	u_SBS_CAN_FD_2/CANFD2_RX	0x0c10_089c[5:4]=2'b10	u_GLUE_top/dbg_pin_out[18]
X_Boot_Fail	0x0c10_089c[7:6]=2'b00	u_GLUE_top/Boot_Fail	0x0c10_089c[7:6]=2'b01	u_FTGPIO010/GPIO[19]	0x0c10_089c[7:6]=2'b10	u_GLUE_top/dbg_pin_out[19]
X_P0_LED0	0x0c10_089c[9:8]=2'b00	u_SBS_GMAC/P0_LED0	0x0c10_089c[9:8]=2'b01	u_FTGPIO010/GPIO[20]		
X_P0_LED1	0x0c10_089c[11:10]=2'b00	u_SBS_GMAC/P0_LED1	0x0c10_089c[11:10]=2'b01	u_FTGPIO010/GPIO[21]		
X_P1_LED0	0x0c10_089c[13:12]=2'b00	u_SBS_GMAC_1/P1_LED0	0x0c10_089c[13:12]=2'b01	u_FTGPIO010/GPIO[22]		
X_P1_LED1	0x0c10_089c[15:14]=2'b00	u_SBS_GMAC_1/P1_LED1	0x0c10_089c[15:14]=2'b01	u_FTGPIO010/GPIO[23]		
X_UART2_TX	0x0c10_089c[17:16]=2'b00	u_FTUART010_2/UART2_TX	0x0c10_089c[17:16]=2'b01	u_FTGPIO010/GPIO[24]		
X_UART2_RX	0x0c10_089c[19:18]=2'b00	u_FTUART010_2/UART2_RX	0x0c10_089c[19:18]=2'b01	u_FTGPIO010/GPIO[25]		
X_UART2_CTSN	0x0c10_089c[21:20]=2'b00	u_FTUART010_2/UART2_CTSN	0x0c10_089c[21:20]=2'b01	u_FTGPIO010/GPIO[26]		
X_UART2_RTSN	0x0c10_089c[23:22]=2'b00	u_FTUART010_2/UART2_RTSN	0x0c10_089c[23:22]=2'b01	u_FTGPIO010/GPIO[27]		
X_I2C2_SCL	0x0c10_089c[25:24]=2'b00	u_FTIIC010_2/I2C2_SCL	0x0c10_089c[25:24]=2'b01	u_FTGPIO010/GPIO[28]		
X_I2C2_SDA	0x0c10_089c[27:26]=2'b00	u_FTIIC010_2/I2C2_SDA	0x0c10_089c[27:26]=2'b01	u_FTGPIO010/GPIO[29]		
X_I2C3_SCL	0x0c10_089c[29:28]=2'b00	u_FTIIC010_3/I2C3_SCL	0x0c10_089c[29:28]=2'b01	u_FTGPIO010/GPIO[30]		
X_I2C3_SDA	0x0c10_089c[31:30]=2'b00	u_FTIIC010_3/I2C3_SDA	0x0c10_089c[31:30]=2'b01	u_FTGPIO010/GPIO[31]		
X_SD1_CMD_RSP	0x0c10_08a0[1:0]=2'b00	u_FTSDC021_1/SD1_CMD_RSP	0x0c10_08a0[1:0]=2'b01	u_FTGPIO010/GPIO[0]		
X_SD1_DAT0	0x0c10_08a0[3:2]=2'b00	u_FTSDC021_1/SD1_DAT0	0x0c10_08a0[3:2]=2'b01	u_FTGPIO010/GPIO[1]		
X_SD1_DAT1	0x0c10_08a0[5:4]=2'b00	u_FTSDC021_1/SD1_DAT1	0x0c10_08a0[5:4]=2'b01	u_FTGPIO010/GPIO[2]		
X_SD1_DAT2	0x0c10_08a0[7:6]=2'b00	u_FTSDC021_1/SD1_DAT2	0x0c10_08a0[7:6]=2'b01	u_FTGPIO010/GPIO[3]		
X_SD1_DAT3	0x0c10_08a0[9:8]=2'b00	u_FTSDC021_1/SD1_DAT3	0x0c10_08a0[9:8]=2'b01	u_FTGPIO010/GPIO[4]		
X_SD1_WP	0x0c10_08a0[11:10]=2'b00	u_FTSDC021_1/SD1_WP	0x0c10_08a0[11:10]=2'b01	u_FTGPIO010/GPIO[5]		
X_SD1_CD	0x0c10_08a0[13:12]=2'b00	u_FTSDC021_1/SD1_CD	0x0c10_08a0[13:12]=2'b01	u_FTGPIO010/GPIO[6]		
X_SD1_RSTN	0x0c10_08a0[15:14]=2'b00	u_FTSDC021_1/SD1_RSTN	0x0c10_08a0[15:14]=2'b01	u_FTGPIO010/GPIO[7]		
X_SD0_CMD_RSP	0x0c10_08a0[17:16]=2'b00	u_FTSDC021/SD0_CMD_RSP	0x0c10_08a0[17:16]=2'b01	u_Motor_Top/ePWM7A		
X_SD0_DAT0	0x0c10_08a0[19:18]=2'b00	u_FTSDC021/SD0_DAT0	0x0c10_08a0[19:18]=2'b01	u_Motor_Top/ePWM7B		
X_SD0_DAT1	0x0c10_08a0[21:20]=2'b00	u_FTSDC021/SD0_DAT1	0x0c10_08a0[21:20]=2'b01	u_Motor_Top/ePWM8A		
X_SD0_DAT2	0x0c10_08a0[23:22]=2'b00	u_FTSDC021/SD0_DAT2	0x0c10_08a0[23:22]=2'b01	u_Motor_Top/ePWM8B		
X_SD0_DAT3	0x0c10_08a0[25:24]=2'b00	u_FTSDC021/SD0_DAT3	0x0c10_08a0[25:24]=2'b01	u_Motor_Top/ePWM9A		
X_SD0_WP	0x0c10_08a0[27:26]=2'b00	u_FTSDC021/SD0_WP	0x0c10_08a0[27:26]=2'b01	u_Motor_Top/ePWM9B		
X_SEC_CPU_TRSTn	0x0c10_08a0[29:28]=2'b00	u_SOTERIA2/SEC_CPU_TRSTn	0x0c10_08a0[29:28]=2'b01	u_FTUART010_3/UART3_RTSN		
X_SEC_CPU_TMS	0x0c10_08a0[31:30]=2'b00	u_SOTERIA2/SEC_CPU_TMS	0x0c10_08a0[31:30]=2'b01	u_FTUART010_3/UART3_CTSN		
X_SEC_CPU_TDI	0x0c10_08a4[1:0]=2'b00	u_SOTERIA2/SEC_CPU_TDI	0x0c10_08a4[1:0]=2'b01	u_FTUART010_3/UART3_RX		
X_SEC_CPU_TDO	0x0c10_08a4[3:2]=2'b00	u_SOTERIA2/SEC_CPU_TDO	0x0c10_08a4[3:2]=2'b01	u_FTUART010_3/UART3_TX		
X_EQEP_AIN[0]	0x0c10_08a4[5:4]=2'b00	u_Motor_Top/EQEP_AIN[0]	0x0c10_08a4[5:4]=2'b01	u_FTSSP010_3/SPI3_SCLK		
X_EQEP_AIN[1]	0x0c10_08a4[7:6]=2'b00	u_Motor_Top/EQEP_AIN[1]	0x0c10_08a4[7:6]=2'b01	u_FTSSP010_3/SPI3_FS		

X_EQEP_AIN[2]	0x0d10_08a4[9:8]=2'b00	u_Motor_Top/EQEP_AIN[2]	0x0d10_08a4[9:8]=2'b01	u_FTSSP010_3/SPI3_TX		
X_EQEP_AIN[3]	0x0d10_08a4[11:10]=2'b00	u_Motor_Top/EQEP_AIN[3]	0x0d10_08a4[11:10]=2'b01	u_FTSSP010_3/SPI3_RX		
X_EQEP_BIN[0]	0x0d10_08a4[13:12]=2'b00	u_Motor_Top/EQEP_BIN[0]	0x0d10_08a4[13:12]=2'b01	u_FTSSP010_2/SPI2_SCLK		
X_EQEP_BIN[1]	0x0d10_08a4[15:14]=2'b00	u_Motor_Top/EQEP_BIN[1]	0x0d10_08a4[15:14]=2'b01	u_FTSSP010_2/SPI2_FS		
X_EQEP_BIN[2]	0x0d10_08a4[17:16]=2'b00	u_Motor_Top/EQEP_BIN[2]	0x0d10_08a4[17:16]=2'b01	u_FTSSP010_2/SPI2_TX		
X_EQEP_BIN[3]	0x0d10_08a4[19:18]=2'b00	u_Motor_Top/EQEP_BIN[3]	0x0d10_08a4[19:18]=2'b01	u_FTSSP010_2/SPI2_RX		
X_EQEP_I[0]	0x0d10_08a4[21:20]=2'b00	u_Motor_Top/EQEP_I[0]	0x0d10_08a4[21:20]=2'b01	u_FTUART010_2/UART2_TX		
X_EQEP_I[1]	0x0d10_08a4[23:22]=2'b00	u_Motor_Top/EQEP_I[1]	0x0d10_08a4[23:22]=2'b01	u_FTUART010_2/UART2_RX		
X_EQEP_I[2]	0x0d10_08a4[25:24]=2'b00	u_Motor_Top/EQEP_I[2]	0x0d10_08a4[25:24]=2'b01	u_FTUART010_2/UART2_CTSN		
X_EQEP_I[3]	0x0d10_08a4[27:26]=2'b00	u_Motor_Top/EQEP_I[3]	0x0d10_08a4[27:26]=2'b01	u_FTUART010_2/UART2_RTSN		
X_EQEP_S[0]	0x0d10_08a4[29:28]=2'b00	u_Motor_Top/EQEP_S[0]	0x0d10_08a4[29:28]=2'b01	u_FTUART010_3/UART3_TX		
X_EQEP_S[1]	0x0d10_08a4[31:30]=2'b00	u_Motor_Top/EQEP_S[1]	0x0d10_08a4[31:30]=2'b01	u_FTUART010_3/UART3_RX		
X_EQEP_S[2]	0x0d10_08a8[1:0]=2'b00	u_Motor_Top/EQEP_S[2]	0x0d10_08a8[1:0]=2'b01	u_FTUART010_3/UART3_CTSN		
X_EQEP_S[3]	0x0d10_08a8[3:2]=2'b00	u_Motor_Top/EQEP_S[3]	0x0d10_08a8[3:2]=2'b01	u_FTUART010_3/UART3_RTSN		
X_EQEP_AOUT[0]	0x0d10_08a8[5:4]=2'b00	u_Motor_Top/EQEP_AOUT[0]	0x0d10_08a8[5:4]=2'b01	u_FTUSART010_1/USART1_CLK		
X_EQEP_AOUT[1]	0x0d10_08a8[7:6]=2'b00	u_Motor_Top/EQEP_AOUT[1]	0x0d10_08a8[7:6]=2'b01	u_FTUSART010_1/USART1_TX		
X_EQEP_AOUT[2]	0x0d10_08a8[9:8]=2'b00	u_Motor_Top/EQEP_AOUT[2]	0x0d10_08a8[9:8]=2'b01	u_FTUSART010_1/USART1_RX		
X_EQEP_AOUT[3]	0x0d10_08a8[11:10]=2'b00	u_Motor_Top/EQEP_AOUT[3]	0x0d10_08a8[11:10]=2'b01	u_FTUSART010_1/USART1_CTSN		
X_EQEP_BOUT[0]	0x0d10_08a8[13:12]=2'b00	u_Motor_Top/EQEP_BOUT[0]	0x0d10_08a8[13:12]=2'b01	u_FTUSART010_1/USART1_RTSN		
X_EQEP_BOUT[1]	0x0d10_08a8[15:14]=2'b00	u_Motor_Top/EQEP_BOUT[1]	0x0d10_08a8[15:14]=2'b01	u_FTPWMTMR010/tmr1_out		
X_EQEP_BOUT[2]	0x0d10_08a8[17:16]=2'b00	u_Motor_Top/EQEP_BOUT[2]	0x0d10_08a8[17:16]=2'b01	u_FTPWMTMR010/tmr1_out_n		
X_EQEP_BOUT[3]	0x0d10_08a8[19:18]=2'b00	u_Motor_Top/EQEP_BOUT[3]	0x0d10_08a8[19:18]=2'b01	u_FTPWMTMR010/tmr2_out		
X_ENDAT_TCLK[2]	0x0d10_08a8[21:20]=2'b00	u_Motor_Top/ENDAT_TCLK[2]	0x0d10_08a8[21:20]=2'b01	u_FTPWMTMR010/tmr2_out_n		
X_ENDAT_DOUT[2]	0x0d10_08a8[23:22]=2'b00	u_Motor_Top/ENDAT_DOUT[2]	0x0d10_08a8[23:22]=2'b01	u_FTPWMTMR010/tmr3_out		
X_EXADC3_SCLK	0x0d10_08a8[25:24]=2'b00	u_Motor_Top/EXADC3_SCLK	0x0d10_08a8[25:24]=2'b01	u_FTPWMTMR010/tmr3_out_n		
X_EXADC3_CSN	0x0d10_08a8[27:26]=2'b00	u_Motor_Top/EXADC3_CSN	0x0d10_08a8[27:26]=2'b01	u_FTPWMTMR010/tmr4_out		
X_EXADC3_A[0]	0x0d10_08a8[29:28]=2'b00	u_Motor_Top/EXADC3_A[0]	0x0d10_08a8[29:28]=2'b01	u_FTPWMTMR010/tmr4_out_n		
X_EXADC3_A[1]	0x0d10_08a8[31:30]=2'b00	u_Motor_Top/EXADC3_A[1]	0x0d10_08a8[31:30]=2'b01	u_FTPWMTMR010/tmr5_out		
X_EXADC3_A[2]	0x0d10_08ac[1:0]=2'b00	u_Motor_Top/EXADC3_A[2]	0x0d10_08ac[1:0]=2'b01	u_FTPWMTMR010/tmr5_out_n		
X_EXADC3_DOUTA	0x0d10_08ac[3:2]=2'b00	u_Motor_Top/EXADC3_DOUTA	0x0d10_08ac[3:2]=2'b01	u_FTPWMTMR010/tmr6_out		
X_EXADC3_DOUTB	0x0d10_08ac[5:4]=2'b00	u_Motor_Top/EXADC3_DOUTB	0x0d10_08ac[5:4]=2'b01	u_FTPWMTMR010/tmr6_out_n		
X_XMC_CSN[0]	0x0d10_08ac[7:6]=2'b00	u_Motor_Top/XMC_CSN	0x0d10_08ac[7:6]=2'b01	u_FTSSP010_2/SPI2_FS		
X_XMC_OEN	0x0d10_08ac[9:8]=2'b00	u_Motor_Top/XMC_OEN	0x0d10_08ac[9:8]=2'b01	u_FTSSP010_2/SPI2_SCLK		
X_XMC_WEN	0x0d10_08ac[11:10]=2'b00	u_Motor_Top/XMC_WEN	0x0d10_08ac[11:10]=2'b01	u_FTSSP010_2/SPI2_RX		
X_XMC_ADDR[0]	0x0d10_08ac[13:12]=2'b00	u_Motor_Top/XMC_ADDR[0]	0x0d10_08ac[13:12]=2'b01	u_FTSSP010_2/SPI2_TX		
X_XMC_ADDR[4]	0x0d10_08ac[15:14]=2'b00	u_Motor_Top/XMC_ADDR[4]	0x0d10_08ac[15:14]=2'b01	u_FTPWMTMR010/tmr1_out		
X_XMC_ADDR[5]	0x0d10_08ac[17:16]=2'b00	u_Motor_Top/XMC_ADDR[5]	0x0d10_08ac[17:16]=2'b01	u_FTPWMTMR010/tmr1_out_n		
X_XMC_ADDR[6]	0x0d10_08ac[19:18]=2'b00	u_Motor_Top/XMC_ADDR[6]	0x0d10_08ac[19:18]=2'b01	u_FTPWMTMR010/tmr2_out		
X_XMC_ADDR[7]	0x0d10_08ac[21:20]=2'b00	u_Motor_Top/XMC_ADDR[7]	0x0d10_08ac[21:20]=2'b01	u_FTPWMTMR010/tmr2_out_n		
X_XMC_ADDR[8]	0x0d10_08ac[23:22]=2'b00	u_Motor_Top/XMC_ADDR[8]	0x0d10_08ac[23:22]=2'b01	u_FTPWMTMR010/tmr3_out		
X_XMC_ADDR[9]	0x0d10_08ac[25:24]=2'b00	u_Motor_Top/XMC_ADDR[9]	0x0d10_08ac[25:24]=2'b01	u_FTPWMTMR010/tmr3_out_n		
X_XMC_ADDR[10]	0x0d10_08ac[27:26]=2'b00	u_Motor_Top/XMC_ADDR[10]	0x0d10_08ac[27:26]=2'b01	u_FTPWMTMR010/tmr4_out		
X_XMC_ADDR[11]	0x0d10_08ac[29:28]=2'b00	u_Motor_Top/XMC_ADDR[11]	0x0d10_08ac[29:28]=2'b01	u_FTPWMTMR010/tmr4_out_n		
X_XMC_ADDR[12]	0x0d10_08ac[31:30]=2'b00	u_Motor_Top/XMC_ADDR[12]	0x0d10_08ac[31:30]=2'b01	u_FTPWMTMR010/tmr5_out		
X_XMC_ADDR[13]	0x0d10_08b0[1:0]=2'b00	u_Motor_Top/XMC_ADDR[13]	0x0d10_08b0[1:0]=2'b01	u_FTPWMTMR010/tmr5_out_n		
X_XMC_ADDR[14]	0x0d10_08b0[3:2]=2'b00	u_Motor_Top/XMC_ADDR[14]	0x0d10_08b0[3:2]=2'b01	u_FTPWMTMR010/tmr6_out		
X_XMC_ADDR[15]	0x0d10_08b0[5:4]=2'b00	u_Motor_Top/XMC_ADDR[15]	0x0d10_08b0[5:4]=2'b01	u_FTPWMTMR010/tmr6_out_n		
X_XMC_DATA[0]	0x0d10_08b0[7:6]=2'b00	u_Motor_Top/XMC_DATA[0]	0x0d10_08b0[7:6]=2'b01	u_FTGPIO010/GPIO[16]		
X_XMC_DATA[1]	0x0d10_08b0[9:8]=2'b00	u_Motor_Top/XMC_DATA[1]	0x0d10_08b0[9:8]=2'b01	u_FTGPIO010/GPIO[17]		
X_XMC_DATA[2]	0x0d10_08b0[11:10]=2'b00	u_Motor_Top/XMC_DATA[2]	0x0d10_08b0[11:10]=2'b01	u_FTGPIO010/GPIO[18]		
X_XMC_DATA[3]	0x0d10_08b0[13:12]=2'b00	u_Motor_Top/XMC_DATA[3]	0x0d10_08b0[13:12]=2'b01	u_FTGPIO010/GPIO[19]		
X_XMC_DATA[4]	0x0d10_08b0[15:14]=2'b00	u_Motor_Top/XMC_DATA[4]	0x0d10_08b0[15:14]=2'b01	u_FTGPIO010/GPIO[20]		
X_XMC_DATA[5]	0x0d10_08b0[17:16]=2'b00	u_Motor_Top/XMC_DATA[5]	0x0d10_08b0[17:16]=2'b01	u_FTGPIO010/GPIO[21]		
X_XMC_DATA[6]	0x0d10_08b0[19:18]=2'b00	u_Motor_Top/XMC_DATA[6]	0x0d10_08b0[19:18]=2'b01	u_FTGPIO010/GPIO[22]		
X_XMC_DATA[7]	0x0d10_08b0[21:20]=2'b00	u_Motor_Top/XMC_DATA[7]	0x0d10_08b0[21:20]=2'b01	u_FTGPIO010/GPIO[23]		
X_XMC_DATA[8]	0x0d10_08b0[23:22]=2'b00	u_Motor_Top/XMC_DATA[8]	0x0d10_08b0[23:22]=2'b01	u_FTGPIO010/GPIO[24]		
X_XMC_DATA[9]	0x0d10_08b0[25:24]=2'b00	u_Motor_Top/XMC_DATA[9]	0x0d10_08b0[25:24]=2'b01	u_FTGPIO010/GPIO[25]		

X_XMC_DATA[10]	0x0c10_08b0[27:26]=2'b00	u_Motor_Top/XMC_DATA[10]	0x0c10_08b0[27:26]=2'b01	u_FTGPIO010/GPIO[26]		
X_XMC_DATA[11]	0x0c10_08b0[29:28]=2'b00	u_Motor_Top/XMC_DATA[11]	0x0c10_08b0[29:28]=2'b01	u_FTGPIO010/GPIO[27]		
X_XMC_DATA[12]	0x0c10_08b0[31:30]=2'b00	u_Motor_Top/XMC_DATA[12]	0x0c10_08b0[31:30]=2'b01	u_FTGPIO010/GPIO[28]		
X_XMC_DATA[13]	0x0c10_08b4[1:0]=2'b00	u_Motor_Top/XMC_DATA[13]	0x0c10_08b4[1:0]=2'b01	u_FTGPIO010/GPIO[29]		
X_XMC_DATA[14]	0x0c10_08b4[3:2]=2'b00	u_Motor_Top/XMC_DATA[14]	0x0c10_08b4[3:2]=2'b01	u_FTGPIO010/GPIO[30]		
X_XMC_DATA[15]	0x0c10_08b4[5:4]=2'b00	u_Motor_Top/XMC_DATA[15]	0x0c10_08b4[5:4]=2'b01	u_FTGPIO010/GPIO[31]		
X_UART0_CTSN	0x0c10_08b4[7:6]=2'b00	u_FTUART010/UART0_CTSN	0x0c10_08b4[7:6]=2'b01	u_DAC_TOP_1/DAC1_EXTRG		
X_UART1_CTSN	0x0c10_08b4[9:8]=2'b00	u_FTUART010_1/UART1_CTSN	0x0c10_08b4[9:8]=2'b01	u_DAC_TOP_2/DAC2_EXTRG		
X_USART0_TX	0x0c10_08b4[11:10]=2'b00	u_FTUSART010/USART0_TX	0x0c10_08b4[11:10]=2'b01	u_DAC_TOP/DAC0_EXTRG		
X_USART0_RX	0x0c10_08b4[13:12]=2'b00	u_FTUSART010/USART0_RX	0x0c10_08b4[13:12]=2'b01	u_DAC_TOP_1/DAC1_EXTRG		
X_USART0_CTSN	0x0c10_08b4[15:14]=2'b00	u_FTUSART010/USART0_CTSN	0x0c10_08b4[15:14]=2'b01	u_DAC_TOP_2/DAC2_EXTRG		
X_USART0_RTSN	0x0c10_08b4[17:16]=2'b00	u_FTUSART010/USART0_RTSN	0x0c10_08b4[17:16]=2'b01	u_DAC_TOP_3/DAC3_EXTRG		
X_CA53_TRACECLK	0x0c10_08b4[18]=1'b0	u_CA53Q_subsystem/traceclk	0x0c10_08b4[18]=1'b1	u_Motor_Top/DS_MCLK[0]		
X_CA53_TRACECTL	0x0c10_08b4[18]=1'b0	u_CA53Q_subsystem/tracectl	0x0c10_08b4[18]=1'b1	u_Motor_Top/DS_MCLK[1]		
X_CA53_TRACED[0]	0x0c10_08b4[18]=1'b0	u_CA53Q_subsystem/tracedata[0]	0x0c10_08b4[18]=1'b1	u_Motor_Top/DS_MCLK[2]		
X_CA53_TRACED[1]	0x0c10_08b4[18]=1'b0	u_CA53Q_subsystem/tracedata[1]	0x0c10_08b4[18]=1'b1	u_Motor_Top/DS_MCLK[3]		
X_CA53_TRACED[2]	0x0c10_08b4[18]=1'b0	u_CA53Q_subsystem/tracedata[2]	0x0c10_08b4[18]=1'b1	u_Motor_Top/DS_MCLK[4]		
X_CA53_TRACED[3]	0x0c10_08b4[18]=1'b0	u_CA53Q_subsystem/tracedata[3]	0x0c10_08b4[18]=1'b1	u_Motor_Top/DS_MCLK[5]		

Figure 0-3. Pin-mux function

15. Power and Ground Pin Descriptions

Table 0-2. Power and Ground Pins

Name	Minimum Voltage[1]	Nominal Voltage.	Maximum Voltage	Description
VCCK	0.81	0.9	0.99	Core power/ground
GNDK		0		
VCC3IO_0	2.97	3.3	3.63	3.3V I/O power
VCC3IO_1	1.62/2.97	1.8/3.3	1.98/3.63	1.8V/3.3V IO power for dual voltage group 1
VCC18IO_2	1.62	1.8	1.98	1.8V I/O power for eMMC I/O
VCC3IO_3	1.62/2.97	1.8/3.3	1.98/3.63	1.8V/3.3V I/O power for dual voltage group 0
VCC3IO_4	2.97	3.3	3.63	3.3V I/O power
VCC3IO_5	2.97	3.3	3.63	3.3V I/O power
VCC18IO_6	1.62	1.8	1.98	1.8V I/O power for efuse I/O
GNDIO_*		0		I/O ground
VCC18UD_*	1.62	1.8	1.98	I/O converter power
VCC33A_U3	2.97	3.3	3.63	USB3 analog power/ground
VCC18A_U3	1.62	1.8	1.98	
VCC09A_TX_U3	0.81	0.9	0.99	
VCC09A_RX_U3	0.81	0.9	0.99	
GNDA_U3		0		
VCC33A_U20	2.97	3.3	3.63	USB2_0 analog power/ground
VCC18A_U20	1.62	1.8	1.98	
GNDA_U20		0		
VCC33A_U21	2.97	3.3	3.63	USB2_1 analog power/ground

VCC18A_U21	1.62	1.8	1.98	
GNDA_U21		0		
VCC18A_RX_GMAC0	1.62	1.8	1.98	
VCC18A_TX_GMAC0	1.62	1.8	1.98	GPHY_0 analog power/ground
VCC33A_GMAC0	2.97	3.3	3.63	
GNDA_GMAC0		0		
VCC18A_RX_GMAC1	1.62	1.8	1.98	
VCC18A_TX_GMAC1	1.62	1.8	1.98	GPHY_1 analog power/ground
VCC33A_GMAC1	2.97	3.3	3.63	
GNDA_GMAC1		0		
VCC33A_LVTX_E	2.97	3.3	3.63	
VCC09D_LVTX_E	0.81	0.9	0.99	LVTX Even part power/ground
GND33A_LVTX_E		0		
GND09D_LVTX_E		0		
VCC33A_LVTX_O	2.97	3.3	3.63	
VCC09D_LVTX_O	0.81	0.9	0.99	LVTX Odd part power/ground
GND33A_LVTX_O		0		
GND09D_LVTX_O		0		
VCC18A_ADC	1.62	1.8	1.98	ADC analog power/ground
GND18A_ADC		0		
VCC18A_DAC0	1.62	1.8	1.98	DAC_0 analog power/ground
GND18A_DAC0		0		
VCC18A_DAC1	1.62	1.8	1.98	DAC_1 analog power/ground
GND18A_DAC1		0		

VCC18A_DAC2	1.62	1.8	1.98	DAC_2 analog power/ground
GND18A_DAC2		0		
VCC18A_DAC3	1.62	1.8	1.98	DAC_3 analog power/ground
GND18A_DAC3		0		
VCC18A_TDC	1.62	1.8	1.98	TDC analog power/ground
GND18A_TDC		0		
VCC09K_DDR_PHY	0.837	0.9	0.99	DDR analog power
VCC18A_DDR_PLL	1.71	1.8	1.89	
VCC12O_DDR0	1.14	1.2	1.26	
VCC12O_DDR1	1.14	1.2	1.26	
VCC12O_DDRCK	1.14	1.2	1.26	
VCC18_VPH_PCIE	1.62	1.8	1.98	PCIE analog power/ground
VCC09_VP_PCIE	0.837	0.9	0.99	
VCC09_VPTX0_PCIE	0.837	0.9	0.99	
VCC09_VPTX1_PCIE	0.837	0.9	0.99	
VCC09_VPTX2_PCIE	0.837	0.9	0.99	
VCC09_VPTX3_PCIE	0.837	0.9	0.99	
VCC09_VPDIG_PCIE	0.837	0.9	0.99	
GND_PCIE		0		
VCC09A_PLL0	0.81	0.9	0.99	PLL 0 analog power/ground
GND09A_PLL0		0		
VCC09A_PLL1	0.81	0.9	0.99	PLL 1 analog power/ground
GND09A_PLL1		0		
VCC09A_PLL2	0.81	0.9	0.99	PLL 2 analog power/ground

GND09A_PLL2		0		
VCC09A_PLL3	0.81	0.9	0.99	PLL 3 analog power/ground
GND09A_PLL3		0		
VCC09A_PLL4	0.81	0.9	0.99	PLL 4 analog power/ground
GND09A_PLL4		0		
VCC09A_DLL	0.81	0.9	0.99	DLL analog power/ground
GND09A_PDLL		0		
VCC18_RTC	1.62	1.8	1.98	RTC power
VCC18_VDD2_OTP	1.62	1.8	1.98	OTP analog power
X_FSOURCE	1.62	1.8	1.98	eFuse program power

[1] Note: This is IP simulation Junction Voltage specification, not final chip supply voltage specification.

16. Dedicate Selection Pin

Table 0-3. Dedicate Selection Pin

Name	Width	Dir.	Voltage	Default Setting	Pin Description
X_OM	1	I	1.8 V	PD	0: Function mode 1: Test mode Note: test mode is used for tester and chip debug only. It is not allowed for adding pull-up resistor at X_OM pin or normal function is not workable.

17. Hardware Strap Pin

Table 0-4. Hardware strap Pin

Strap function	Pin Name	Description
X_ISP	X_UART2_TX	1'b0 : Normal Mode, Secure Boot 1'b1 : ISP Mode, Flash programming
X_ISP_IF	X_UART2_RTSN	1'b0 : UART Mode, Baud rate :115200 1'b1 : USB, Speed: USB2.0. FTOG210 is used
X_BOOT_0	X_UART1_TX	{X_BOOT_1, X_BOOT_0} 2'b00 : Boot from SPI NOR Flash 2'b01 : Boot from SPI NAND Flash [1] 2'b10 : Boot from eMMC 2'b11 : Boot from SD Card
X_BOOT_1	X_UART1_RTSN	

[1] If system boot up with Winbond NAND flash, TX5 only support the flash device with "default BUF=1".

For example: W25N01GVxx**IG** / W25M02GVxx**IG**.

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LIFE SUPPORT APPLICATIONS

These products are not designed for use in life support appliances, devices, or systems where malfunction of these products can reasonably be expected to result in personal injury. Customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify Unimap for any damages resulting from such improper use or sales.

18. Package Information

The Unimap-TX5 package information can be found in *Table 0-5* and *Table 0-6*.

Table 0-5. HMI Package Summary Information

Package Parameter	Value	Unit	Description
Package Type	FCCSP		-
Package Dimensions	19x19	mm	-
Package Pins	784	pins	-
Package Pin Pitch	0.65	mm	-

Table 0-6. Motor Package Summary Information

Package Parameter	Value	Unit	Description
Package Type	FCCSP		-
Package Dimensions	17x17	mm	-
Package Pins	625	pins	-
Package Pin Pitch	0.65	mm	-

Power Domain

This chapter contains the following sections:

19 Power domain overview
20 Power Mode

19. Power domain overview

There are two internal power domains in the Unimap-TX5 and it includes core power domain and alive power domain. The power domains will be turned on/off using the external power switches controlled by the software through X_PWRCTRL pin. Figure 0-1 depicts the logic circuit partition by the power domain. Vcore domain has its own voltage monitor, POR will detect the POR reset signal of Vcore domain to check if the power of Vcore domain is ready or not. FTSCU100 PCU circuit will also detect the POR reset signal of Alive domain to check if the power of Alive domain is ready or not.

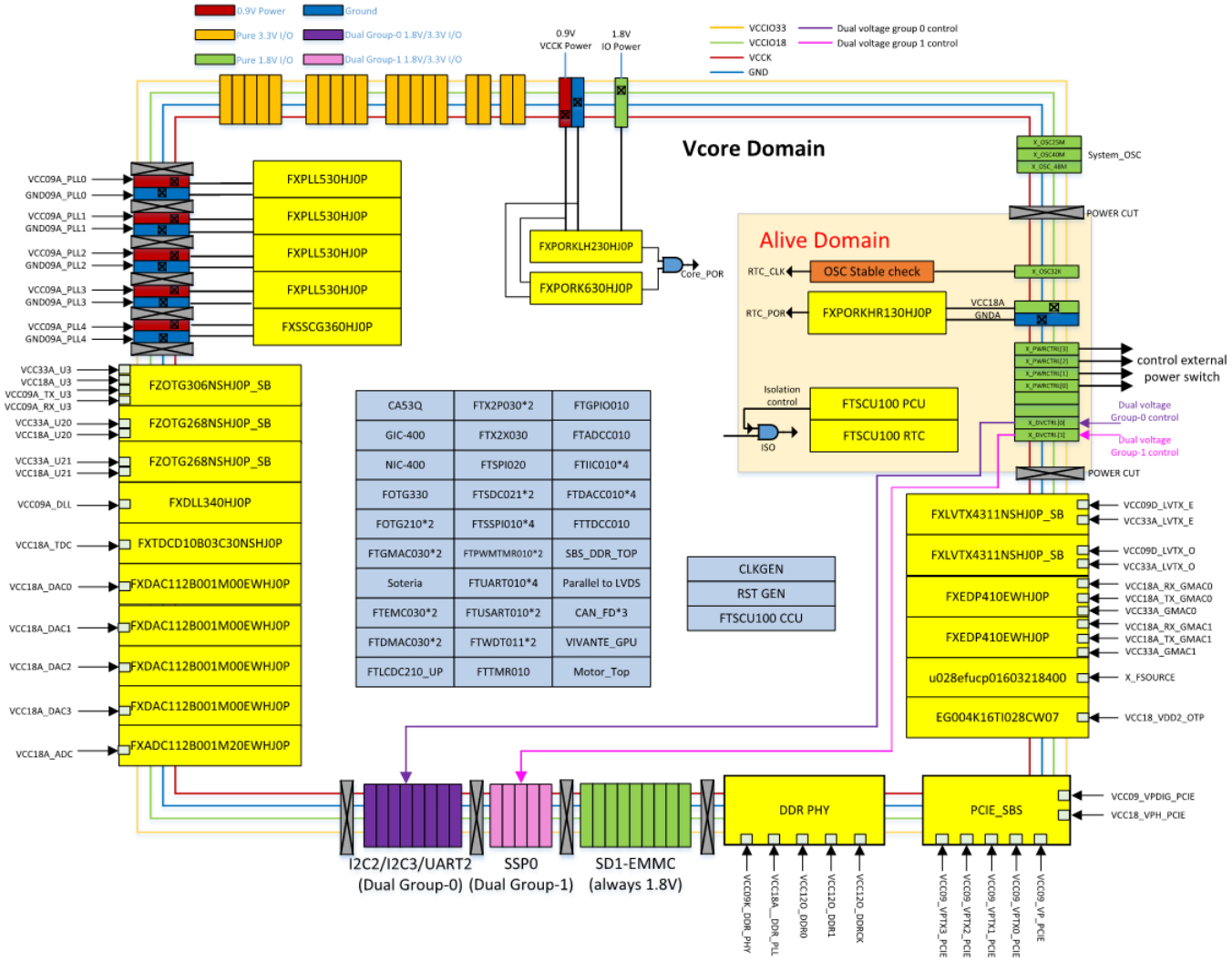


Figure 0-1. Power domain block diagram of Unimap-TX5

19.1 Power domain “Alive”

This power domain is powered by 1.8 V external power supply (or backup power supply) and it is always on. There are four Alive GPIOs are used for wake up system to turn on core power domain when entering soft-off mode.

19.2 Power domain “Vcore”

The power to Vcore power domain is controlled by software depending on low power requirement. When software issues the soft-off command to SCU, the external power switch is turned off. There are 12 events to wake up system from sleep mode to normal mode.

20. Power Mode

There are four power modes for Unimap-TX5: Soft-off mode, Sleep mode, Low speed mode, and Normal mode. The power mode relationship shows in *Figure 0-2*.

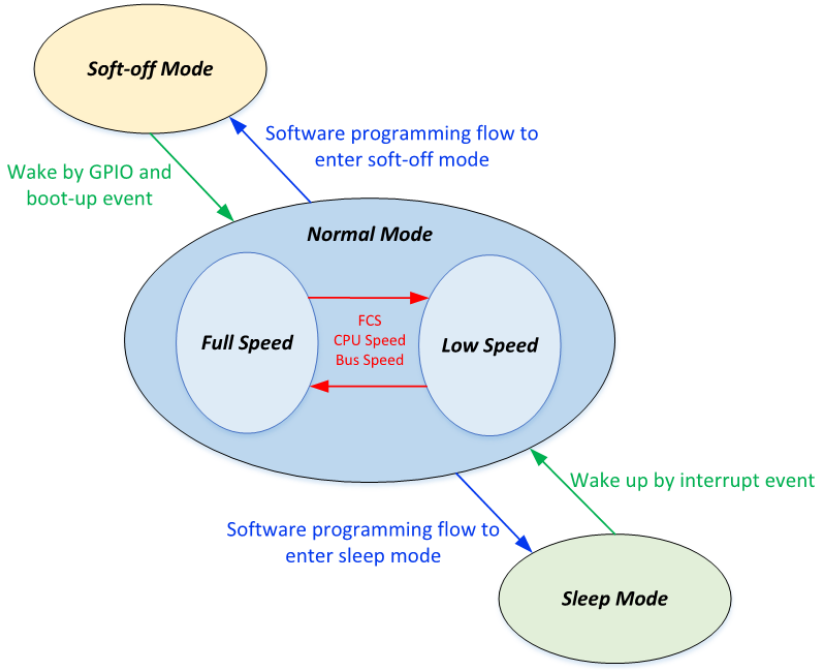


Figure 0-2. Power Mode Relationship diagram

20.1 Soft-off Mode

The soft-off mode is used to shut down the supply powers of all blocks of Unimap-TX5, except the logic in the alive domain. The system power consumption is the lowest, but it needs longer recovery time for system re-boot. *Figure 0-5* show the soft-off programming flow when the system enters the soft-off mode.
Unimap-TX5 provide 4 GPIOs and RTC alarm function to wake up system from soft-off mode. When system accepts wake up event in soft-off mode, system will assert X_PWRCTRL[3:0] to enable external power switch and power up the supply power of Vcore domain.

20.2 Sleep Mode

In the sleep mode, all clocks can be gated except the scu clock for saving the power consumption, and the most power consumption is from the leakage power. The scu clock is used to monitor the wakeup event. The IP to be clock gated can be selected, the control registers are shown in 0x0c10_0050 ~ 0x0c10_0088 of FTSCU100.

20.3 Low Speed Mode

In the low speed mode, chip is able at running all applications and CPU and bus clock can be switched to the lower frequency for saving the dynamic power. User could change those clock

frequency with FCS / CPU-speed / BUS-speed commands.

- CPU clock speed is switched to 600 MHz.
- AXI bus clock speed is switched to 200 MHz.
- AHB bus clock speed is switched to 50 MHz.
- APB bus clock speed is switched to 25 MHz.

20.4 Normal Mode

In the normal mode, chip is able at running all applications. Below lists the maximum clock speed of CPU and bus clock:

- CPU clock speed is up to 1200 MHz.
- AXI bus clock speed is up to 400 MHz.
- AHB bus clock speed is up to 100 MHz.
- APB bus clock speed is up to 50 MHz.

20.5 Power Mode Summary

X_PWRCTRL[0] is used to turn on/turn off of the core power of Vcore power domain.
X_PWRCTRL[1] is used to turn on/turn off IO power of Vcore power domain. X_PWRCTRL[3:2] are used to turn on/turn off other analog IP power of Vcore power domain.

Table 0-1. Power Mode Summary Table

Power Domain	Modules	Power Controlled Signal	Power Mode			
			Softoff	Sleep	Low Speed	Normal
Vcore	All modules are in Vcore domain except for alive_grp.	X_PWRCTRL[3:0]	Off	On	On	On
Alive	alive_grp	Always on	On	On	On	On

System Control and Boot-up Sequence

This chapter contains the following sections:

錯誤! 找不到參照來源。 錯誤! 找不到參照來源。
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錯誤! 找不到參照來源。 錯誤! 找不到參照來源。

21. System Control Unit

The System Control Unit (SCU) of Unimap-TX5 will require the following functionalities:

1. Hardware controlled boot-up state machine to perform system warm-start / cold-start sequence.
2. Software controlled clock gating for each IP block
3. Software controlled reset signals for each IP block
4. Software controlled power gating for core power domain
5. Frequency change sequence. (FCS command)
6. CPU/BUS speed change command.
7. Boot-up strap pin latch
8. Re-boot function.

21.1 Summary of Control Register

Table 0-1 lists and describes the summary of the control registers. Because FTSCU100 is standard IP, some registers exist that will not be used in Unimap-TX5. Additionally, detailed register information is contained in the following subsections.

Table 0-1. Summary of FTSCU100 Control Register (Base-address : 0x0C10_0000)

Offset	Name	Description
0x000	BTUP_STS	Boot-up status register
0x004	BTUP_CTRL	Boot-up control register
0x008	PWR_CTRL	Power control register
0x00C	PWRUP_SEQ	Power-up sequence control register
0x010	CHIPID	Chip ID register
0x014	VERID	Status control unit version register
0x018	STRAP	Strap value register
0x020	PWR_MOD	Power mode register
0x024	INT_STS	Interrupt status register
0x028	INT_EN	Interrupt enable register
0x02C	SWRST_CTRL	Software reset control register
0x030	PLL_CTRL	CPU PLL Control Register

Offset	Name	Description
0x034	PLL_USER_CTRL0	BUS PLL Control Register
0x038	PLL_USER_CTRL1	DDR SSCG PLL Control Register
0x03C	PLL_USER_CTRL2	DDR SSCG PLL Control Register
0x040	PLL2_CTRL	DDR SSCG PLL Control Register
0x050 0x054	AHBCLKG	AHB clock control register
0x058 0x05C	SLP_AHBCLKG	AHB clock sleep control register
0x060 0x064	APBCLKG	APB clock control register
0x068 0x06C	SLP_APBCLKG	APB clock sleep control register
0x080 0x084	AXICLKG	AXI clock gated register
0x088 0x08C	SLP_AXICLKG	AXI clock sleep control register
0x0B0	MPCORE_CTRL	Multi-CPU and Multi-Core control register
0x0B4	MCORE_WFI_CTRL	Multi-core WFI/WFE control register
0x0C0	SLP_WAKUP_ST	Sleep wakeup events status
0x0C4	SLP_WAKUP_EN	Sleep wakeup events enable
0x0F0	BUS_CTRL_STS	Bus Clock Control Status Register
0x0F8	Feature_0	Feature 0 Register
0x0FC	Feature_1	Feature 1 Register
0x120	EXT_CMD	Extend Command Register

Offset	Name	Description
0x124	SYS_STS	System Status Register
0x128	EXTEND_EN_CTRL	Extend enable Register
0x200	RTC_TIME1	RTC timer register 1
0x204	RTC_TIME2	RTC timer register 2
0x208	RTC_ALM1	RTC alarm time register 1
0x20C	RTC_ALM2	RTC alarm time register 2
0x210	RTC_CTRL	RTC control register
0x214	RTC_TRIM	RTC tick trim control register
0x220	BCD_RTC_TIME1	RTC timer register 1 with BCD code
0x224	BCD_RTC_TIME2	RTC timer register 2 with BCD code
0x228	BCD_RTC_ALM1	RTC alarm time register 1 with BCD code
0x22C	BCD_RTC_ALM2	RTC alarm time register 2 with BCD code
0x280 ~ 0x2BC	RTC_PSPR	RTC scratchpad registers

21.2 Boot-up Status Register (0x000)

Unimap-TX5 can be booted up from several events, which include the external boot-up ports of btup_event, RTC alarm, power button, and so on. These boot-up events are recorded in the boot-up status register when the system receives them at the soft-off mode. The BOOTUP_STS is used to record the events that are triggered from the external pins. These boot-up events can be individually masked by the boot-up control register on the customer applications. The PMR bit of the boot-up status register is re-booted from the soft-off mode. If the watchdog issues a reset, the WDR bit will be set. The HWR bit is set when re-booted by using the hardware reset.

Table 0-2. Boot-up status Register (0x000)

Bit	Symbol	Access	Default	Description
[31:22]	-	R	0x0	-
[21]	EXT_BOOTUP_STS X_WAKE[3]	RW1C	0x0	System was awakened by an external source. boot-up event triggered from X_WAKE[3]
[20]	EXT_BOOTUP_STS X_WAKE[2]	RW1C	0x0	System was awakened by an external source. boot-up event triggered from X_WAKE[2]
[19]	EXT_BOOTUP_STS X_WAKE[1]	RW1C	0x0	System was awakened by an external source. boot-up event triggered from X_WAKE[1]
[18]	EXT_BOOTUP_STS X_WAKE[0]	RW1C	0x0	System was awakened by an external source. boot-up event triggered from X_WAKE[0]
[17]	RTC_BTUP_STS	RW1C	0x0	Boot up from the RTC alarm event 1: Boot up by RTC alarm (Internal or external) events 0: No RTC alarm boot-up event
[16]	PWRBTN_STS	RW1C	0x0	Boot up by the power button 1: Boot up from the power button 0: No power button event
[15:14]	-	R	0x0	-
[13]	BROWN_OUT	RW1C	0x0	Enter the dormant mode due to the brown-out detect event 1: Power fault event triggered to enter the dormant mode 0: No power fault reset event
[12]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[11]	PMR	RW1C	0x0	Boot up from the power-off mode 1: Boot up from the power-off mode 0: No boot-up from the power-off mode
[10]	SMR	RW1C	0x0	Boot up from the dormant mode 1: Boot-up from the dormant mode 0: No boot-up from the dormant mode
[9]	WDR	RW1C	0x0	Re-boot by the watchdog reset 1: Watchdog reset occurred. 0: Watchdog reset does not occur.
[8]	HWR	RW1C	0x0	Re-boot by hardware reset 1: Hardware reset occurred. 0: Hardware reset does not occur.
[7:0]	-	R	0x0	-

21.3 Boot-up Control Register (0x004)

These boot-up events can be individually masked by the boot-up control register on the customer applications.

Table 0-3. Boot-up Control Register (0x004)

Bit	Symbol	Access	Default	Description
[31:22]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[21]	EXT_BOOTUP_EN X_WAKE[3]	RW	0x1	These bits selectively enable the usage of the boot-up events that are triggered from the “X_WAKE[3]” pins.
[20]	EXT_BOOTUP_EN X_WAKE[2]	RW	0x1	These bits selectively enable the usage of the boot-up events that are triggered from the “X_WAKE[2]” pins.
[19]	EXT_BOOTUP_EN X_WAKE[1]	RW	0x1	These bits selectively enable the usage of the boot-up events that are triggered from the “X_WAKE[1]” pins.
[18]	EXT_BOOTUP_EN X_WAKE[0]	RW	0x1	These bits selectively enable the usage of the boot-up events that are triggered from the “X_WAKE[0]” pins.
[17]	RTC_BTUP_EN	RW	0x1	This bit allows the RTC alarm signal to contribute to the boot-up of the power control unit. 1’b1: RTC alarm output is enabled and a transition from low to high on the RTC alarm output will cause a boot-up event to the power control unit. 1’b0: RTC alarm output is disabled; transition on the RTC alarm output will not cause a wakeup event to the power control unit.
[16]	PWRBTN_EN	RW	0x1	This bit allows the power button signal to contribute to the boot-up of the power control unit. 1’b1: Power button output, after the debounce circuit, this signal is enabled and a transition from low to high on the power button debounce output will cause a boot-up event to the power control unit.

Bit	Symbol	Access	Default	Description
				1’b0: Power button output is disabled; the transition on the power button debounce output will not cause a wakeup event to the power control unit.
[15:0]	-	RW	0x0	-

21.4 Power Control Register (0x008)

Unimap-TX5 provides four programmable power control pins for the external regulators control.
The PWRDN_CTRL bits of the power control register are set to be the expected power control pins when the system enters the soft-off mode. The PWRUP_CTRL bits are able to directly set the current power control pins. Users can program these bits and then set the PWRCTRL_UPDATE bit. An additional function sets the power control pins in advance when the system is re-booted from the soft-off mode. Users can also program the PWRUP_CTRL bits without setting PWRCTRL_UPDATE before entering the soft-off mode.

The power control pins are not immediately changed, and will be updated when the system comes back to the working state from the dormant mode. The PWRCTRL_DCSR bits are used to set the I/O driving capability for the power control pins.

Table 0-4. Power Control Register (0x008)

Bit	Symbol	Access	Default	Description
[31:28]	PWRCTRL_DCSR	RW	0x1	Power pins I/O driving capability For Unimap standard I/O, it is allowed to adjust the I/O driving capability. The register provides to set the programmable pins by software.
[27]	pwrbtn_offchk	RW	0x0	Power button release check in the power off procedure 1: Check the power button release when entering the power-off or dormant mode. 0: Do not check the power button status when entering the power-off or dormant mode
[26:25]	brown_out_mode	RW	0x0	Core power domain (CCU power domain) brown-out detected to enter the power mode selection 2'b11: Brown-out enters the power-off mode 2'b10: Brown-out enters the dormant mode 2'b0x: Disable the Core power domain (CCU domain) brown-out check function
[24]	PWRCTRL_UPDATE	RWS	0x0	Power control pins update command The bit is set to directly set the power control pins value by the PWRUP_CTRL, and it will be cleared after finishing the command.
[23:16]	debounce_mode	RW	0x5E	The register used to configure the debounce time for the power

Bit	Symbol	Access	Default	Description
				button.
[15:12]	-	R	0x0	-
[11:8]	PWRUP_CTRL	RW	0xF	Power control pins at the working state
[7:4]	-	R	0x0	-
[3:0]	PWRDN_CTRL	RW	0x0	Power control pins at soft-off state

21.5 Power Start-up Sequence Control Register (0x00C)

A programmable power-up sequence is provided by Unimap-TX5, and it allows the power control pins that do not need to be simultaneously enabled. It provide a time division to in-order power enable to avoid in-rush current when simultaneously turning on the large logical. Unimap-TX5 has extra four timeslots and programmable registers for the power pins enable sequence setting.

Table 0-5. Power Start-up Sequence Control Register (0x00C)

Bit	Symbol	Access	Default	Description
[31:28]	-	R	0x0	-
[27:24]	PWRUP_ENSLOT4	RW	0xF	Power control pins at timeslot4 in the power start-up sequence
[23:20]	-	R	0x0	-
[19:16]	PWRUP_ENSLOT3	RW	0x7	Power control pins at timeslot3 in the power start-up sequence
[15:12]	-	R	0x0	-
[11:8]	PWRUP_ENSLOT2	RW	0x3	Power control pins at timeslot2 in the power start-up sequence
[7:4]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[3:0]	PWRUP_ENSLOT1	RW	0x1	Power control pins at timeslot1 in the power start-up sequence

21.6 Chip ID Register (0x010)

This registers configured by SoC integrator to provide the chip ID. The register can be programmed by software though a unlock sequence.

Table 0-6. Chip ID Register (0x010)

Bit	Symbol	Access	Default	Description
[31:0]	CHIP ID	R or R/PW	0xFFFF_FFFF	Chip ID. The register can be programmed through a unlock procedure; otherwise, it will be read-only mode.

21.7 System Control Unit Version Register (0x014)

This register identifies the version of FTSCU100.

Table 0-7. System Control Unit Version Register (0x014)

Bit	Symbol	Access	Default	Description
[31:0]	IP VERSION	R	0x0003_0100	IP version

21.8 Strap Value Register (0x018)

The strapping pins are used to decide the initial value through the external pull-up/pull-down resister at the reset interval. These pins generally should be selected for the pure output function in

Table 0-8. Strap Value Register (0x018)

Bit	Symbol	Access	Default	Description
[31:20]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[19:18]	Strap Pin : X_BOOT[1:0]	R	STRAP	Strap pin X_BOOT[0] is from X_UART1_TX. Strap pin X_BOOT[1] is from X_UART1_RTSN. 2'b00: boot from SPI NOR flash. 2'b01: boot from SPI NAND flash. 2'b10: boot from eMMC. 2'b11: boot from SD Card.
[17]	Strap Pin : X_ISP_IF	R	STRAP	Strap pin is from X_UART2_RTSN. 1: boot from USB2.0. FOTG210 is used. 0: boot from UART, baud rate 115200.
[16]	Strap Pin : X_ISP	R	STRAP	Strap pin is from X_UART2_TX. 1: Secure boot. 0: Flash programming
[15:0]	-	R	0x0	-

21.9 Power Mode Register (0x020)

The power mode register provides commands and parameters of the operation sequence at the working state. FTSCU100 allows the combination of several commands for transition in one operation mode.

Table 0-9. Power Mode Register (0x020)

Bit	Symbol	Access	Default	Description
[31]	SELFR_CMD_OFF	RW	0x0	DDR/SDR self-refresh command off.

Bit	Symbol	Access	Default	Description
				If this bit is set, FTSCU100 will NOT issue the self-refresh command to DDR/SDR in all power modes and frequency change sequences. If system does not have DDR/SDR, please always set to 1'b1.
[30]	FCS_PLL2_RSTn	RW	0x0	PLL2 resets in the frequency change sequence. If the PLL2 output frequency is not satisfied, PLL2 should be reset in the FCS mode. The bit is valid, when the hardware parameter, FTSCU100_DDR3PLL_ON, is defined. 0: Reset PLL2 1: Keep PLL2 active in FCS
[29]	-	RW	0x0	-
[28]	FCS_PLL_RSTn	RW	0x0	PLL resets in the frequency change sequence. If the PLL output frequency is not satisfied, PLL should be reset in the FCS mode. 0: Reset PLL 1: Keep PLL active in FCS
[27:24]	-	RW	0x0	-
[23:20]	BUS_MUX	RW	0x0	Bus target frequency programmable register. It is used to set the output value of scu_busclk_mux for the controlled multiplexer of the BUS clock after issuing bus_speed or FCS command. The value should be held before setting the

Bit	Symbol	Access	Default	Description
				BUS_SPEED command. [0]: 1'b0: aclk = 400MHz . 1'b1: aclk = 200MHz . [1]: 1'b0: hclk = 100MHz . 1'b1: hclk = 50MHz . [2]: 1'b0: pclk = 50MHz . 1'b1: pclk = 25MHz .
[19:16]	FCLK_MUX	RW	0x0	CPU target frequency programmable register. It is used to set the scu_fclk_mux output value for the controlled multiplexer of the BUS clock after issuing cpu_speed or FCS command. The value should be held before setting the CPU_SPEED command. [0]: 1'b0: cpu_fclk/cpu_aclk = 1.2G/400MHz . 1'b1: cpu_fclk/cpu_aclk = 600M/200MHz .
[15:14]	-	R	0x0	-
[13:12]	CPUSPEED_MSK	RW	0x0	Disable the CPU speed function for multi-CPU. It is used to select which CPU will be executed and which CPU speed command for multi-CPU, and users can program 1 to disable the CPU speed function. [0] : mask wfi from CA53 [1] : mask wfi from CA53 L2
[11]	-	RW	0x0	-
[10]	SW_RST	RWS	0x0	Software reset command. The bit is set to issue a reset signal that the interval cab be

Bit	Symbol	Access	Default	Description
				adjusted by programming SWRST_CTRL (offset 0x02C), and be cleared by hardware after the reset is done. Each IP sw_rst mask register is referred to 0x0c10_0920 ~ 0x0c10_092C.
[9]	CURR_RESTORE	RWS	0x0	Restore current setting frequency in programming register. All PLL setting values, including CLKIN_MUX, and the power mode setting values, FCLK_MUX, BUS_MUX, and REMAP will be restored to the PLL control (PLL_CTRL) and power mode (PWR_MOD) registers.
[8]	PLL_UPDATE	RWS	0x0	PLL update command. PLL setting value is directly updated by the PLL programming register.
[7]	REBOOT	RWS	0x0	Reboot command. The bit is set to issue a reset to initiate the CPU and bus state. In the combined command, the reset will be de-asserted after another operation sequence is done. However, it is prohibited to perform the sleep command. Each IP reboot mask register is referred to 0x0c10_0930 ~ 0x0c10_093C.
[6]	FCS	RWS	0x0	Frequency Change Sequence (FCS). The bit is set to enter the frequency change sequence, and will

Bit	Symbol	Access	Default	Description
				be automatically cleared after finishing FCS.
[5]	BUS_SPEED	RWS	0x0	BUS speed change command. The bit is set to change the BUS speed and will be automatically cleared after finishing the BUS speed mode. The behavior is similar to the FCS command when the SOFTOFF bit is simultaneously set. The BUS speed command will be done after booting-up from the dormant mode.
[4]	CPU_SPEED	RWS	0x0	CPU speed change command. The bit is set to change the CPU speed and will be automatically cleared after finishing the CPU speed mode. The behavior is similar to the FCS command when the SOFTOFF bit is simultaneously set. The CPU speed command will be done after booting-up from the dormant mode
[3]	SLEEP	RWS	0x0	Sleep-mode sequence. This bit will be automatically cleared after waking up. The command does not execute with the soft-off or power-off command.
[2]	-	R	0x0	-
[1]	SOFTOFF	RWS	0x0	Soft-off command. To enter dormant (deep-sleep) mode When the bit is set, the system enters the dormant state, and will be reset when the

Bit	Symbol	Access	Default	Description
				system comes back from the working state.
[0]	-	R	0x0	-

21.10 Interrupt Status Register (0x024)

The system control unit status register presents the interrupt status and some system operating mode flags.

Table 0-10. Interrupt Status Register (0x024)

Bit	Symbol	Access	Default	Description
[31:19]	-	R	0x0	-
[18]	INT_RTC_SEC	RW1C	0x0	RTC second out interrupt status
[17]	INT_RTC_PER	RW1C	0x0	RTC periodic interrupt status
[16]	INT_RTC_ALARM	RW1C	0x0	RTC alarm interrupt status
[15:13]	-	R	0x0	-
[12]	-	RW1C	0x0	-
[11:9]	-	R	0x0	-
[8]	INT_PLL_UPDATE	RW1C	0x0	PLL update finish interrupt status
[7]	-	R	0x0	-
[6]	INT_FCS	RW1C	0x0	FCS command finish interrupt status
[5]	INT_BUSSPEED	RW1C	0x0	BUS speed finish interrupt status
[4]	INT_CPUSPEED	RW1C	0x0	CPU speed finish interrupt status

Bit	Symbol	Access	Default	Description
[3]	INT_WAKEUP	RW1C	0x0	Wake up event status in the sleep mode
[2]	-	R	0x0	-
[1]	INT_PWRBTN_RISE	RW1C	0x0	Power button rising interrupt status at the working state
[0]	INT_PWRBTN_FALL	RW1C	0x0	Power button falling interrupt status at the working state

21.11 Interrupt Enable Register (0x028)

The system control unit status register presents the interrupt status and some system operating mode flags.

Table 0-11. Interrupt Enable Register (0x028)

Bit	Symbol	Access	Default	Description
[31:19]	-	R	0x0	-
[18]	RTC_SEC_EINT	RW	0x1	RTC second out interrupt enable
[17]	RTC_PER_EINT	RW	0x1	RTC periodic interrupt enable
[16]	RTC_ALARM_EINT	RW	0x1	RTC alarm interrupt enable
[15:13]	-	R	0x0	-
[12]	-	RW	0x0	-
[11:9]	-	R	0x0	-
[8]	PLL_UPDATE_EINT	RW	0x1	PLL update finish interrupt enable
[7]	-	R	0x1	-

Bit	Symbol	Access	Default	Description
[6]	FCS_EINT	RW	0x1	FCS command finish interrupt enable
[5]	BUSSPEED_EINT	RW	0x1	BUS speed finish interrupt enable
[4]	CPUSPEED_EINT	RW	0x1	CPU speed finish interrupt enable
[3]	WAKEUP_EINT	RW	0x1	Wake up event enable in the sleep mode
[2]	-	R	0x0	-
[1]	PWRBTN_RISE_EINT	RW	0x1	Power button rising interrupt enable at the working state
[0]	PWRBTN_FALL_EINT	RW	0x1	Power button falling interrupt enable at the working state

21.12 Software Reset Control Register (0x02C)

The manual-controlled software method can be used to assert and de-assert the reset signal through programming SWRST_CTRL. If SWRST_WAIT is all-set, all bits are written to one, and SWRST_ACTIVE is all-clear, all bits are written to zero, the software reset signal will be directly controlled by SW_RST. When SW_RST is set, the software reset signal is asserted, and SW_RST writes 0 to de-assert the signal. The reset period is fully controlled by software programming.

Table 0-12. Software Reset Control Register (0x02C)

Bit	Symbol	Access	Default	Description
[31:21]	-	R	0x0	-
[20:16]	SWRST_WAIT	RW	0x0	Software reset delay cycles of APB bus clock
[15:5]	-	R	0x0	-
[4:0]	SWRST_ACTIVE	RW	0xF	Software reset active cycles of APB bus clock

21.13 PLL Control Register (0x030 ~ 0x040)

The PLL control register is used to set the PLL values (0x030 ~0x040). These programmed PLL setting values does not directly write to the PLL control pins, it needs to set the FCS or PLL_UPDATE command in the power-mode register. If users want to know the current setting of PLL, users can set the PLL_RESTORE command to resume the register content.

Table 0-13. CPU PLL Control Register (0x030)

Bit	Symbol	Access	Default	Description
[31:25]	-	RW	0x0	-
[24:16]	CPU_PLL_NS	RW	0x64	control the CPU PLL NS[8:0]
[15:12]	CPU_PLL_PS	RW	0x0	control the CPU PLL PS[3:0]
[11]	-	RW	0x0	-
[10:8]	CPU_PLL_MS	RW	0x1	control the CPU PLL MS[2:0]
[7]	-	R	0x0	-
[6:4]	-	RW	0x0	-
[3:2]	-	R	0x0	-
[1]	PLLSTABLE	R	0x0	PLL locking and stability status 1: PLL is stable. 0: PLL is not stable
[0]	PLLEN	RW	0x0	PLL enable control, the PLL cannot directly be turned ON/OFF when the bit is set. The procedure should accompany a FCS command to stop or enable PLL.

Bit	Symbol	Access	Default	Description
				1: Enable all CPU PLL/ BUS PLL/ LVDS PLL/ MAC PLL 0: Disable all CPU PLL/ BUS PLL/ LVDS PLL/ MAC PLL

Table 0-14. BUS PLL Control Register (0x034)

Bit	Symbol	Access	Default	Description
[31:17]	-	RW	0x0	-
[16:8]	BUS_PLL_NS	RW	0x190	control the BUS PLL NS[8:0]
[7:4]	BUS_PLL_PS	RW	0x2	control the BUS PLL PS[3:0]
[3]	-	RW	0x0	-
[2:0]	BUS_PLL_MS	RW	0x3	control the BUS PLL MS[2:0]

Table 0-15. DDR SSCG PLL Control Register (0x038)

Bit	Symbol	Access	Default	Description
[31:29]	-	RW	0x0	-
[28]	DDR_PLL_SSEN	RW	0x0	control the DDR PLL SSEN
[27:25]	-	RW	0x0	-
[24]	DDR_PLL_FIR	RW	0x0	control the DDR PLL FIR
[23:21]	-	RW	0x0	-
[20]	DDR_PLL_MP	RW	0x0	control the DDR PLL MP
[19:18]	-	RW	0x0	-
[17:16]	DDR_PLL_TYPE	RW	0x0	control the DDR PLL TYPE[1:0]
[15]	-	RW	0x0	-
[14:12]	DDR_PLL_IS	RW	0x0	control the DDR PLL IS[2:0]

Bit	Symbol	Access	Default	Description
[11:0]	DDR_PLL_NS	RW	0x0	control the DDR PLL NS[11:0]

Table 0-16. DDR SSCG PLL Control Register (0x03C)

Bit	Symbol	Access	Default	Description
[31:24]	-	RW	0x0	-
[23:0]	DDR_PLL_NV	RW	0x0	control the DDR PLL NV[23:0]

Table 0-17. DDR SSCG PLL Control Register (0x040)

Bit	Symbol	Access	Default	Description
[31:8]	DDR_PLL_NF	RW	0x0C8000	control the CPU PLL NF[23:0]
[7:1]	-	R	0x0	-
[0]	PLL2EN	RW	0x0	PLL2 enable control The PLL2 cannot be directly turned ON/OFF when the bit is set. The procedure should accompany a FCS command to stop or enable PLL2. 1: Enable DDR PLL 0: Disable DDR PLL

21.14 HCLK & other clock gating control (0x050 ~ 0x05C)

The register is used to control an output port, scu_hclkg_n, which is designed to block the clocks of the AHB bus for the low-power consumption requirement. It usually connects to clock gated cells in the clock generation module (CLK_GEN) for the SoC integration. Register 0x50 and 0x54 present the clock gating setting when system works in normal mode. Register 0x58 and 0x5C present the clock gating setting when system works in sleep mode.

Table 0-18. hclk & other clock gating control (0x050)

Bit	Symbol	Access	Default	Description
[31]	scu_hclkg_n[31:0]	RW	0x0	usb3_coreclkin clock gating control 1'b1 : enable clock, 1'b0 : disable clock
[30]		RW	0x0	otg_1_coreclkin clock gating control
[29]		RW	0x1	otg_0_coreclkin clock gating control

Bit	Symbol	Access	Default	Description
[28]		RW	0x1	ddr_clk clock gating control
[27]		RW	0x0	lvds_lc_scaler_clk clock gating control
[26]		RW	0x0	lvds_tv_clk clock gating control
[25]		RW	0x0	lvds_lc_clk clock gating control
[24]		RW	0x0	gpu_aclk clock gating control
[23]		RW	0x0	gpu_clk2x clock gating control
[22]		RW	0x0	gmac1_ptpclk clock gating control
[21]		RW	0x0	gmac1_clk_2pt5_i clock gating control
[20]		RW	0x0	gmac1_clk_25_i clock gating control
[19]		RW	0x0	gmac1_clk_125_i clock gating control
[18]		RW	0x0	gmac0_ptpclk clock gating control
[17]		RW	0x0	gmac0_clk_2pt5_i clock gating control
[16]		RW	0x0	gmac0_clk_25_i clock gating control
[15]		RW	0x0	gmac0_clk_125_i clock gating control
[14]		RW	0x0	canfd_degli_clk clock gating control
[13]		RW	0x0	canfd_ext_clk clock gating control
[12]		RW	0x0	can2_osc_clk clock gating control
[11]		RW	0x0	can1_osc_clk clock gating control
[10]		RW	0x0	can0_osc_clk clock gating control

Bit	Symbol	Access	Default	Description
[9]		RW	0x0	tdc_ctrl_xclk clock gating control
[8]		RW	0x0	adc_mclk clock gating control
[7]		RW	0x1	sec_hclk clock gating control
[6]		RW	0x1	sd1_sys_hclk clock gating control
[5]		RW	0x1	sd0_sys_hclk clock gating control
[4]		RW	0x1	nic_hclk clock gating control
[3]		RW	0x0	can2_hclk clock gating control
[2]		RW	0x0	can1_hclk clock gating control
[1]		RW	0x0	can0_hclk clock gating control
[0]		RW	0x0	gpu_hclk clock gating control

Table 0-19. hclk & other clock gating control (0x054)

Bit	Symbol	Access	Default	Description
[31:26]		RW	0x3F	-
[25]	scu_hclkg_n[63:32]	RW	0x0	motor_exADC_sys40mclk clock gating control 1'b1 : enable clock, 1'b0 : disable clock
[24]		RW	0x0	motor_eqep_sysclk clock gating control
[23]		RW	0x0	motor_epwm_sysclk clock gating control
[22]		RW	0x0	motor_endat_sysclk clock gating control

Bit	Symbol	Access	Default	Description
[21]		RW	0x0	pcie auxclk clock gating control
[20]		RW	0x0	usart1_clk clock gating control
[19]		RW	0x0	usart0_clk clock gating control
[18]		RW	0x0	uart3_uclk clock gating control
[17]		RW	0x0	uart2_uclk clock gating control
[16]		RW	0x1	uart1_uclk clock gating control
[15]		RW	0x1	uart0_uclk clock gating control
[14]		RW	0x0	ssp3_sspclk clock gating control
[13]		RW	0x0	ssp2_sspclk clock gating control
[12]		RW	0x1	ssp1_sspclk clock gating control
[11]		RW	0x0	ssp0_sspclk clock gating control
[10]		RW	0x1	sec_rsa_clk clock gating control
[9]		RW	0x1	spi_clk clock gating control
[8]		RW	0x1	sd1_clk clock gating control
[7]		RW	0x1	sd0_clk clock gating control
[6]		RW	0x1	wdtn_extclk clock gating control (n=0~1)
[5]		RW	0x0	tmr_extclkn clock gating control (n=1~3)
[4]		RW	0x0	pwmtmr1_extclkn clock gating control (n=1~8)
[3]		RW	0x0	pwmtmr0_extclkn clock gating control (n=1~8)

Bit	Symbol	Access	Default	Description
[2]		RW	0x0	usb3_aclk clock gating control
[1]		RW	0x0	usb3_lpclk clock gating control
[0]		RW	0x0	usb3_clk60 clock gating control

Table 0-20. hclk & other clock gating control in sleep mode (0x058)

Bit	Symbol	Access	Default	Description
[31]	slp_hclkg_n[31:0]	RW	0x0	usb3_coreclkin clock gating control in sleep mode 1'b1 : enable clock, 1'b0 : disable clock
[30]		RW	0x0	otg_1_coreclkin clock gating control in sleep mode
[29]		RW	0x1	otg_0_coreclkin clock gating control in sleep mode
[28]		RW	0x1	ddr_clk clock gating control in sleep mode
[27]		RW	0x0	lvds_lc_scaler_clk clock gating control in sleep mode
[26]		RW	0x0	lvds_tv_clk clock gating control in sleep mode
[25]		RW	0x0	lvds_lc_clk clock gating control in sleep mode
[24]		RW	0x0	gpu_aclk clock gating control in sleep mode
[23]		RW	0x0	gpu_clk2x clock gating control in sleep mode
[22]		RW	0x0	gmac1_ptpclk clock gating control in sleep mode
[21]		RW	0x0	gmac1_clk_2pt5_i clock gating control in sleep mode
[20]		RW	0x0	gmac1_clk_25_i clock gating control in sleep mode

Bit	Symbol	Access	Default	Description
[19]		RW	0x0	gmac1_clk_125_i clock gating control in sleep mode
[18]		RW	0x0	gmac0_ptpclk clock gating control in sleep mode
[17]		RW	0x0	gmac0_clk_2pt5_i clock gating control in sleep mode
[16]		RW	0x0	gmac0_clk_25_i clock gating control in sleep mode
[15]		RW	0x0	gmac0_clk_125_i clock gating control in sleep mode
[14]		RW	0x0	canfd_degli_clk clock gating control in sleep mode
[13]		RW	0x0	canfd_ext_clk clock gating control in sleep mode
[12]		RW	0x0	can2_osc_clk clock gating control in sleep mode
[11]		RW	0x0	can1_osc_clk clock gating control in sleep mode
[10]		RW	0x0	can0_osc_clk clock gating control in sleep mode
[9]		RW	0x0	tdc_ctrl_xclk clock gating control in sleep mode
[8]		RW	0x0	adc_mclk clock gating control in sleep mode
[7]		RW	0x1	sec_hclk clock gating control in sleep mode
[6]		RW	0x1	sd1_sys_hclk clock gating control in sleep mode
[5]		RW	0x1	sd0_sys_hclk clock gating control in sleep mode
[4]		RW	0x1	nic_hclk clock gating control in sleep mode
[3]		RW	0x0	can2_hclk clock gating control in sleep mode
[2]		RW	0x0	can1_hclk clock gating control in sleep mode
[1]		RW	0x0	can0_hclk clock gating control in sleep mode

Bit	Symbol	Access	Default	Description
[0]		RW	0x0	gpu_hclk clock gating control in sleep mode

Table 0-21. hclk & other clock gating control in sleep mode (0x05C)

Bit	Symbol	Access	Default	Description
[31:26]		RW	0x3F	-
[25]		RW	0x0	motor_exADC_sys40mclk clock gating control in sleep mode 1'b1 : enable clock, 1'b0 : disable clock
[24]		RW	0x0	motor_eqep_sysclk clock gating control in sleep mode
[23]		RW	0x0	motor_epwm_sysclk clock gating control in sleep mode
[22]		RW	0x0	motor_endat_sysclk clock gating control in sleep mode
[21]		RW	0x0	pcie auxclk clock gating control in sleep mode
[20]	slp_hclkg_n[63:32]	RW	0x0	usart1_clk clock gating control in sleep mode
[19]		RW	0x0	usart0_clk clock gating control in sleep mode
[18]		RW	0x0	uart3_uclk clock gating control in sleep mode
[17]		RW	0x0	uart2_uclk clock gating control in sleep mode
[16]		RW	0x1	uart1_uclk clock gating control in sleep mode
[15]		RW	0x1	uart0_uclk clock gating control in sleep mode
[14]		RW	0x0	ssp3_sspclk clock gating control in sleep mode
[13]		RW	0x0	ssp2_sspclk clock gating control in sleep mode

Bit	Symbol	Access	Default	Description
[12]		RW	0x1	ssp1_sspclk clock gating control in sleep mode
[11]		RW	0x0	ssp0_sspclk clock gating control in sleep mode
[10]		RW	0x1	sec_rsa_clk clock gating control in sleep mode
[9]		RW	0x1	spi_clk clock gating control in sleep mode
[8]		RW	0x1	sd1_clk clock gating control in sleep mode
[7]		RW	0x1	sd0_clk clock gating control in sleep mode
[6]		RW	0x1	wdtn_extclk clock gating control in sleep mode (n=0~1)
[5]		RW	0x0	tmr_extclkn clock gating control in sleep mode (n=1~3)
[4]		RW	0x0	pwmtmr1_extclkn clock gating control in sleep mode (n=1~8)
[3]		RW	0x0	pwmtmr0_extclkn clock gating control in sleep mode (n=1~8)
[2]		RW	0x0	usb3_aclk clock gating control in sleep mode
[1]		RW	0x0	usb3_lpclk clock gating control in sleep mode
[0]		RW	0x0	usb3_clk60 clock gating control in sleep mode

21.15 PCLK clock gating control (0x060 ~ 0x06C)

Register 0x60 and 0x64 present the clock gating setting in pclk domain when system works in normal mode.

Register 0x68 and 0x6C present the clock gating setting in pclk domain when system works in sleep mode.

Table 0-22. pclk clock gating control (0x060)

Bit	Symbol	Access	Default	Description
[31]	scu_pclkg_n[31:0]	RW	0x0	tmr_pclk clock gating control 1'b1 : enable clock, 1'b0 : disable clock
[30]		RW	0x0	tdc_pclk clock gating control
[29]		RW	0x0	ssp3_pclk clock gating control
[28]		RW	0x0	ssp2_pclk clock gating control
[27]		RW	0x1	ssp1_pclk clock gating control
[26]		RW	0x0	ssp0_pclk clock gating control
[25]		RW	0x0	pwmtmr1_pclk clock gating control
[24]		RW	0x1	pwmtmr0_pclk clock gating control
[23]		RW	0x0	lvds_pclk clock gating control
[22]		RW	0x0	i2c3_pclk clock gating control
[21]		RW	0x0	i2c2_pclk clock gating control
[20]		RW	0x1	i2c1_pclk clock gating control
[19]		RW	0x1	i2c0_pclk clock gating control
[18]		RW	0x1	gpio_pclk clock gating control
[17]		RW	0x0	gphy1_pclk clock gating control
[16]		RW	0x0	gphy0_pclk clock gating control
[15]		RW	0x0	gmac1_pclk clock gating control
[14]		RW	0x0	gmac0_pclk clock gating control

Bit	Symbol	Access	Default	Description
[13]		RW	0x1	emc_ram1_pclk clock gating control
[12]		RW	0x1	emc_ram0_pclk clock gating control
[11]		RW	0x1	dmac030_1_pclk clock gating control
[10]		RW	0x1	dmac030_0_pclk clock gating control
[9]		RW	0x1	ddr_pclk clock gating control
[8]		RW	0x0	pcie_i_apb_clk clock gating control (pcie cr interface control clock)
[7]		RW	0x0	pcie_ext_pclk clock gating control (pcie side band control clock)
[6]		RW	0x0	usb3_apb_clk clock gating control
[5]		RW	0x0	otg_1_apb_clk clock gating control
[4]		RW	0x1	otg_0_apb_clk clock gating control
[3]		RW	0x0	dac3_clk clock gating control
[2]		RW	0x0	dac2_clk clock gating control
[1]		RW	0x0	dac1_clk clock gating control
[0]		RW	0x0	dac0_clk clock gating control

Table 0-23. pclk clock gating control (0x064)

Bit	Symbol	Access	Default	Description
[31:16]		RW	0xFFFF	-
[15]		RW	0x0	motor_xmc_ctl_pclk clock gating control 1'b1 : enable clock, 1'b0 : disable clock
[14]		RW	0x0	motor_exadc_pclk clock gating control
[13]		RW	0x0	motor_eqep_pclk clock gating control
[12]		RW	0x0	motor_epwm_pclk clock gating control
[11]		RW	0x0	adc_pclk clock gating control
[10]		RW	0x0	x2x_pclk clock gating control
[9]	scu_pclkg_n[63:32]	RW	0x1	x2p1_pclk clock gating control
[8]		RW	0x1	x2p0_pclk clock gating control
[7]		RW	0x0	wdt1_pclk clock gating control
[6]		RW	0x1	wdt0_pclk clock gating control
[5]		RW	0x0	usart1_pclk clock gating control
[4]		RW	0x0	usart0_pclk clock gating control
[3]		RW	0x0	uart3_pclk clock gating control
[2]		RW	0x0	uart2_pclk clock gating control
[1]		RW	0x1	uart1_pclk clock gating control
[0]		RW	0x1	uart0_pclk clock gating control

Table 0-24. pclk clock gating control in sleep mode (0x068)

Bit	Symbol	Access	Default	Description
[31]	slp_pclkg_n[31:0]	RW	0x0	tmr_pclk clock gating control in sleep mode 1'b1 : enable clock, 1'b0 : disable clock
[30]		RW	0x0	tdc_pclk clock gating control in sleep mode
[29]		RW	0x0	ssp3_pclk clock gating control in sleep mode
[28]		RW	0x0	ssp2_pclk clock gating control in sleep mode
[27]		RW	0x1	ssp1_pclk clock gating control in sleep mode
[26]		RW	0x0	ssp0_pclk clock gating control in sleep mode
[25]		RW	0x0	pwmtmr1_pclk clock gating control in sleep mode
[24]		RW	0x1	pwmtmr0_pclk clock gating control in sleep mode
[23]		RW	0x0	lvds_pclk clock gating control in sleep mode
[22]		RW	0x0	i2c3_pclk clock gating control in sleep mode
[21]		RW	0x0	i2c2_pclk clock gating control in sleep mode
[20]		RW	0x1	i2c1_pclk clock gating control in sleep mode
[19]		RW	0x1	i2c0_pclk clock gating control in sleep mode
[18]		RW	0x1	gpio_pclk clock gating control in sleep mode
[17]		RW	0x0	gphy1_pclk clock gating control in sleep mode
[16]		RW	0x0	gphy0_pclk clock gating control in sleep mode
[15]		RW	0x0	gmac1_pclk clock gating control in sleep mode

Bit	Symbol	Access	Default	Description
[14]		RW	0x0	gmac0_pclk clock gating control in sleep mode
[13]		RW	0x1	emc_ram1_pclk clock gating control in sleep mode
[12]		RW	0x1	emc_ram0_pclk clock gating control in sleep mode
[11]		RW	0x1	dmac030_1_pclk clock gating control in sleep mode
[10]		RW	0x1	dmac030_0_pclk clock gating control in sleep mode
[9]		RW	0x1	ddr_pclk clock gating control in sleep mode
[8]		RW	0x0	pcie_i_apb_clk clock gating control in sleep mode (pcie cr interface control clock)
[7]		RW	0x0	pcie_ext_pclk clock gating control in sleep mode (pcie side band control clock)
[6]		RW	0x0	usb3_apb_clk clock gating control in sleep mode
[5]		RW	0x0	otg_1_apb_clk clock gating control in sleep mode
[4]		RW	0x1	otg_0_apb_clk clock gating control in sleep mode
[3]		RW	0x0	dac3_clk clock gating control in sleep mode
[2]		RW	0x0	dac2_clk clock gating control in sleep mode
[1]		RW	0x0	dac1_clk clock gating control in sleep mode
[0]		RW	0x0	dac0_clk clock gating control in sleep mode

Table 0-25. pclk clock gating control in sleep mode (0x06C)

Bit	Symbol	Access	Default	Description
[31:16]		RW	0xFFFF	-
[15]		RW	0x0	motor_xmc_ctl_pclk clock gating control in sleep mode 1'b1 : enable clock, 1'b0 : disable clock
[14]		RW	0x0	motor_exadc_pclk clock gating control in sleep mode
[13]		RW	0x0	motor_eqep_pclk clock gating control in sleep mode
[12]		RW	0x0	motor_epwm_pclk clock gating control in sleep mode
[11]		RW	0x0	adc_pclk clock gating control in sleep mode
[10]		RW	0x0	x2x_pclk clock gating control in sleep mode
[9]	slp_pclkg_n[63:32]	RW	0x1	x2p1_pclk clock gating control in sleep mode
[8]		RW	0x1	x2p0_pclk clock gating control in sleep mode
[7]		RW	0x0	wdt1_pclk clock gating control in sleep mode
[6]		RW	0x1	wdt0_pclk clock gating control in sleep mode
[5]		RW	0x0	usart1_pclk clock gating control in sleep mode
[4]		RW	0x0	usart0_pclk clock gating control in sleep mode
[3]		RW	0x0	uart3_pclk clock gating control in sleep mode
[2]		RW	0x0	uart2_pclk clock gating control in sleep mode
[1]		RW	0x1	uart1_pclk clock gating control in sleep mode
[0]		RW	0x1	uart0_pclk clock gating control in sleep mode

21.16 AXI-CLK clock gating control (0x080, 0x088)

Register 0x80 present the clock gating setting in aclk domain when system works in normal mode. Register 0x88 present the clock gating setting in aclk domain when system works in sleep mode.

Table 0-26. axi-clk clock gating control (0x080)

Bit	Symbol	Access	Default	Description
[31:22]	scu_aclkg_n[31:0]	RW	0x3FF	-
[21]		RW	0x0	x2x_aclks clock gating control 1'b1 : enable clock, 1'b0 : disable clock
[20]		RW	0x0	x2x_aclkm clock gating control
[19]		RW	0x1	x2p1_aclk clock gating control
[18]		RW	0x1	x2p0_aclk clock gating control
[17]		RW	0x1	spi_aclk clock gating control
[16]		RW	0x0	pcie_slv_aclk clock gating control
[15]		RW	0x0	pcie_mstr_aclk clock gating control
[14]		RW	0x0	pcie_dbi_aclk clock gating control
[13]		RW	0x1	nic_aclk clock gating control
[12]		RW	0x0	lvds_aclk clock gating control
[11]		RW	0x0	gmac1_aclk clock gating control
[10]		RW	0x0	gmac0_aclk clock gating control
[9]		RW	0x1	emc_ram1_aclk clock gating control

Bit	Symbol	Access	Default	Description
[8]		RW	0x1	emc_ram0_aclk clock gating control
[7]		RW	0x1	dmac030_1_aclk clock gating control
[6]		RW	0x1	dmac030_0_aclk clock gating control
[5]		RW	0x0	ddr_aclk3 clock gating control
[4]		RW	0x1	ddr_aclk1 clock gating control
[3]		RW	0x1	ddr_aclk0 clock gating control
[2]		RW	0x1	gic400_clk clock gating control
[1]		RW	0x0	otg_1_sclk clock gating control
[0]		RW	0x1	otg_0_sclk clock gating control

Table 0-27. axi-clk clock gating control in sleep mode (0x088)

Bit	Symbol	Access	Default	Description
[31:22]		RW	0x3FF	-
[21]		RW	0x0	x2x_aclks clock gating control in sleep mode 1'b1 : enable clock, 1'b0 : disable clock
[20]		RW	0x0	x2x_aclkm clock gating control in sleep mode
[19]	slp_aclkg_n[31:0]	RW	0x1	x2p1_aclk clock gating control in sleep mode
[18]		RW	0x1	x2p0_aclk clock gating control in sleep mode
[17]		RW	0x1	spi_aclk clock gating control in sleep mode

Bit	Symbol	Access	Default	Description
[16]		RW	0x0	pcie_slv_aclk clock gating control in sleep mode
[15]		RW	0x0	pcie_mstr_aclk clock gating control in sleep mode
[14]		RW	0x0	pcie_dbi_aclk clock gating control in sleep mode
[13]		RW	0x1	nic_aclk clock gating control in sleep mode
[12]		RW	0x0	lvds_aclk clock gating control in sleep mode
[11]		RW	0x0	gmac1_aclk clock gating control in sleep mode
[10]		RW	0x0	gmac0_aclk clock gating control in sleep mode
[9]		RW	0x1	emc_ram1_aclk clock gating control in sleep mode
[8]		RW	0x1	emc_ram0_aclk clock gating control in sleep mode
[7]		RW	0x1	dmac030_1_aclk clock gating control in sleep mode
[6]		RW	0x1	dmac030_0_aclk clock gating control in sleep mode
[5]		RW	0x0	ddr_aclk3 clock gating control in sleep mode
[4]		RW	0x1	ddr_aclk1 clock gating control in sleep mode
[3]		RW	0x1	ddr_aclk0 clock gating control in sleep mode
[2]		RW	0x1	gic400_clk clock gating control in sleep mode
[1]		RW	0x0	otg_1_sclk clock gating control in sleep mode
[0]		RW	0x1	otg_0_sclk clock gating control in sleep mode

21.17 Multi-CPU Control Register (0x0B0, 0x0B4)

The register of MPCORE_CTRL is used to directly set CPU clock ON/OFF without any power mode command. The field of CPU_DISABLE for the MPCORE_CTRL register is programmed to disable the cluster clock when finishing the CPU SPEED command or waking up from the Sleep mode. CPU0_MUX show the current Mux values for scn_fclk_mux.

Table 0-28. Multi-CPU Control Register (0x0B0)

Bit	Symbol	Access	Default	Description
[31:30]	-	R	0x0	-
[29:28]	CPU_TURNON	W	0x0	Turn ON CPU clock The bit is used to directly turn on the CPU clock and will not wait other CPU to enter the idle state. Write 1 to turn on CPU
[27:26]	-	R	0x0	-
[25:24]	CPU_TURNOFF	RW	0x0	For write: Write 1 to turn off the CPU clock The bit is used to directly turn off the CPU clock without waiting CPU to enter the idle state. For read to check the CPU clock gated status 1: CPU is ON. 0: CPU is OFF
[23:18]	-	R	0x0	-
[17:16]	CPU_DISABLE	RW	0x0	Disable CPU that enters the WFI, WFE, or idle state when the CPU Speed or Sleep command is finished [0] : CA53 , [1]: CA53 L2
[15:8]	-	R	0x0	-
[7:4]	CPU1_MUX	R	0x0	Not used in Unimap
[3:0]	CPU0_MUX	R	0x0	CPU0 current clock frequency setting value.

Bit	Symbol	Access	Default	Description
				Value of scu_fclk_mux.

Table 0-29. Multi-core WFI control register (0x0B4)

Bit	Symbol	Access	Default	Description
[31:9]	-	R	0x0	-
[8]	CPU1_WFI_MSK	RW	0x0	Specify checking by standby_wfi or standby_wfe when executing the SCU commands for each core of Cluster 1 CPU. [0]: CA53 L2 cache 1: Checked by standby_wfe 0: Checked by standby_wfi
[7:4]	-	R	0x0	-
[3:0]	CPU0_WFI_MSK	RW	0x0	Specify checking by standby_wfi or standby_wfe when executing the SCU commands for each core of Cluster 0 CPU. [0]: Core 0 of CA53, [1]: Core 1 of CA53, [2]: Core 2 of CA53, [3]: Core 2 of CA53, 1: Checked by standby_wfe. 0: Checked by standby_wfi

21.18 Wakeup Event Status and control Register (0x0C0, 0x0C4)

The register (0x0C4) is to control enable bit of the wakeup event when system is wakeup from sleep mode to normal state. The register (0x0C0) shows the status of the wakeup events.

Table 0-30. Wakeup Event Status Register (0x0C0)

Bit	Symbol	Access	Default	Description
[31:13]	-	R	0x0	-
[12]	SLP_WAKUP_ST12	RW1C	0x0	Wakeup from sleep by PCIE interrupt
[11]	SLP_WAKUP_ST11	RW1C	0x0	Wakeup from sleep by FTGMAC030_1 interrupt
[10]	SLP_WAKUP_ST10	RW1C	0x0	Wakeup from sleep by FTGMAC030 interrupt
[9]	SLP_WAKUP_ST9	RW1C	0x0	Wakeup from sleep by PCIe beacon (PCIE/WAKE#)
[8]	SLP_WAKUP_ST8	RW1C	0x0	Wakeup from sleep by PCIe/WAKE#
[7]	SLP_WAKUP_ST7	RW1C	0x0	Wakeup from sleep by PCIe/CLKREQ#
[6]	SLP_WAKUP_ST6	RW1C	0x0	Wakeup from sleep by GPIO Interrupt
[5]	SLP_WAKUP_ST5	RW1C	0x0	Wakeup from sleep by FOTG330 interrupt
[4]	SLP_WAKUP_ST4	RW1C	0x0	Wakeup from sleep by FOTG210_1 interrupt
[3]	SLP_WAKUP_ST3	RW1C	0x0	Wakeup from sleep by FOTG210 interrupt
[2]	SLP_WAKUP_ST2	RW1C	0x0	Wakeup from sleep by FTGMAC030_1/WOL
[1]	SLP_WAKUP_ST1	RW1C	0x0	Wakeup from sleep by FTGMAC030/WOL
[0]	SLP_WAKUP_ST0	RW1C	0x0	Wakeup from sleep by the boot-up events

Table 0-31. Wakeup Event Enable Register (0x0C4)

Bit	Symbol	Access	Default	Description
[31:13]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[12]	SLP_WAKUP_EN12	RW	0x1	Wakeup enable for PCIE interrupt
[11]	SLP_WAKUP_EN11	RW	0x1	Wakeup enable for FTGMAC030_1 interrupt
[10]	SLP_WAKUP_EN10	RW	0x1	Wakeup enable for FTGMAC030 interrupt
[9]	SLP_WAKUP_EN9	RW	0x1	Wakeup enable for PCIe beacon (PCIE/WAKE#)
[8]	SLP_WAKUP_EN8	RW	0x1	Wakeup enable for PCIe/WAKE#
[7]	SLP_WAKUP_EN7	RW	0x1	Wakeup enable for PCIe/CLKREQ#
[6]	SLP_WAKUP_EN6	RW	0x1	Wakeup enable for GPIO Interrupt
[5]	SLP_WAKUP_EN5	RW	0x1	Wakeup enable for FOTG330 interrupt
[4]	SLP_WAKUP_EN4	RW	0x1	Wakeup enable for FOTG210_1 interrupt
[3]	SLP_WAKUP_EN3	RW	0x1	Wakeup enable for FOTG210 interrupt
[2]	SLP_WAKUP_EN2	RW	0x1	Wakeup enable for FTGMAC030_1/WOL
[1]	SLP_WAKUP_EN1	RW	0x1	Wakeup enable for FTGMAC030/WOL
[0]	SLP_WAKUP_EN0	RW	0x0	Wakeup enable for the boot-up events

21.19 Extend Command Register (0x120 ~ 0x128)

Registers (0x120 ~ 0x128) show FTSCU100 extended register setting.

Table 0-32. Extend Command Register (0x120)

Bit	Symbol	Access	Default	Description
[31:5]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[4]	-	RW	0x0	Not used in Unimap
[3]	-	R	0x0	-
[2]	ddr_mode	RW	0x1	1: DDR self-refresh ack is a level signal. 0: DDR self-refresh ack is a pulse signal.
[1]	WDT1_REBOOT_EN	RW	0x0	1'b0 : WDT1 will reboot all IP and CPU include SCU 1'b1 : WDT1 will reboot all IP and CPU except SCU (not support in Tx5B)
[0]	WDT_REBOOT_EN	RW	0x0	1'b0 : WDT will reboot all IP and CPU include SCU 1'b1 : WDT will reboot all IP and CPU except SCU (not support in Tx5B)

Table 0-33. Extend Command Register (0x124)

Bit	Symbol	Access	Default	Description
[31:16]	-	R	0x0	-
[15]	-	RW1C	0x0	Not used in Unimap
[14:8]	-	R	0x0	-
[7]	REBOOT_STS	RW1C	0x0	1: system is reboot by reboot command.
[6:2]	-	R	0x0	-
[1]	WDT1_RST_STS	RW1C	0x0	1: SCU is reboot by WDT1 reset.

Bit	Symbol	Access	Default	Description
[0]	WDT0_RST_STS	RW1C	0x0	1: SCU is reboot by WDT reset.

Table 0-34. Extend Command Register (0x128)

Bit	Symbol	Access	Default	Description
[31:13]	-	R	0x0	-
[12]	tmr_dis	RW	0x1	cpuoff_tmr disable
[11:0]	tmr_init	RW	0xFFF	cpuoff_tmr counter

21.20 RTC Time Register (0x200, 0x204)

The current time can be read from two time registers defined as below. These time registers are updated every second.

Table 0-35. RTC Time1 Register (0x200)

Bit	Symbol	Access	Default	Description
[31:27]	-	R	0x0	-
[26:24]	RTC_WEEKDAY	RW	0x0	Current weekday. The valid value is ranging from 0 to 6. 0: Sunday 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday
[23:21]	-	R	0x0	-
[20:16]	RTC_HOUR	RW	0x0	Current hour. The valid value is ranging from 0 to 23.
[15:14]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[13:8]	RTC_MIN	RW	0x0	Current minute. The valid value is ranging from 0 to 59
[7:6]	-	R	0x0	-
[5:0]	RTC_SEC	RW	0x0	Current second. The valid value is ranging from 0 to 59

Table 0-36. RTC Time2 Register (0x204)

Bit	Symbol	Access	Default	Description
[31]	-	R	0x0	-
[30:24]	RTC_CENTURY	RW	0x14	Current century. The valid value is ranging from 0 to 99.
[23]	-	R	0x0	-
[22:16]	RTC_YEAR	RW	0xA	Current year. The valid value is ranging from 0 to 99.
[15:12]	-	R	0x0	-
[11:8]	RTC_MONTH	RW	0x1	Current month. The valid value is ranging from 1 to 12.
[7:5]	-	R	0x0	-
[4:0]	RTC_DATE	RW	0x1	Current day. The valid value is ranging from 1 to 31.

21.21 RTC Alarm Time Register (0x208, 0x20C)

Two alarm registers are used to set the time-of-day/date alarm time. The alarm can be programmed by the alarm enable (ALARM_EN) of the control register (Offset 0x210) to activate the wake event on an alarm match condition when the chip is in the soft-off mode. The alarm also can issue an interrupt when the system is at the working state.

Table 0-37. RTC Alarm Time1 Register (0x208)

Bit	Symbol	Access	Default	Description
[31:27]	-	R	0x0	-
[26:24]	ALM_WEEKDAY	RW	0x7	Alarm weekday. The valid value is ranging from 0 to 6. When all bits are set to '1', this field will be "don't care" in the alarm matching.
[23:21]	-	R	0x0	-
[20:16]	ALM_HOUR	RW	0x1F	Alarm hour. The valid value is ranging from 0 to 23. When all bits are set to '1', this field will be "don't care" in the alarm matching.
[15:14]	-	R	0x0	-
[13:8]	ALM_MIN	RW	0x3F	Alarm minute. The valid value is ranging from 0 to 59. When all bits are set to '1', this field will be "don't care" in the alarm matching
[7:6]	-	R	0x0	-
[5:0]	ALM_SEC	RW	0x3F	Alarm second. The valid value is ranging from 0 to 59. When all bits are set to '1', this field will be "don't care" in the alarm matching

Table 0-38. RTC Alarm Time2 Register (0x20C)

Bit	Symbol	Access	Default	Description
[31:12]	-	R	0x0	-
[11:8]	ALM_MONTH	RW	0x0	Alarm Month. The valid value is ranging from 1 to 12. When all bits are set to '0', the field is "don't care" in alarm matching.
[7:5]	-	R	0x0	-
[4:0]	ALM_DATE	RW	0x0	Alarm Date. The valid value is ranging from 1 to 31. When all bits are set to '0', the field is "don't care" in alarm matching.

21.22 RTC Control Register (0x210)

The RTC control register (0x210) provide the function of enabling RTC timer and RTC alarm interrupt. And it also provides some registers to monitor RTC status.

Table 0-39. RTC Control Register (0x210)

Bit	Symbol	Access	Default	Description
[31]	-	R	0x0	-
[30]	-	RW	0x0	Not used in Unimap
[29:16]	-	R	0x0	-
[15]	RTC_CLK_READY	R	0x0	RTC clock stable status
[14:12]	-	R	0x0	-
[11]	PWUTen_sts	R	0x0	Periodic wakeup timer enable status
[10]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[9]	RTC_ALMEN_STS	R	0x0	RTC alarm enable status. The bit will be set when RTC accepts the command and is cleared when RTC_ALARM_EN is disabled.
[8]	RTCEN_STS	R	0x0	RTC enable status. The bit will be set when RTC accepts the command and is cleared when RTC_EN is disabled
[7]	SECOUT_EN	RW	0x0	Enable an event in each second
[6:4]	PERINT_SEL	RW	0x0	Periodic interrupt output signal select 3'b111: Each second 3'b110: Each minute and triggered at 0 second of every minute 3'b101: Each hour and triggered at 0 second of every hour. 3'b100: Each day and triggered at 0 second of every day 3'b011: Each month and triggered at 0 second of every month Others: Disable
[3]	-	R	0x0	-
[2]	LOCK_EN	RW	0x0	Lock the Time2 register after reading the Time1 register, the lock will be released by the following conditions: 1. Time2 is read. 2. Time1 or Time2 is written, 3. RTC is disabled.

Bit	Symbol	Access	Default	Description
[1]	RTC_ALARM_EN	RW	0x0	RTC alarm enable
[0]	RTC_EN	RW	0x0	RTC enable. The RTC clock should be confirmed stable through monitoring bit 2 of the OSC control register (0x1C) before turning on RTC.

21.23 RTC Tick Trim Control Register (0x214)

RTC includes a tick generation block to provide precise tick for counting the time in each second. The tick can be trimmed by the RTC tick trim control register to adjust the counting value of the tick generation block and compensate the frequency drift caused by the crystal inaccuracies. Unimap-TX5 uses 32.768 KHz crystal oscillator, and the initial value of TRIM_P register is consistent with the frequency of OSC.

Table 0-40. RTC Tick Trim Control Register (0x214)

Bit	Symbol	Access	Default	Description
[31:16]	TRIM_P	R/PW	0x8000	RTC tick dividing factor
[15:0]	-	R	0x0	-

21.24 RTC Time BCD Register (0x220, 0x224)

RTC time control register (0x200, 0x204) and RTC time BCD register (0x220, 0x224) share the same entity register. The difference between those registers are the decoding and encoding circuits for BCD are included to set or read the RTC current time/date.

Table 0-41. RTC Time1 BCD Register (0x220)

Bit	Symbol	Access	Default	Description
[31:28]	-	R	0x0	-
[27:24]	BCD_RTC_WEEKDAY	RW	0x0	Current weekday. The valid value is ranging from 0 to 6. 0: Sunday 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday
[23:20]	BCD_RTC_HOUR_DEC	RW	0x0	Current hour in decimal and unit values. The decimal valid value is ranging from 0 to 2, and the unit valid value is ranging from 0 to 9. The legal programmed value is ranging from 0 to 0x23.
[19:16]	BCD_RTC_HOUR	RW	0x0	
[15:12]	BCD_RTC_MIN_DEC	RW	0x0	Current minute in decimal and unit values. The decimal valid value is ranging from 0 to 5, and the unit valid value is ranging from 0 to 9. The legal programmed value is ranging from 0 to 0x59.
[11:8]	BCD_RTC_MIN	RW	0x0	
[7:4]	BCD_RTC_SEC_DEC	RW	0x0	Current second in decimal and unit values. The decimal valid value is ranging from 0 to 5, and the unit valid value is ranging from 0 to 9. The legal programmed value is ranging from 0 to 0x59.
[3:0]	BCD_RTC_SEC	RW	0x0	

Table 0-42. RTC Time2 BCD Register (0x224)

Bit	Symbol	Access	Default	Description
[31:28]	BCD_RTC_CENTURY_DEC	RW	0x2	Current century in decimal and unit values.
[27:24]	BCD_RTC_CENTURY	RW	0x0	The decimal valid value is ranging from 0 to 9, and the unit valid value is ranging from 0 to 9.
[23:20]	BCD_RTC_YEAR_DEC	RW	0x1	Current year in decimal and unit values.
[19:16]	BCD_RTC_YEAR	RW	0x0	The decimal valid value is ranging from 0 to 9, and the unit valid value is ranging from 0 to 9.
[15:12]	BCD_RTC_MONTH_DEC	RW	0x0	Current month in decimal and unit values.
[11:8]	BCD_RTC_MONTH	RW	0x1	The decimal valid value is ranging from 0 to 1, and the unit valid value is ranging from 1 to 9.
[7:4]	BCD_RTC_DATE_DEC	RW	0x0	Current day in decimal and unit values.
[3:0]	BCD_RTC_DATE	RW	0x1	The decimal valid value is ranging from 0 to 3, and the unit valid value is ranging from 1 to 9.

21.25 RTC Alarm Time BCD Register (0x228, 0x22C)

RTC alarm time control register (0x208, 0x20C) and RTC alarm time BCD register (0x228, 0x22C) share the same entity register. RTC alarm time BCD registers are used to set the RTC alarm by BCD encoding for RTC alarm.

Table 0-43. RTC Alarm Time1 BCD Register (0x228)

Bit	Symbol	Access	Default	Description
[31:28]	-	R	0x0	-

Bit	Symbol	Access	Default	Description
[27:24]	BCD_ALM_WEEKDAY	RW	0x7	Alarm weekday setting by BCD. The valid value is ranging from 0 to 6, and set 7 to disable the weekday setting for the RTC alarm.
[23:20]	BCD_ALM_HOUR_DEC	RW	0x3	Alarm hour setting by BCD in decimal and unit values. The decimal valid value is ranging from 0 to 2, and the unit valid value is ranging from 0 to 9. The legal programmed value is up to 0x23, and set to 0x31 to disable the hour setting for the RTC alarm.
[19:16]	BCD_ALM_HOUR	RW	0x1	
[15:12]	BCD_ALM_MIN_DEC	RW	0x6	Alarm minute setting by BCD in decimal and unit values. The decimal valid value is ranging from 0 to 5, and the unit valid value is ranging from 0 to 9. The legal programmed value is up to 0x59, and set to 0x63 to disable the minute setting for the RTC alarm.
[11:8]	BCD_ALM_MIN	RW	0x3	
[7:4]	BCD_ALM_SEC_DEC	RW	0x6	Alarm second setting by BCD in decimal values. The decimal valid value is ranging from 0 to 5, and the unit valid value is ranging from 0 to 9. The legal programmed value is up to 0x59, and set to 0x63 to disable the second setting for the RTC alarm.
[3:0]	BCD_ALM_SEC	RW	0x3	

Table 0-44. RTC Alarm Time2 BCD Register (0x22C)

Bit	Symbol	Access	Default	Description
[31:16]	-	R	0x0	-
[15:12]	BCD_ALM_MONTH_DEC	RW	0x0	Alarm month setting by BCD in decimal and unit values. The decimal valid value is ranging from 0 to 1, and the valid value is ranging from 0 to 9. The legal programmed value is from 0x1 to 0x12, and set to 0x0 to disable the month setting for the RTC alarm.
[11:8]	BCD_ALM_MONTH	RW	0x0	
[7:4]	BCD_ALM_DATE_DEC	RW	0x0	Alarm date setting by BCD in decimal & unit values. The decimal valid value is ranging from 0 to 3, and the valid value is ranging from 0 to 9. The legal programmed value is from 0x1 to 0x31, and set to 0x0 to disable the date setting for the RTC alarm.
[3:0]	BCD_ALM_DATE	RW	0x0	

21.26 RTC Scratchpad Register (0x280 ~ 0x2BC)

The scratchpad registers are usually used to store the backup data of the system when entering the soft-off mode. Unimap-TX5 provides 16x32 bits register for scratchpad memory.

Table 0-45. RTC Scratchpad Register (0x280 ~ 2x2BC)

Bit	Symbol	Access	Default	Description
[31:0]	RTC_PSPR_0	RW	0x0	Scratchpad register. (0x280)
[31:0]	RTC_PSPR_1	RW	0x0	Scratchpad register. (0x284)
[31:0]	RTC_PSPR_2	RW	0x0	Scratchpad register. (0x288)
[31:0]	RTC_PSPR_3	RW	0x0	Scratchpad register. (0x28C)
[31:0]	RTC_PSPR_4	RW	0x0	Scratchpad register. (0x290)
[31:0]	RTC_PSPR_5	RW	0x0	Scratchpad register. (0x294)
[31:0]	RTC_PSPR_6	RW	0x0	Scratchpad register. (0x298)
[31:0]	RTC_PSPR_7	RW	0x0	Scratchpad register. (0x29C)
[31:0]	RTC_PSPR_8	RW	0x0	Scratchpad register. (0x2A0)
[31:0]	RTC_PSPR_9	RW	0x0	Scratchpad register. (0x2A4)
[31:0]	RTC_PSPR_10	RW	0x0	Scratchpad register. (0x2A8)
[31:0]	RTC_PSPR_11	RW	0x0	Scratchpad register. (0x2AC)
[31:0]	RTC_PSPR_12	RW	0x0	Scratchpad register. (0x2B0)
[31:0]	RTC_PSPR_13	RW	0x0	Scratchpad register. (0x2B4)
[31:0]	RTC_PSPR_14	RW	0x0	Scratchpad register. (0x2B8)

Bit	Symbol	Access	Default	Description
[31:0]	RTC_PSPR_15	RW	0x0	Scratchpad register. (0x2BC)

22. Boot-up Sequence

The POR analog IP is usually integrated to the same power domain with the Real Time Clock Unit. When the backup power (From a battery most of the time) is applied, the power-on reset will be asserted to initialize all functions of the Real Time Clock Unit and stabilize the OSCL clock from a 32.768 kHz crystal. Other power domains will stay at the power-off state before the asserted power enable event from the T_pwrbtn pins that being connected to a power button.

After pressing the power button, the system will start to execute the power enable sequence. It will enable the power control unit, and issue a scu_por_n to the reset related PCU logical circuit. A hardware reset input port is in the power control unit, which is connected to a RC circuit with a reset button in PCB to automatically trigger a reset event and then, release it after the interval of the RC charging time when power is attached. In the power enable sequence, PCU will enable the power of the clock control unit and start the boot-up sequence after inactivating the reset.

The clock control unit is implemented in the core power domain, and the most blocks and peripherals are integrated into this power domain. Consequently, it is possible to cause the in-rush current when the power is turned on. FTSCU100 provides four time-slot power start-up sequences in the power control unit. After finishing the power-up sequence, the core power will be checked stable from the ccu_vccok signal, and the isolation control signal, scu_isopwr_n, will be de-asserted to build up the communication between the backup and core power domains. To stabilize the internal clocks, PLL and DLL must be stable and are set to a specified frequency before the clock gating signals are de-asserted. When the boot-up sequence is complete, all clocks are unblocked and Soteria engine starts the secure boot flow. After finishing the secure boot flow, Soteria will release the reset control of CPU and CPU starts to fetch the program code.

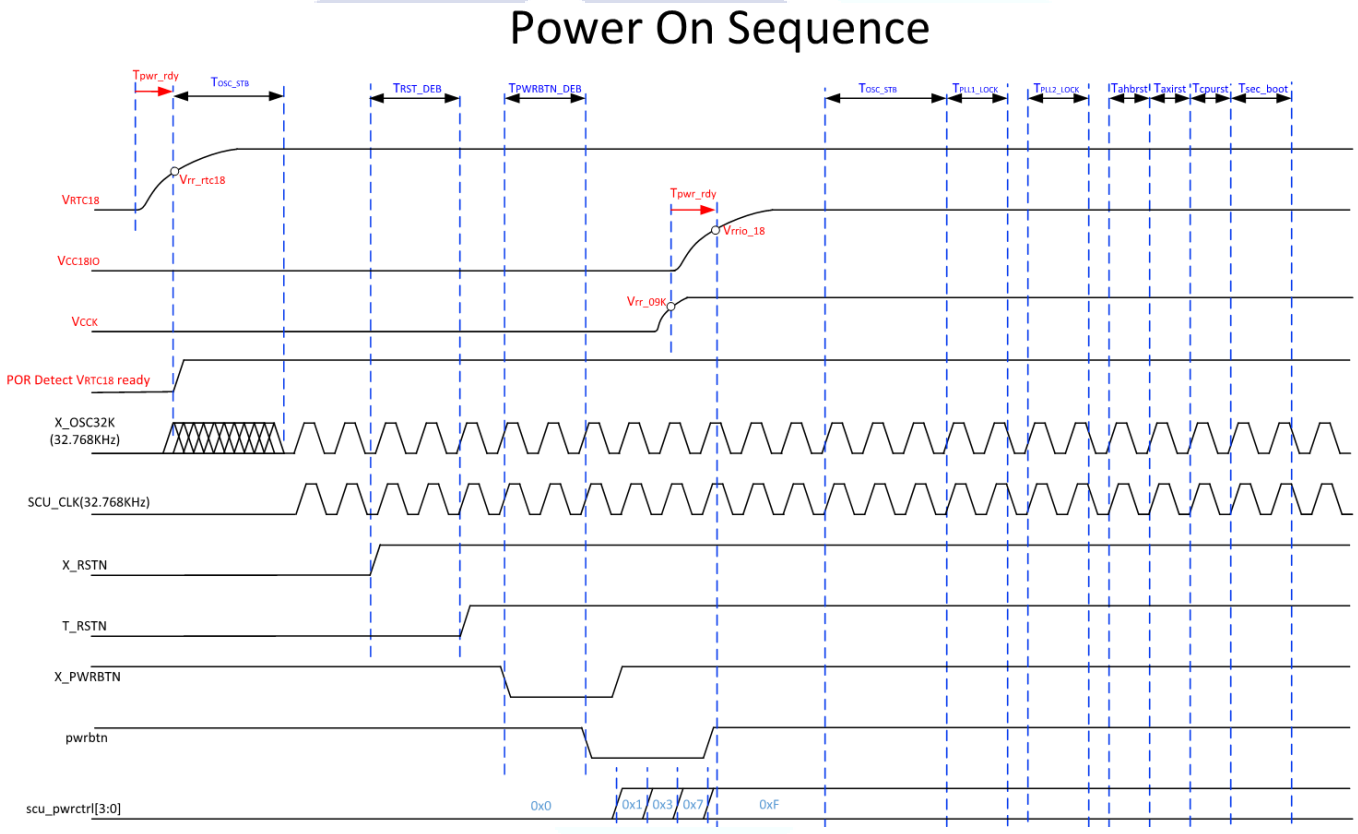
Figure 0-1 and *Figure 0-2* show the power-on parameter and diagram of Unimap-TX5.

1. VRTC18, VCC18IO and VCCK are from the system board
2. The VCC18IO and VCCK rise to about"60%*VCC"and POR reset is relea
and system start to boot up sequence.

Symbol	Description and Condition	Min.	Typ.	Max.	Unit
Vrr_rtc18	VCC18A rise relax voltage (POR low → high)	1	1.25	1.45	V
Vfr_rtc18	VCC18A fall release voltage (POR high → low)	0.7	1	1.25	V
Vrr_io18	VCC18A rise relax voltage (POR low → high)	1	1.24	1.45	V
Vfr_io18	VCC18A fall release voltage (POR high → low)	0.72	1	1.24	V
Vrr_09K	VCC09A rise relax voltage (POR Low → High)	0.53	0.61	0.73	V
Vfr_09K	VCC09A fall release voltage (POR High → Low)	0.46	0.53	0.66	V

Symbol	Description and Condition	Value	Unit	Notice
Tpwr_rdy	power rising to stable time	2 (max)	ms	Recommend that VCC3IO and VCCK rising time has to be less than 2ms
TOSC_STB	Crystal pad power up clock stable time	10 (max)	ms	Crystal oscillator component must be stable within 10ms after power up
TRST_DEB	Hardware reset signal debounce time	3 (min)	ms	Reset signal is active low and has to be longer than 3ms to trigger system hardware reset
TPWRBTN_DEB	power button signal debounce time	15 (min)	ms	power button is active low and has to be longer than 15ms to trigger system hardware reset
TPLL1_LOCK	PLL locking time	100 (max)	us	FXPLL530HJ0P stable time
TPLL2_LOCK	DDR PLL locking time	80 (max)	us	FXSSCG360HJ0P stable time
Tahbrst		4	pclk cycle	-
Taxirst		4	pclk cycle	-
Tcpurst		4	pclk cycle	-
Tsec_boot	Seteria security boot time	-	-	

Figure 0-1. Power-on Sequence timing parameter in Unimap-TX5



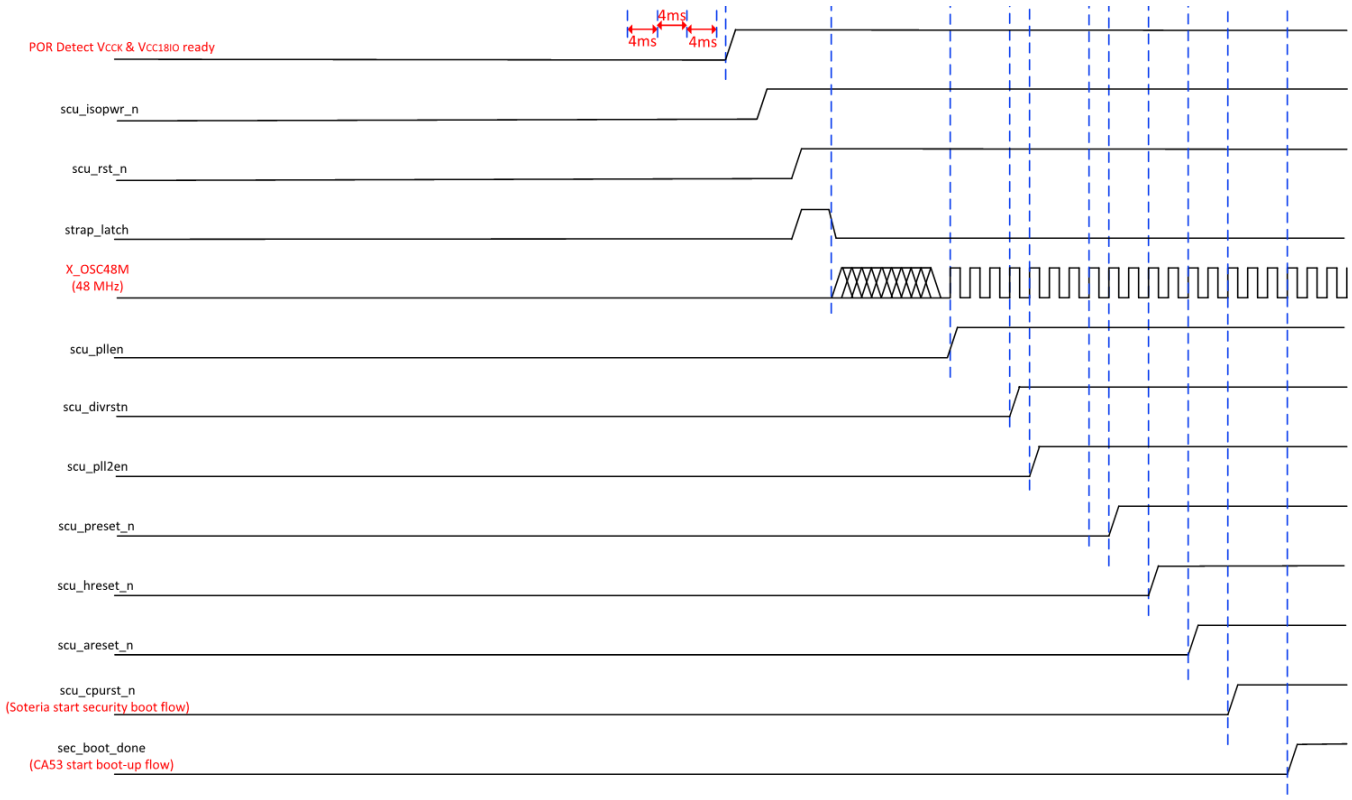


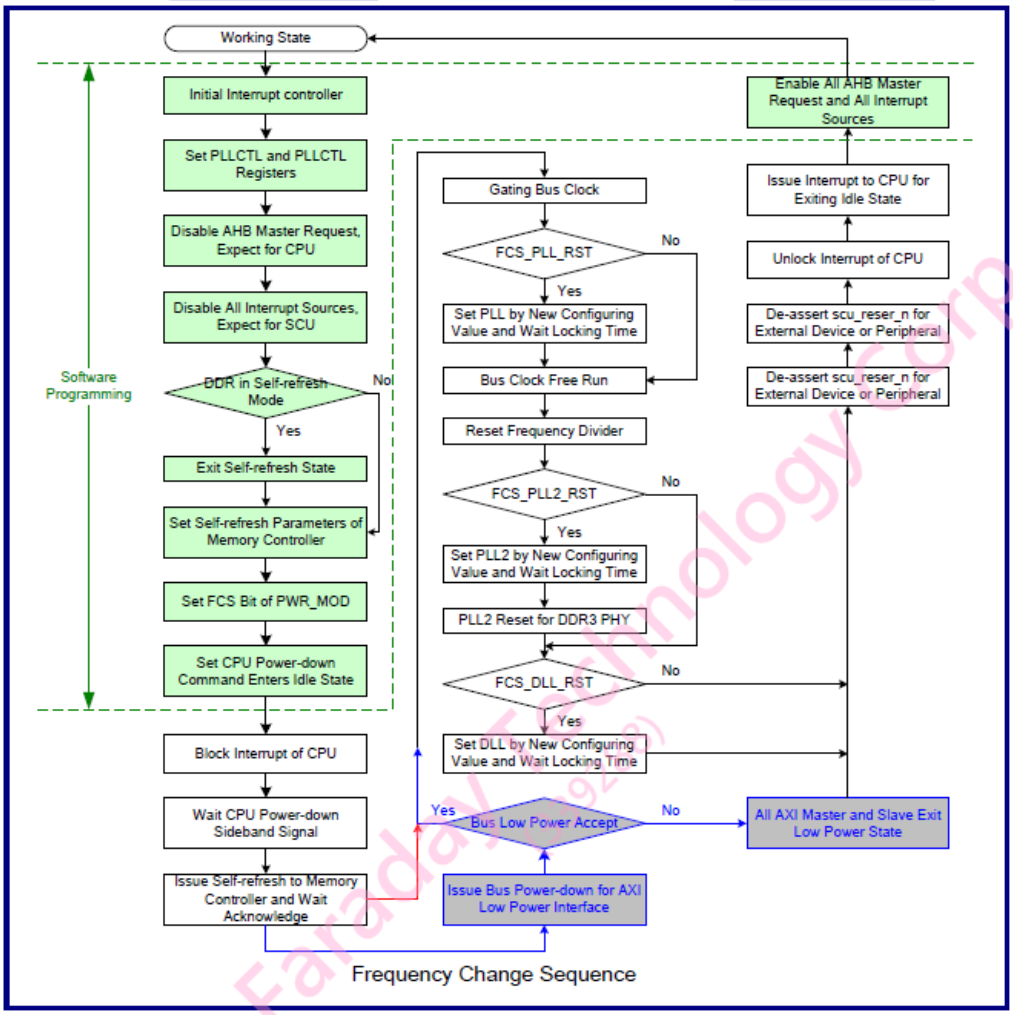
Figure 0-2. Power on Sequence Diagram in Unimap-TX5

23. System operation Sequence

At the working state, SCU provides several clock management functions to adjust the system operation speed and bus frequency according to the performance and the power requirement. These functions include the frequency change, and clock gating to provide a safe controlling procedure for the dynamic frequency change. For the power saving, the clocks of the unused blocks can be gated through the register programming at the working state. For further power saving requirement, the soft-off sequence could turn off all power except backup power.

23.1 Frequency Change Sequence (FCS)

The Frequency Change Sequence (FCS) is a way to change the operation clock without re-booting. It can dynamically increase the bus frequency that allows the system to operate at a higher frequency for a peak load. An apparatus is designed to adjust the bus and CPU operation clocks. In a SoC, the chip main clock is generated by PLL. However, if the configured values of PLL is changed, they need to reset and execute the re-lock procedure. The main clock will be stopped in the interval to regress the lock process for PLL. SCU provides the safe sequence to complete the PLL re-lock procedure, and the sequence is shown in *Figure 0-3*.



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Figure 0-3. FCS programming flow

23.2 Sleep Operation sequence

For saving the power consumption, users should stop CPU when not executing any task and users can expect to resume from a short latency without re-booting. The sleeping operation sequence is used to gate the CPU, and partial IP clocks and keep the system context in memory. The operation is shown in *Figure 0-4*. Before entering the sleeping state, the sleep clock control registers should be programmed to specify which IP is clock gated in the sleeping state. When the SLEEP bit is programmed and an idle instruction is issued, SCU will gate the CPU clocks until an interrupt event is received for waking up.

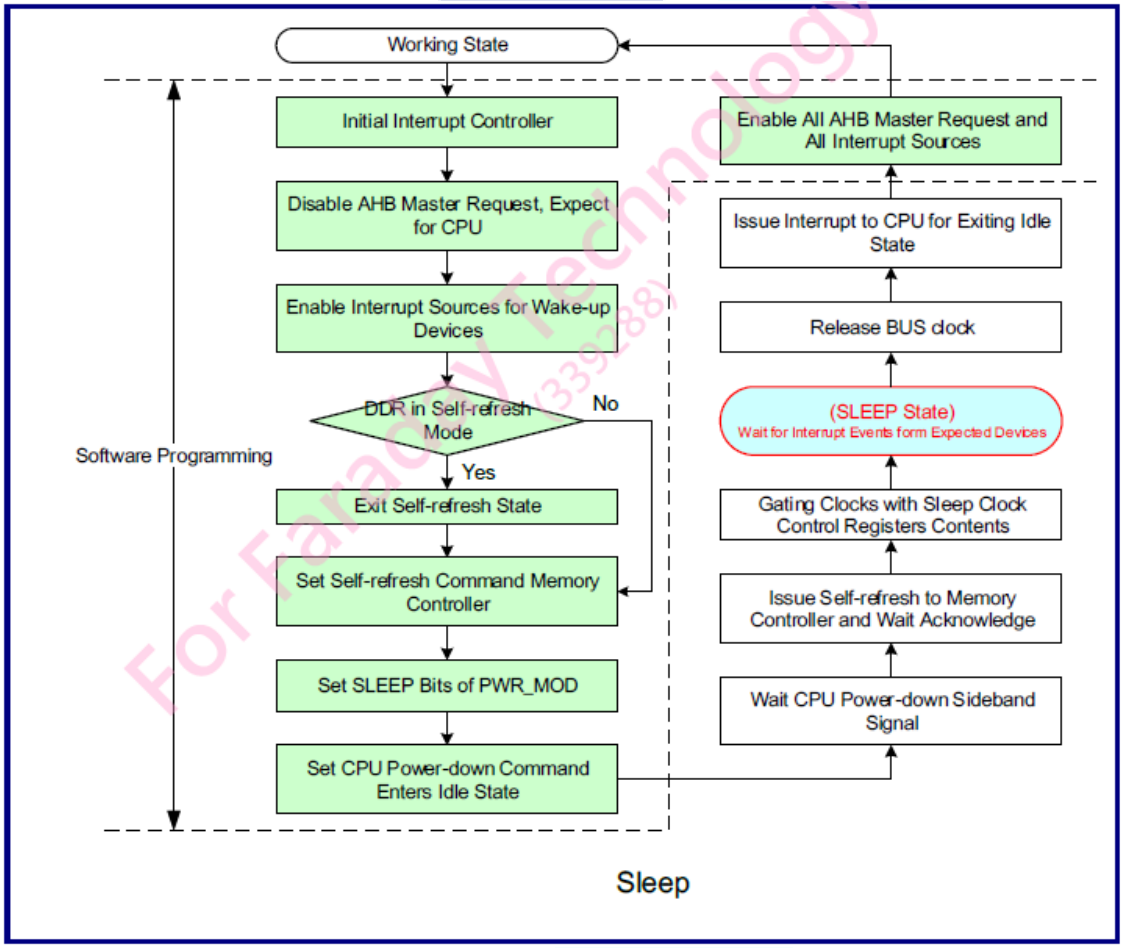


Figure 0-4. Sleep mode programming flow

23.3 Soft-off Operation Sequence

The soft-off mode is used to shut down the supply powers of core domain of Unimap-TX5, except the logic in the alive domain. The system power consumption is the lowest, but it needs longer recovery time for system re-boot. The system should follow an orderly sequence to enter soft-off state, and return to work state by boot-up event. *Figure 0-5* show the soft-off programming flow.

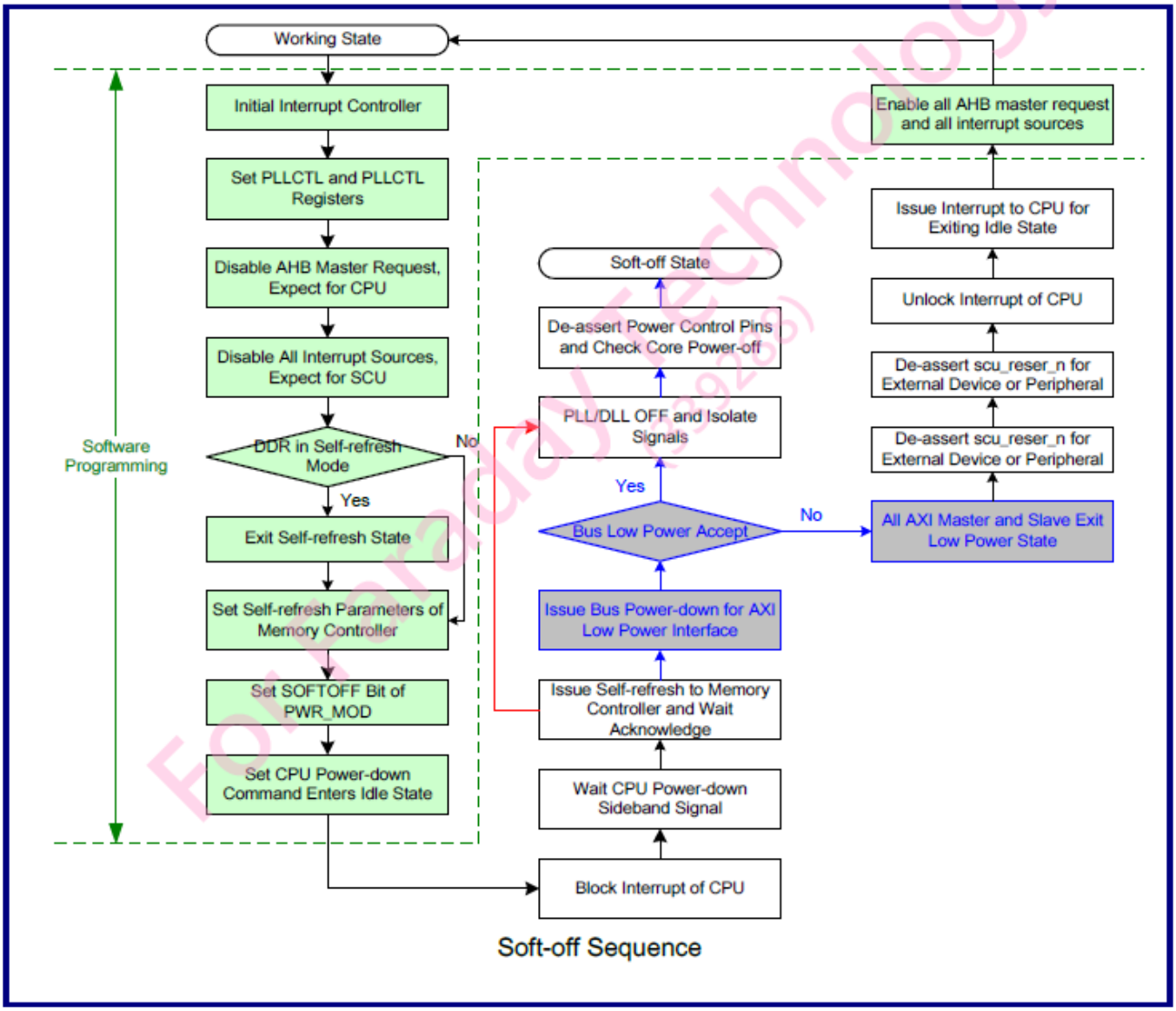


Figure 0-5. Soft-off programming flow

23.4 SCU Event summary

FTSCU100 of Unimap-TX5 provide six boot-up events to trigger alive domain to start power-up sequence of core domain, and return the system from soft-off mode to normal mode. Each event could be enable or disable individually by programming control register (0x0C100004[21:16]) in FTSCU100. *Table 0-46* shows all boot-up event list in Unimap-TX5.

Table 0-46. FTSCU100 Boot-up Event list

Boot-up Event	Bit offset	Description
X_PWRBTN	0	Enable control register : 0x0C100004[16] Status check register : 0x0C100000[16]
RTC_alarm	1	Enable control register : 0x0C100004[17] Status check register : 0x0C100000[17]
X_WAKE[0]	2	Enable control register : 0x0C100004[18] Status check register : 0x0C100000[18]
X_WAKE[1]	3	Enable control register : 0x0C100004[19] Status check register : 0x0C100000[19]
X_WAKE[2]	4	Enable control register : 0x0C100004[20] Status check register : 0x0C100000[20]
X_WAKE[3]	5	Enable control register : 0x0C100004[21] Status check register : 0x0C100000[21]

There are thirteen events used to wake up CPU from sleep mode back to normal mode. The 1st event is boot-up event, *Table 0-46* describes detail control information of boot-up events. *Table 0-47* shows all sleep mode wake-up event list of Unimap-TX5. Each event could be controlled by SCU register (0x0C1000C4[12:0]) individually, and it decides which wakeup event to use to return the system from sleep

mode to normal mode.

Table 0-47. FTSCU100 Sleep mode Wake-up Event list

Sleep mode Wake-up Event	Bit offset	Description
boot-up event	0	Enable control register : 0x0C1000C4[0] Status check register : 0x0C1000C0[0]
FTGMAC030/WOL	1	Enable control register : 0x0C1000C4[1] Status check register : 0x0C1000C0[1]
FTGMAC030_1/WOL	2	Enable control register : 0x0C1000C4[2] Status check register : 0x0C1000C0[2]
FOTG210 interrupt	3	Enable control register : 0x0C1000C4[3] Status check register : 0x0C1000C0[3]
FOTG210_1 interrupt	4	Enable control register : 0x0C1000C4[4] Status check register : 0x0C1000C0[4]
FOTG330 interrupt	5	Enable control register : 0x0C1000C4[5] Status check register : 0x0C1000C0[5]
GPIO Interrupt	6	Enable control register : 0x0C1000C4[6] Status check register : 0x0C1000C0[6]
PCIe/CLKREQ#	7	Enable control register : 0x0C1000C4[7] Status check register : 0x0C1000C0[7]
PCIe/WAKE#	8	Enable control register : 0x0C1000C4[8] Status check register : 0x0C1000C0[8]

Sleep mode Wake-up Event	Bit offset	Description
PCIe beacon (PCIE/WAKE#)	9	Enable control register : 0x0C1000C4[9] Status check register : 0x0C1000C0[9]
FTGMAC030 interrupt	10	Enable control register : 0x0C1000C4[10] Status check register : 0x0C1000C0[10]
FTGMAC030_1 interrupt	11	Enable control register : 0x0C1000C4[11] Status check register : 0x0C1000C0[11]
PCIE interrupt	12	Enable control register : 0x0C1000C4[12] Status check register : 0x0C1000C0[12]

24. System External Control register

Unimap-TX5 includes many different digital IPs and analog IPs, and each IP has its own sideband signal. All sidebands are collected and sorted into the external register table. *Table 0-48* shows the summary list of system external register.

Table 0-48. Summary of System External Register

Address	Description
0x0C10_0800	CAN_FD clock control
0x0C10_0804	CAN_FD_0 side-band control
0x0C10_0808	CAN_FD_0 side-band control
0x0C10_080C	CAN_FD_1 side-band control
0x0C10_0810	CAN_FD_1 side-band control
0x0C10_0814	CAN_FD_2 side-band control
0x0C10_0818	CAN_FD_2 side-band control
0x0C10_081C	USB2_0 side-band control
0x0C10_0820	USB2_1 side-band control
0x0C10_0824	USB3 side-band control
0x0C10_0828	DDR side-band control
0x0C10_082C	DDR/GPU/ADC/DAC/TDC/Soteria side-band control
0x0C10_0830	CA53 side-band control
0x0C10_0834	CA53 side-band control
0x0C10_0838	CA53 side-band control
0x0C10_083C	CA53 side-band control
0x0C10_0840	CA53 side-band control
0x0C10_0844	CA53 side-band control

Address	Description
0x0C10_0848	CA53 side-band control
0x0C10_084C	CA53 side-band control
0x0C10_0850	CA53 side-band control
0x0C10_0854	CA53 side-band control
0x0C10_0858	WDT/SDC/EMMC/DLL side-band control
0x0C10_085C	GMAC 0 / FXEDP412EWHJ0P_0 side-band control
0x0C10_0860	GMAC 0 / FXEDP412EWHJ0P_0 side-band control
0x0C10_0864	GMAC 0 / FXEDP412EWHJ0P_0 side-band control
0x0C10_0868	GMAC 1 / FXEDP412EWHJ0P_1 side-band control
0x0C10_086C	GMAC 1 / FXEDP412EWHJ0P_1 side-band control
0x0C10_0870	GMAC 1 / FXEDP412EWHJ0P_1 side-band control
0x0C10_0874	LVTX side-band control
0x0C10_0878	GIC400 / GLUE side-band control
0x0C10_087C	GLUE side-band control
0x0C10_0880	PCIE side-band control
0x0C10_0884	PLL_LVDS control
0x0C10_0888	PLL_GMAC control
0x0C10_088C	clkgen side-band control
0x0C10_0890	SRAM side-band control
0x0C10_0894	Reserved
0x0C10_0898	pad pin mux
0x0C10_089C	pad pin mux
0x0C10_08A0	pad pin mux
0x0C10_08A4	pad pin mux

Address	Description
0x0C10_08A8	pad pin mux
0x0C10_08AC	pad pin mux
0x0C10_08B0	pad pin mux
0x0C10_08B4	pad pin mux
0x0C10_08B8	pad PU control
0x0C10_08BC	pad PU control
0x0C10_08C0	pad PU control
0x0C10_08C4	pad PU control
0x0C10_08C8	pad PU control
0x0C10_08CC	pad PU control
0x0C10_08D0	pad PU control
0x0C10_08D4	pad PU control
0x0C10_08D8	pad PU control
0x0C10_08DC	pad PD control
0x0C10_08E0	pad PD control
0x0C10_08E4	pad PD control
0x0C10_08E8	pad PD control
0x0C10_08EC	pad PD control
0x0C10_08F0	pad PD control
0x0C10_08F4	pad PD control
0x0C10_08F8	pad PD control
0x0C10_08FC	pad PD control
0x0C10_0920	FTSCU100 Software reset mask control
0x0C10_0924	FTSCU100 Software reset mask control

Address	Description
0x0C10_0928	FTSCU100 Software reset mask control
0x0C10_092C	FTSCU100 Software reset mask control
0x0C10_0930	FTSCU100 reboot mask control
0x0C10_0934	FTSCU100 reboot mask control
0x0C10_0938	FTSCU100 reboot mask control
0x0C10_093C	FTSCU100 reboot mask control
0x0C10_0940	DDR reset control
0x0C10_0944	System WFI control
0x0C10_0948	IP MSB address extension
0x0C10_094C	pad dcsr control
0x0C10_0950	pad dcsr control
0x0C10_0954	pad dcsr control
0x0C10_0958	pad dcsr control
0x0C10_095C	DDR mode register read
0x0C10_0960	DDR mode register read
0x0C10_0964	DDR mode register read
0x0C10_0968	DDR mode register read
0x0C10_096C	DDR mode register read
0x0C10_0970	DDR mode register read
0x0C10_0974	DDR mode register read
0x0C10_0978	DDR mode register read

24.1 CAN_FD Sideband Control (0x0C10_0800 ~ 0x0C10_0818)

Table 0-49. CAN_FD clock control (0x0C10_0800)

Bit	Symbol	Access	Default	Description
[12:8]	reg_canfd_extclk_div	RW	0x0	CANFD external clock divider factor
[4:0]	reg_canfd_degli_div	RW	0x0	CANFD de-glitch clock divider factor

Table 0-50. CAN_FD_0 side-band control (0x0C10_0804)

Bit	Symbol	Access	Default	Description
[18]	reg_canfd0_ecc_uncorr_flag	R		flag asserted due to non-correctable error detected in memory read access by either the hot (CPU) or by the FlexCAN
[17]	reg_canfd0_ecc_corr_flag	R		flag asserted due to correctable error detected in memory read access.
[16]	reg_canfd0_stop_ack	R		stop mode acknowledge
[13:8]	reg_canfd0_degli_count	RW	0x2	deglitch time counter
[6]	reg_canfd0_test	RW	0x0	indicates the current access has test mode privileges
[5]	reg_canfd0_supervisor	RW	0x1	supervisor mode request
[4]	reg_canfd0_stop	RW	0x0	stop mode request
[3]	reg_canfd0_doze	RW	0x0	doze mode request
[2]	reg_canfd0_debug	RW	0x0	debug mode request
[1]	reg_canfd0_sw_rstn	RW	0x1	soft reset
[0]	reg_canfd0_ipt_se_async	RW	0x0	Scan enable for asynchronous logic

Table 0-51. CAN_FD_0 side-band control (0x0C10_0808)

Bit	Symbol	Access	Default	Description
[31:0]	reg_canfd0_ecc_err_addr	R		RAM address related to the detection of a correctable or non-correctable error in memory read access.

Table 0-52. CAN_FD_1 side-band control (0x0C10_080C)

Bit	Symbol	Access	Default	Description
[18]	reg_canfd1_ecc_uncorr_flag	R		flag asserted due to non-correctable error detected in memory read access by either the hot (CPU) or by the FlexCAN
[17]	reg_canfd1_ecc_corr_flag	R		flag asserted due to correctable error detected in memory read access.
[16]	reg_canfd1_stop_ack	R		stop mode acknowledge
[13:8]	reg_canfd1_degli_count	RW	0x2	deglitch time counter
[6]	reg_canfd1_test	RW	0x0	indicates the current access has test mode privileges
[5]	reg_canfd1_supervisor	RW	0x1	supervisor mode request
[4]	reg_canfd1_stop	RW	0x0	stop mode request
[3]	reg_canfd1_doze	RW	0x0	doze mode request
[2]	reg_canfd1_debug	RW	0x0	debug mode request
[1]	reg_canfd1_sw_rstn	RW	0x1	soft reset
[0]	reg_canfd1_ipt_se_async	RW	0x0	Scan enable for asynchronous logic

Table 0-53. CAN_FD_1 side-band control (0x0C10_0810)

Bit	Symbol	Access	Default	Description
[31:0]	reg_canfd1_ecc_err_addr	R		RAM address related to the detection of a correctable or non-correctable error in memory read access.

Table 0-54. CAN_FD_2 side-band control (0x0C10_0814)

Bit	Symbol	Access	Default	Description
[18]	reg_canfd2_ecc_uncorr_flag	R		flag asserted due to non-correctable error detected in memory read access by either the hot (CPU) or by the FlexCAN
[17]	reg_canfd2_ecc_corr_flag	R		flag asserted due to correctable error detected in memory read access.
[16]	reg_canfd2_stop_ack	R		stop mode acknowledge
[13:8]	reg_canfd2_degli_count	RW	0x2	deglitch time counter
[6]	reg_canfd2_test	RW	0x0	indicates the current access has test mode privileges
[5]	reg_canfd2_supervisor	RW	0x1	supervisor mode request
[4]	reg_canfd2_stop	RW	0x0	stop mode request
[3]	reg_canfd2_doze	RW	0x0	doze mode request
[2]	reg_canfd2_debug	RW	0x0	debug mode request
[1]	reg_canfd2_sw_rstn	RW	0x1	soft reset
[0]	reg_canfd2_ipt_se_async	RW	0x0	Scan enable for asynchronous logic

Table 0-55. CAN_FD_2 side-band control (0x0C10_0818)

Bit	Symbol	Access	Default	Description
[31:0]	reg_canfd2_ecc_err_addr	R		RAM address related to the detection of a correctable or non-correctable error in memory read access.

24.2 USB 2 / USB 3 sideband Control (0x0C10_081C ~ 0x0C10_0824)

Table 0-56. USB2_0 side-band control (0x0C10_081C)

Bit	Symbol	Access	Default	Description
[25]	usbh_conn_det_n	R	0x1	Combiantional Flag. HC Attach indication with low active.
[24]	usbh_wkp_det_n	R	0x1	Combiantional Flag. HC RESUME indication with low active.
[18]	usb20_rstn_ctrl[2]	RW	0x1	1'b0 : PONRST = 1'b0 1'b1 : PONRST = system reset control
[17]	usb20_rstn_ctrl[1]	RW	0x1	1'b0 : presetn = 1'b0 1'b1 : presetn = system reset control
[16]	usb20_rstn_ctrl[0]	RW	0x1	1'b0 : rst_n = 1'b0 1'b1 : rst_n = system reset control
[14]	ISOEN	RW	0x0	Power isolation enable 0: Disable power isolation mode 1: Enable power isolation mode
[10]	HSTX_SW_FOC_EN	RW	0x0	enable high-speed TX driver output swing level adjustment
[6]	u_iddig	RW	0x1	ID Dig
[5]	PLLALIV	RW	0x0	FZOTG268NSHJ0P PLLALIV control

Bit	Symbol	Access	Default	Description
[4]	OSCOUTEN	RW	0x0	CLKOSCO clock output selection 0: CLKOSCO is disabled. 1: CLKOSCO is enabled.
[3]	XTLSEL	RW	0x0	Reference clock frequency selection 0: Select the 12 MHz input reference clock 1: Select the 30 MHz input reference clock
[2]	EXTCTRL_SUSPENDM	RW	0x1	FZOTG268NSHJ0P external SUSPENDM control
[1]	l1_wakeup	RW	0x0	L1 wakeup sideband signal
[0]	wakeup	RW	0x0	Wakeup sideband signal

Table 0-57. USB2_1 side-band control (0x0C10_0820)

Bit	Symbol	Access	Default	Description
[25]	usbh_conn_det_n	R	0x1	Combiantional Flag. HC Attach indication with low active.
[24]	usbh_wkp_det_n	R	0x1	Combiantional Flag. HC RESUME indication with low active.
[18]	usb21_rstn_ctrl[2]	RW	0x1	1'b0 : PONRST = 1'b0 1'b1 : PONRST = system reset control
[17]	usb21_rstn_ctrl[1]	RW	0x1	1'b0 : presetn = 1'b0 1'b1 : presetn = system reset control
[16]	usb21_rstn_ctrl[0]	RW	0x1	1'b0 : rst_n = 1'b0

Bit	Symbol	Access	Default	Description
				1'b1 : rst_n = system reset control
[14]	ISOEN	RW	0x0	Power isolation enable 0: Disable power isolation mode 1: Enable power isolation mode
[10]	HSTX_SW_FOC_EN	RW	0x0	enable high-speed TX driver output swing level adjustment
[6]	u_iddig	RW	0x1	ID Dig
[5]	PLLALIV	RW	0x0	FZOTG268NSHJ0P PLLALIV control
[4]	OSCOUTEN	RW	0x0	CLKOSCO clock output selection 0: CLKOSCO is disabled. 1: CLKOSCO is enabled.
[3]	XTLSEL	RW	0x0	Reference clock frequency selection 0: Select the 12 MHz input reference clock 1: Select the 30 MHz input reference clock
[2]	EXTCTRL_SUSPENDM	RW	0x1	FZOTG268NSHJ0P external SUSPENDM control
[1]	l1_wakeup	RW	0x0	L1 wakeup sideband signal
[0]	wakeup	RW	0x0	Wakeup sideband signal

Table 0-58. USB3 side-band control (0x0C10_0824)

Bit	Symbol	Access	Default	Description
[28:24]	DebugSel	RW	0x0	FZOTG306 debug status output selection

Bit	Symbol	Access	Default	Description
				this signal is only used for debugging and should be set to 5'b0 for normal operation
[16]	LPBK_TEST_PASS	R		Loopback test pass, Pin2Pin loopback test pass
[14]	usb3_rstn_ctrl[2]	RW	0x1	1'b0 : PONRST = 1'b0 1'b1 : PONRST = system reset control
[13]	usb3_rstn_ctrl[1]	RW	0x1	1'b0 : APB_RST_N = 1'b0 1'b1 : APB_RST_N = system reset control
[12]	usb3_rstn_ctrl[0]	RW	0x1	1'b0 : hwrst_n = 1'b0 1'b1 : hwrst_n = system reset control
[6]	LPBK_TEST_EN	RW	0x0	Enable the Pin2Pin loopback test
[5]	FORCE_SUSPENDM	RW	0x1	Reserve system register control for debug purpose. Suspend control: 1'b0: Draw the suspend current 1'b1: Draw the normal current
[4]	FORCE_LPM_EN	RW	0x0	Reserve system register control for debug purpose. CLKLPM control: 1'b0: Disable CLKLPM in the L1 mode 1'b1: Enable CLKLPM in the L1 mode
[3]	ISO_EN	RW	0x0	Power isolation 1'b0: Normal operation

Bit	Symbol	Access	Default	Description
				1'b1: When analog powers are floating (off), VCC09A/VCC09D leaks the least current.
[2]	SSCG_Bypass	RW	0x0	Bypass SSCG. If the control signals are set by using the I2C registers inside PHY, SSCG will be directly disabled or enabled by using this pin. 1'b0: Enable SSCG in the TX output 1'b1: Disable SSCG in the TX output
[1]	DEVICE_PHY	RW	0x1	USB 3.2 Gen1 differential pin selection If DEVICE_PHY is set to 1'b0 (Host), SSTXA will be the positive pin (SSTXP), SSTXB will be the negative pin (SSTXM), SSRXA will be the positive pin (SSRXP), and SSRXB will be the negative pin (SSRXM). If DEVICE_PHY is set to 1'b1 (Device), SSTXA will be the negative pin (SSTXM), SSTXB will be the positive pin (SSTXP), SSRXA will be the negative pin (SSRXM), and SSRXB will be the positive pin (SSRXP).
[0]	REF_FREQSEL	RW	0x0	Reference clock frequency selection 1'b0: Select the 12-MHz input reference clock 1'b1: Select the 30-MHz input reference clock

24.3 DDR/GPU/ADC/DAC/TDC/Soteria side-band control (0x0C10_0828 ~ 0x0C10_082C)

Table 0-59. DDR side-band control (0x0C10_0828)

Bit	Symbol	Access	Default	Description
[31:28]	arqos_3	RW	0x0	DDR channel 3 read QoS control
[27:24]	awqos_3	RW	0x0	DDR channel 3 write QoS control
[23:20]	arqos_2	RW	0x0	DDR channel 2 read QoS control
[19:16]	awqos_2	RW	0x0	DDR channel 2 write QoS control
[15:12]	arqos_1	RW	0x0	DDR channel 1 read QoS control
[11:8]	awqos_1	RW	0x0	DDR channel 1 write QoS control
[7:4]	arqos_0	RW	0x0	DDR channel 0 read QoS control
[3:0]	awqos_0	RW	0x1	DDR channel 0 write QoS control

Table 0-60. DDR/GPU/ADC/DAC/TDC/Soteria side-band control (0x0C10_082C)

Bit	Symbol	Access	Default	Description
[31]	adc_xtp1	RW	0x0	FXADC112B001M20EWHJ0P XTP[1] control
[30]	adc_xtp0	RW	0x0	FXADC112B001M20EWHJ0P XTP[0] control
[29]	adc_xsampsel	RW	0x1	FXADC112B001M20EWHJ0P XSAMPSEL control
[28]	adc_xmod	RW	0x0	FXADC112B001M20EWHJ0P XMOD control
[27]	dac_done_3	R		DAC 3 done
[26]	dac_done_2	R		DAC 2 done
[25]	dac_done_1	R		DAC 1 done
[24]	dac_done_0	R		DAC 0 done

Bit	Symbol	Access	Default	Description
[23]	tdc_done	R		TDC done
[22]	adc_done	R		ADC done
[20]	unlock_dbg	R		Soteria unlock_dbg
[17]	disableRamClockGating	RW	0x0	VIVANTE_GPU clock gating control
[16]	scanMode	RW	0x0	VIVANTE_GPU Scan mode
[15:8]	DEBUG_OUT	R		VIVANTE_GPU debug out
[5:4]	stat_ddrc_reg_selfref_type	R		DDR stat_ddrc_reg_selfref_type
[2]	rd_mrr_data_valid	RW1C		DDR mode register read data valid
[0]	dfi_ctrlupd_ack2_ch0	RW	0x0	DDR dfi_ctrlupd_ack2_ch0 control

24.4 CA53 side-band control (0x0C10_0830 ~ 0x0C10_0854)

Table 0-61. CA53 side-band control (0x0C10_0830)

Bit	Symbol	Access	Default	Description
[31]	CA53_tpctl	RW	0x1	
[30]	CA53_L2FLUSHDONE	R		
[29]	CA53_L2FLUSHREQ	RW	0x0	
[28]	CA53_DBGL1RSTDISABLE	RW	0x0	
[27]	CA53_BROADCASTCACHEMAINT	RW	0x0	
[26]	CA53_BROADCASTINNER	RW	0x0	

Bit	Symbol	Access	Default	Description
[25]	CA53_BROADCASTOUTER	RW	0x0	
[24]	CA53_DBGROMADDRV	RW	0x1	
[23]	CA53_EVENTI	RW	0x0	
[22]	CA53_GICCDISABLE	RW	0x1	
[21]	CA53_L2RSTDISABLE	RW	0x0	
[20]	CA53_SYSBARDISABLE	RW	0x1	
[19]	CA53_dbgens	RW	0x1	
[18]	CA53_nidens	RW	0x1	
[17]	CA53_spidens	RW	0x1	
[16]	CA53_spnidens	RW	0x1	
[15:12]	CA53_DBGPWRDUP	RW	0xF	
[11:8]	CA53_EDBGRQ	RW	0x0	
[7:4]	CA53_VINITHI	RW	0x0	
[3:0]	CA53_cryptodisable	RW	0x0	

Table 0-62. CA53 side-band control (0x0C10_0834)

Bit	Symbol	Access	Default	Description
[31:28]	CA53_AA64nAA32	RW	0xF	
[27:24]	CA53_CFGEND	RW	0x0	

Bit	Symbol	Access	Default	Description
[23:20]	CA53_CFGTE	RW	0x0	
[19:16]	CA53_CP15SDISABLE	RW	0x0	
[15:8]	CA53_CLUSTERIDAFF1	RW	0x0	
[7:0]	CA53_CLUSTERIDAFF2	RW	0x0	

Table 0-63. CA53 side-band control (0x0C10_0838)

Bit	Symbol	Access	Default	Description
[31:24]	CA53_RVBARADDR3[39:32]	RW	0x0	
[23:16]	CA53_RVBARADDR2[39:32]	RW	0x0	
[15:8]	CA53_RVBARADDR1[39:32]	RW	0x0	
[7:0]	CA53_RVBARADDR0[39:32]	RW	0x0	

Table 0-64. CA53 side-band control (0x0C10_083C)

Bit	Symbol	Access	Default	Description
[31:2]	CA53_RVBARADDR0[31: 2]	RW	0x0	

Table 0-65. CA53 side-band control (0x0C10_0840)

Bit	Symbol	Access	Default	Description
[31:2]	CA53_RVBARADDR1[31: 2]	RW	0x0	

Table 0-66. CA53 side-band control (0x0C10_0844)

Bit	Symbol	Access	Default	Description
[31:2]	CA53_RVBARADDR2[31: 2]	RW	0x0	

Table 0-67. CA53 side-band control (0x0C10_0848)

Bit	Symbol	Access	Default	Description
[31:2]	CA53_RVBARADDR3[31: 2]	RW	0x0	

Table 0-68. CA53 side-band control (0x0C10_084C)

Bit	Symbol	Access	Default	Description
[23:16]	CA53_DBGROMADDR[39:32]	RW	0x0	
[7:0]	CA53_PERIPHBASE[39:32]	RW	0x0	

Table 0-69. CA53 side-band control (0x0C10_0850)

Bit	Symbol	Access	Default	Description
[31:18]	CA53_PERIPHBASE[31:18]	RW	0xC0	

Table 0-70. CA53 side-band control (0x0C10_0854)

Bit	Symbol	Access	Default	Description
[31:12]	CA53_DBGROMADDR[31:12]	RW	0x8000	

24.5 WDT/SDC/EMMC/DLL side-band control (0x0C10_0858)

Table 0-71. WDT/SDC/EMMC/DLL side-band control (0x0C10_0858)

Bit	Symbol	Access	Default	Description
[30]	DLL_FLDO	R		EMMC DLL Lock detector flag High: Ready after the power-up phase[3] Low: Not ready after the power-up phase
[29]	DLL_PDN	RW	0x0	EMMC DLL Power-down mode setting Active low
[28]	DLL_BYPASS	RW	0x0	EMMC DLL Bypass mode setting Low: Normal mode High: Bypass mode
[27]	sd1_curlimit	RW	0x0	EMMC current limit indication
[26]	sd1_sd_drivest_sel_2	RW	0x0	EMMC pad driving capability (3.3V/1.8V) 3'b000 : 2.2 mA / 3.0 mA 3'b001 : 3.5 mA / 4.5 mA 3'b010 : 4.5 mA / 6.0 mA 3'b011 : 7.0 mA / 9.0 mA 3'b100 : 7.0 mA / 9.0 mA 3'b101 : 8.2 mA / 10 mA 3'b110 : 9.5 mA / 12 mA
[25]	sd1_sd_drivest_sel_1	RW	0x0	
[24]	sd1_sd_drivest_sel_0	RW	0x0	

Bit	Symbol	Access	Default	Description
				3'b111 : 11.5 mA / 15 mA
[23]	sd0_curlimit	RW	0x0	SDC current limit indication
[21]	reg_sd1_cd	RW	0x0	SD1 CD input when pmur_mfs2[13:12]==2'b01
[20]	reg_sd1_wp	RW	0x0	SD1 WP input when pmur_mfs2[11:10]==2'b01
[17]	wdt1_wdtoff_dis	RW	0x0	FTWDT011 1 WDTOFF disable signal 0: Normal function 1: WDT cannot be disabled.
[16]	wdt0_wdtoff_dis	RW	0x0	FTWDT011 0 WDTOFF disable signal 0: Normal function 1: WDT cannot be disabled.
[15:8]	wdt1_tickle	RW	0x0	FTWDT011 1 external kick dog signal
[7:0]	wdt0_tickle	RW	0x0	FTWDT011 0 external kick dog signal

24.6 GMAC / FXEDP412EWHJ0P side-band control (0x0C10_085C ~ 0x0C10_0870)

Table 0-72. GMAC 0 / FXEDP412EWHJ0P_0 side-band control (0x0C10_085C)

Bit	Symbol	Access	Default	Description
[31]	gmac0_pi_suppress_preamble	RW	0x1	Management frame with preamble suppressed 0: PHY will not accept the management frames with suppressed preamble. 1: PHY will accept the management frames with suppressed

Bit	Symbol	Access	Default	Description
				preamble.
[30]	gmac0_pi_turbo_sim	RW	0x0	For simulation, the speed-up mode works to reduce the simulation time. FTC turbo mode only works in two FTC IP simulation environments. 1: Speed up the timer 0: Normal operation
[29]	gmac0_pi_force_tx_lpi	RW	0x0	Only for the internal test mode. Note: Please use the default value '0'.
[28]	gmac0_pi_force_mdix_mode	RW	0x0	For the internal test data input, Note: Please use the default value '0'. 1: mdix mode 0: mdi mode
[27]	gmac0_pi_force_mdix_en	RW	0x0	For the internal test data input, Note: Please use the default value '0'. 1: Decide the mdi/mdix mode with signal "pi_force_mdix_mode" 0: IP auto mdix function, disable "pi_force_mdix_mode" signal
[26]	gmac0_pi_rgmii_en	RW	0x0	0: GMII/MII is active. 1: RGMII is active.
[25]	gmac0_pi_testmd	RW	0x0	Test mode selection bits

Bit	Symbol	Access	Default	Description
				0: Normal mode 1: BIST mode for the loopback self-test of the chip
[24]	gmac0_pi_fiber_en	RW	0x0	This signal forces IP to enter the fiber mode (100BASE-FX). 1: Force fiber mode (100BASE-FX) operation 0: Normal operation
[23]	gmac0_pi_apb_mdi_sel	RW	0x1	APB/MDIO select signal, 0: MDIO 1: APB
[22:20]	gmac0_pi_opmode	RW	0x0	Operation control mode, and only for internal use 3'd0: Devices link through auto-negotiation with their maximum abilities
[19]	gmac0_pi_pwrgate_en	RW	0x0	Only for internal use. Note: Please tie to 0.
[18]	gmac0_pi_pwrdn_pll_alive	RW	0x1	Function is as pi_pwrdn, but the po_pll500 clock will keep alive.
[17]	gmac0_pi_pwrdn_osc_alive	RW	0x1	Function is as pi_pwrdn, but the oscillator clock will keep alive
[16]	gmac0_pi_pwrdn	RW	0x1	Power down signal, active high 1: Force PHY to enter the power-down state, MII register setting will be default setting. 0: Normal operation

Bit	Symbol	Access	Default	Description
[15]	gmac0_pi_pwr_resetn	RW	0x0	Power-on reset for the IP, active low 1: Normal operation 0: Power reset
[14]	gmac0_pi_sys_resetn	RW	0x0	This signal resets PHY to initial mode excluding the register setting, active low 1: PHY system works normally 0: PHY system reset
[9:8]	gmac0_I_PLL_OBD_REFS	RW	0x0	Select the input clock frequency of I_ASICIN 2'b00 : 25 MHz 2'b01 : 50 MHz 2'b10 : 75 MHz 2'b11 : 100 MHz
[7:4]	gmac0_reg_led1_ctrl	RW	0x0	gmac 0 led 1 stretch period [1:0] = 2'b00 : 100 ms [1:0] = 2'b01 : 50 ms [1:0] = 2'b10 : 100 ms [1:0] = 2'b11 : 200 ms
[3:0]	gmac0_reg_led0_ctrl	RW	0x0	gmac 0 led 0 stretch period [1:0] = 2'b00 : 100 ms [1:0] = 2'b01 : 50 ms

Bit	Symbol	Access	Default	Description
				[1:0] = 2'b10 : 100 ms [1:0] = 2'b11 : 200 ms

Table 0-73. GMAC 0 / FXEDP412EWHJ0P_0 side-band control (0x0C10_0860)

Bit	Symbol	Access	Default	Description
[31:16]	gmac0_pi_PHYIDRG3	RW	0x2	For setting the PHY Identifier Register 2 Bits[24:19] of the OUI are mapped to bits[15:10] of PHY Identifier Register 2. Six bits of the model number of Unimap are mapped to bits[9:4]. Four bits of the revision number of Unimap are mapped to bits[3:0].
[15:0]	gmac0_pi_PHYIDRG2	RW	0x1	For setting the PHY Identifier Register 1 PHY Identifier Register 1 stores bits[3:18] of the Organizationally Unique Identifier (OUI).

Table 0-74. GMAC 0 / FXEDP412EWHJ0P_0 side-band control (0x0C10_0864)

Bit	Symbol	Access	Default	Description
[25]	gmac0_po_wol_detect	R		Internal use
[24]	gmac0_po_rx_sop	R		Internal use
[23]	gmac0_po_tx_sop	R		Internal use

Bit	Symbol	Access	Default	Description
[22]	gmac0_po_int_n	R	0x1	The interrupt pins are low-active signals
[21]	gmac0_po_pll500	R		Only for internal use
[20]	gmac0_po_osc25	R		Only for internal use
[19]	gmac0_po_pll25	R		Only for internal use
[18]	gmac0_po_caboff	R		Only for internal use
[17]	gmac0_po_mdix	R		Display mdix function 0: mdi mode 1: mdix mode
[16]	gmac0_po_link_done	R		Link status, 0: Link is down. 1: Link is up.
[4:0]	gmac0_pi_phyadd	RW	0x3	PHY address

Table 0-75. GMAC 1 / FXEDP412EWHJ0P_1 side-band control (0x0C10_0868)

Bit	Symbol	Access	Default	Description
[31]	gmac1_pi_suppress_preamble	RW	0x1	Management frame with preamble suppressed 0: PHY will not accept the management frames with suppressed preamble. 1: PHY will accept the management frames with suppressed preamble.
[30]	gmac1_pi_turbo_sim	RW	0x0	For simulation, the speed-up mode works to reduce the simulation time. FTC turbo mode only works in two FTC IP

Bit	Symbol	Access	Default	Description
				simulation environments. 1: Speed up the timer 0: Normal operation
[29]	gmac1_pi_force_tx_lpi	RW	0x0	Only for the internal test mode. Note: Please use the default value '0'.
[28]	gmac1_pi_force_mdix_mode	RW	0x0	For the internal test data input, Note: Please use the default value '0'. 1: mdix mode 0: mdi mode
[27]	gmac1_pi_force_mdix_en	RW	0x0	For the internal test data input, Note: Please use the default value '0'. 1: Decide the mdi/mdix mode with signal "pi_force_mdix_mode" 0: IP auto mdix function, disable "pi_force_mdix_mode" signal
[26]	gmac1_pi_rgmii_en	RW	0x0	0: GMII/MII is active. 1: RGMII is active.
[25]	gmac1_pi_testmd	RW	0x0	Test mode selection bits 0: Normal mode 1: BIST mode for the loopback self-test of the chip
[24]	gmac1_pi_fiber_en	RW	0x0	This signal forces IP to enter the fiber mode (100BASE-FX).

Bit	Symbol	Access	Default	Description
				1: Force fiber mode (100BASE-FX) operation 0: Normal operation
[23]	gmac1_pi_apb_mdi_sel	RW	0x1	APB/MDIO select signal, 0: MDIO 1: APB
[22:20]	gmac1_pi_opmode	RW	0x0	Operation control mode, and only for internal use 3'd0: Devices link through auto-negotiation with their maximum abilities
[19]	gmac1_pi_pwrgate_en	RW	0x0	Only for internal use. Note: Please tie to 0.
[18]	gmac1_pi_pwrdn_pll_alive	RW	0x1	Function is as pi_pwrdn, but the po_pll500 clock will keep alive.
[17]	gmac1_pi_pwrdn_osc_alive	RW	0x1	Function is as pi_pwrdn, but the oscillator clock will keep alive
[16]	gmac1_pi_pwrdn	RW	0x1	Power down signal, active high 1: Force PHY to enter the power-down state, MII register setting will be default setting. 0: Normal operation
[15]	gmac1_pi_pwr_resetn	RW	0x0	Power-on reset for the IP, active low 1: Normal operation 0: Power reset

Bit	Symbol	Access	Default	Description
[14]	gmac1_pi_sys_resetn	RW	0x0	This signal resets PHY to initial mode excluding the register setting, active low 1: PHY system works normally 0: PHY system reset
[9:8]	gmac1_I_PLL_OBD_REFS	RW	0x0	Select the input clock frequency of I_ASICIN 2'b00 : 25 MHz 2'b01 : 50 MHz 2'b10 : 75 MHz 2'b11 : 100 MHz
[7:4]	gmac1_reg_led1_ctrl	RW	0x0	gmac 0 led 1 stretch period [1:0] = 2'b00 : 100 ms [1:0] = 2'b01 : 50 ms [1:0] = 2'b10 : 100 ms [1:0] = 2'b11 : 200 ms
[3:0]	gmac1_reg_led0_ctrl	RW	0x0	gmac 0 led 0 stretch period [1:0] = 2'b00 : 100 ms [1:0] = 2'b01 : 50 ms [1:0] = 2'b10 : 100 ms [1:0] = 2'b11 : 200 ms

Table 0-76. GMAC 1 / FXEDP412EWHJ0P_1 side-band control (0x0C10_086C)

Bit	Symbol	Access	Default	Description
[31:16]	gmac1_pi_PHYIDRG3	RW	0x5	For setting the PHY Identifier Register 2 Bits[24:19] of the OUI are mapped to bits[15:10] of PHY Identifier Register 2. Six bits of the model number of Unimap are mapped to bits[9:4]. Four bits of the revision number of Unimap are mapped to bits[3:0].
[15:0]	gmac1_pi_PHYIDRG2	RW	0x4	For setting the PHY Identifier Register 1 PHY Identifier Register 1 stores bits[3:18] of the Organizationally Unique Identifier (OUI).

Table 0-77. GMAC 1 / FXEDP412EWHJ0P_1 side-band control (0x0C10_0870)

Bit	Symbol	Access	Default	Description
[25]	gmac1_po_wol_detect	R		Internal use
[24]	gmac1_po_rx_sop	R		Internal use
[23]	gmac1_po_tx_sop	R		Internal use
[22]	gmac1_po_int_n	R	0x1	The interrupt pins are low-active signals
[21]	gmac1_po_pll500	R		Only for internal use
[20]	gmac1_po_osc25	R		Only for internal use
[19]	gmac1_po_pll25	R		Only for internal use

Bit	Symbol	Access	Default	Description
[18]	gmac1_po_caboff	R		Only for internal use
[17]	gmac1_po_mdix	R		Display mdix function 0: mdi mode 1: mdix mode
[16]	gmac1_po_link_done	R		Link status, 0: Link is down. 1: Link is up.
[4:0]	gmac1_pi_phyadd	RW	0x6	PHY address

24.7 LVTX side-band control (0x0C10_0874)

Table 0-78. LVTX side-band control (0x0C10_0874)

Bit	Symbol	Access	Default	Description
[27]	LVTX_FLDO_O	R		Odd side LVTX PLL locked signal
[26]	LVTX_FLDO_E	R		Even side LVTX PLL locked signal
[24]	lvds_even_odd_lvds	RW	0x0	LVTX even/odd control
[23:22]	lvds_LVTX_OPTION_O	RW	0x0	Odd side LVTX OPTION control Option code for setting
[21:20]	lvds_LVTX_CC_O	RW	0x2	Odd side LVTX CC control PLL charge-pump current control pin
[19:18]	lvds_LVTX_FRANGE_O	RW	0x3	Odd side LVTX FRANGE control CLKIN frequency range control 00: 16 MHz ~ 22.5 MHz

Bit	Symbol	Access	Default	Description
				01: 22.5 MHz ~ 45 MHz 10: 45 MHz ~ 90 MHz 11: 90 MHz ~ 100 MHz
[16]	lvds_LVTX_RF_O	RW	0x0	Odd side LVTX RF control 0: Latch data at the falling edge 1: Latch data at the rising edge
[15:14]	lvds_LVTX_OPTION_E	RW	0x0	Even side LVTX OPTION control Option code for setting
[13:12]	lvds_LVTX_CC_E	RW	0x2	Even side LVTX CC control PLL charge-pump current control pin
[11:10]	lvds_LVTX_FRANGE_E	RW	0x3	Even side LVTX FRANGE control CLKIN frequency range control 00: 16 MHz ~ 22.5 MHz 01: 22.5 MHz ~ 45 MHz 10: 45 MHz ~ 90 MHz 11: 90 MHz ~ 100 MHz
[8]	lvds_LVTX_RF_E	RW	0x0	Even side LVTX RF control 0: Latch data at the falling edge 1: Latch data at the rising edge
[7:4]	reg_lvds_scaler_div	RW	0x0	lvds_lc_scaler_clk clock divider factor

Bit	Symbol	Access	Default	Description
[3:0]	reg_lvds_tv_div	RW	0x0	lvds_tv_clk and lvds_lc_clk clock divider factor

24.8 GIC400 / GLUE side-band control (0x0C10_0878 ~ 0x0C10_087C)

Table 0-79. GIC400 / GLUE side-band control (0x0C10_0878)

Bit	Symbol	Access	Default	Description
[31:28]	GLUE_reg_DAC_LAST_SEL[15:12]	RW	0x0	DAC 3 dma_last selection, mapping FTDMAC030_1 dma_last[n] to dac3_dma_last
[27:24]	GLUE_reg_DAC_LAST_SEL[11:8]	RW	0x0	DAC 2 dma_last selection, mapping FTDMAC030_1 dma_last[n] to dac2_dma_last
[23:20]	GLUE_reg_DAC_LAST_SEL[7:4]	RW	0x0	DAC 1 dma_last selection, mapping FTDMAC030_1 dma_last[n] to dac1_dma_last
[19:16]	GLUE_reg_DAC_LAST_SEL[3:0]	RW	0x0	DAC 0 dma_last selection, mapping FTDMAC030_1 dma_last[n] to dac0_dma_last
[15:14]	GLUE_reg_dbg_sel[3:2]	RW	0x0	debug port pad wrap out control 2'b00 : dbg_out[19:0] = dbg_sel[19:0] 2'b01 : dbg_out[19:0] = {dbg_sel[3:0], dbg_sel[19:4]} 2'b10 : dbg_out[19:0] = {dbg_sel[7:0], dbg_sel[19:8]} 2'b11 : dbg_out[19:0] = {dbg_sel[11:0], dbg_sel[19:12]}
[13:12]	GLUE_reg_dbg_sel[1:0]	RW	0x0	debug_sel[19:0] selection 2'b00 : gmac_0 debug port[19:0] 2'b01 : gmac_1 debug port[19:0]

Bit	Symbol	Access	Default	Description
				2'b10 : {4'd0, USB3 debug port[15:0]} 2'b11 : {12'd0, VIVANTE_GPU debug port[7:0]}
[10:8]	-	RW	0x0	Reserved
[6:4]	-	RW	0x0	Reserved
[0]	gic400_CFGSDISABLE	RW	0x0	GIC400 CFGSDISABLE control

Table 0-80. GLUE side-band control (0x0C10_087C)

Bit	Symbol	Access	Default	Description
[31:16]	GLUE_reg_dma_req_sel[31:16]	RW	0x0	FTDMAC030_1 dma request mux [16] : dma_req[0] select 1'b1 : tm5_dma_req_FTPWMTMR010_1, 1'b0 : dac0_dma_req [17] : dma_req[1] select 1'b1 : tm6_dma_req_FTPWMTMR010_1, 1'b0 : dac1_dma_req [18] : dma_req[2] select 1'b1 : tm7_dma_req_FTPWMTMR010_1, 1'b0 : dac2_dma_req [19] : dma_req[3] select 1'b1 : tm8_dma_req_FTPWMTMR010_1,

Bit	Symbol	Access	Default	Description
				1'b0 : dac3_dma_req [27] : dma_req[11] select 1'b1 : sd_dma_req_FTSDC021, 1'b0 : ssp_rxdmareq_u_FTSSP010_2 [29] : dma_req[13] select 1'b1 : sd_dma_req_FTSDC021_1, 1'b0 : ssp_rxdmareq_u_FTSSP010_3
[15:0]	GLUE_reg_dma_req_sel[15:0]	RW	0x0	FTDMAC030_0 dma request mux [5] : dma_req[5] select 1'b1 : sd_dma_req_FTSDC021, 1'b0 : dma_tx_req_FTUSART010 [6] : dma_req[6] select 1'b1 : sd_dma_req_FTSDC021_1, 1'b0 : dma_rx_req_FTUSART010 [11] : dma_req[11] select 1'b1 : tm1_dma_req_FTPWMTMR010_1, 1'b0 : adc_dma_req[0] [12] : dma_req[12] select 1'b1 : tm2_dma_req_FTPWMTMR010_1, 1'b0 : adc_dma_req[1] [13] : dma_req[13] select

Bit	Symbol	Access	Default	Description
				1'b1 : tm3_dma_req_FTPWMTMR010_1, 1'b0 : adc_dma_req[2] [14] : dma_req[14] select 1'b1 : tm4_dma_req_FTPWMTMR010_1, 1'b0 : adc_dma_req[3]

24.9 PCIE side-band control (0x0C10_0880)

Table 0-81. PCIE side-band control (0x0C10_0880)

Bit	Symbol	Access	Default	Description
[31:28]	PCIE_slv_arqos	RW	0x0	pcie slave bus arqos control
[27:24]	PCIE_slv_awqos	RW	0x0	pcie slave bus awqos control
[23:20]	PCIE_dbi_arqos	RW	0x0	pcie dbi bus arqos control
[19:16]	PCIE_dbi_awqos	RW	0x0	pcie dbi bus awqos control
[7]	pcie_FW_update_done	RW	0x0	PCIE FW update done control
[6]	pcie_FW_update_en	RW	0x0	PCIE FW update enable control
[5]	pcie_perstn_out_cfg	RW	0x1	pcie perstn pad output control 1'b0 : output low 1'b1 : output system reset
[4]	pcie_perstn_oe	RW	0x1	pcie perstn pad oe control
[1]	PCIE_clkreqn_sts	RW1C	0x0	pcie clkreqn event status, write one clean
[0]	PCIE_wake_sts	RW1C	0x0	pcie wake event status, write one clean

24.10 PLL_LVDS control (0x0C10_0884)

Table 0-82. PLL_LVDS control (0x0C10_0884)

Bit	Symbol	Access	Default	Description
[20:0]	reg_pll_lvds_ctrl	RW	0x10_6341	PLL_LVDS sideband control [2:0] : MS2 ~ MS0 [7:4] : PS3 ~ PS0 [16:8] : NS8 ~ NS0 [20] : PDN

24.11 PLL_GMAC control (0x0C10_0888)

Table 0-83. PLL_GMAC control (0x0C10_0888)

Bit	Symbol	Access	Default	Description
[20:0]	reg_pll_gmac_ctrl	RW	0x10_7d01	PLL_LVDS sideband control [2:0] : MS2 ~ MS0 [7:4] : PS3 ~ PS0 [16:8] : NS8 ~ NS0 [20] : PDN

24.12 Clkgen side-band control (0x0C10_088C)

Table 0-84. clkgen side-band control (0x0C10_088C)

Bit	Symbol	Access	Default	Description
[31:20]	-	RW	0x0	Reserved
[19]	cpu_rstn_ctrl	RW	0x0	CA53 sub-system nCORERESET[3] control 1'b1 : release control of nCORERESET[0], and it would be controlled by SCU reboot or SW reset function 1'b0 : set nCORERESET[3] == 1'b0, keep CA53 core3 as reset state
[18]		RW	0x0	CA53 sub-system nCORERESET[2] control 1'b1 : release control of nCORERESET[0], and it would be controlled by SCU reboot or SW reset function 1'b0 : set nCORERESET[2] == 1'b0, keep CA53 core2 as reset state
[17]		RW	0x0	CA53 sub-system nCORERESET[1] control 1'b1 : release control of nCORERESET[0], and it would be controlled by SCU reboot or SW reset function 1'b0 : set nCORERESET[1] == 1'b0, keep CA53 core1 as reset state
[16]		RW	0x1	CA53 sub-system nCORERESET[0] control 1'b1 : release control of nCORERESET[0], and it would be controlled by SCU reboot or SW reset function 1'b0 : set nCORERESET[0] == 1'b0, keep CA53 core0 as reset state

Bit	Symbol	Access	Default	Description
[15:11]	-	RW	0x0	Reserved
[9:0]	reg_extclk_div	RW	0x1F3	FTPWMTMR010 / FTPWMTMR010_1 / FTTMR010 / FTWDT011 external clock divider factor

24.13 SRAM side-band control (0x0C10_0890)

Table 0-85. SRAM side-band control (0x0C10_0890)

Bit	Symbol	Access	Default	Description
[30:28]	reg_dmac1_memory_ctrl	RW	0x0	DMAC 1 SRAM cotnrol [0] : SRAM LS pin control [1] : SRAM DS pin control [2] : SRAM SD pin control
[26:24]	reg_dmac0_memory_ctrl	RW	0x0	DMAC 0 SRAM cotnrol
[22:20]	reg_canfd2_memory_ctrl	RW	0x0	CANFD 2 SRAM cotnrol
[18:16]	reg_canfd1_memory_ctrl	RW	0x0	CANFD 1 SRAM cotnrol
[14:12]	reg_canfd0_memory_ctrl	RW	0x0	CANFD 0 SRAM cotnrol
[10:8]	reg_gmac1_memory_ctrl	RW	0x0	GMAC 1 SRAM control
[6:4]	reg_gmac0_memory_ctrl	RW	0x0	GMAC 0 SRAM control
[2:0]	reg_emc_memory_ctrl	RW	0x0	EMC SRAM control

24.14 Reserved Register (0x0C10_0894)

Table 0-86. Reserved Register (0x0C10_0894)

Bit	Symbol	Access	Default	Description
[31:3]	-	RW	0x0	Reserved
[2]	iddq_mode_en	RW	0x1	PCIE iddq mode mux control 1'b0: phy_reset controlled by PCIE controller 1'b1: phy_reset controlled by signal "phy_reset_iddq"
[1]	phy_reset_iddq	RW	0x1	control PCIE phy_reset when "iddq_mode_en" == 1'b1
[0]	phy_test_powerdown	RW	0x1	PCIE iddq mode control, use to control PCIE phy's "phy_test_powerdown"

24.15 Pad pin-mux Selection (0x0C10_0898 ~ 0x0C10_08B4)

Table 0-87. Pad pin mux "pmur_mfs0" (0x0C10_0898)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs0[31:30]	RW	0x0	X_GPIO[15] pad out mux 2'b00 : GPIO[15] 2'b01 : Motor/ds_din[5] 2'b10 : debug_out[15]
[29:28]	pmur_mfs0[29:28]	RW	0x0	X_GPIO[14] pad out mux 2'b00 : GPIO[14] 2'b01 : Motor/ds_din[4] 2'b10 : debug_out[14]

Bit	Symbol	Access	Default	Description
[27:26]	pmur_mfs0[27:26]	RW	0x0	X_GPIO[13] pad out mux 2'b00 : GPIO[13] 2'b01 : Motor/ds_din[3] 2'b10 : debug_out[13]
[25:24]	pmur_mfs0[25:24]	RW	0x0	X_GPIO[12] pad out mux 2'b00 : GPIO[12] 2'b01 : Motor/ds_din[2] 2'b10 : debug_out[12]
[23:22]	pmur_mfs0[23:22]	RW	0x0	X_GPIO[11] pad out mux 2'b00 : GPIO[11] 2'b01 : Motor/ds_din[1] 2'b10 : debug_out[11]
[21:20]	pmur_mfs0[21:20]	RW	0x0	X_GPIO[10] pad out mux 2'b00 : GPIO[10] 2'b01 : Motor/ds_din[0] 2'b10 : debug_out[10]
[19:18]	pmur_mfs0[19:18]	RW	0x0	X_GPIO[9] pad out mux 2'b00 : GPIO[9] 2'b01 : FTPWMTMR010/tmr3_out_n 2'b10 : debug_out[9]

Bit	Symbol	Access	Default	Description
[17:16]	pmur_mfs0[17:16]	RW	0x0	X_GPIO[8] pad out mux 2'b00 : GPIO[8] 2'b01 : FTPWMTMR010/tmr3_out 2'b10 : debug_out[8]
[15:14]	pmur_mfs0[15:14]	RW	0x0	X_GPIO[7] pad out mux 2'b00 : GPIO[7] 2'b01 : FTPWMTMR010/tmr2_out_n 2'b10 : debug_out[7]
[13:12]	pmur_mfs0[13:12]	RW	0x0	X_GPIO[6] pad out mux 2'b00 : GPIO[6] 2'b01 : FTPWMTMR010/tmr2_out 2'b10 : debug_out[6]
[11:10]	pmur_mfs0[11:10]	RW	0x0	X_GPIO[5] pad out mux 2'b00 : GPIO[5] 2'b01 : FTPWMTMR010/tmr1_out_n 2'b10 : debug_out[5]
[9:8]	pmur_mfs0[9:8]	RW	0x0	X_GPIO[4] pad out mux 2'b00 : GPIO[4] 2'b01 : FTPWMTMR010/tmr1_out 2'b10 : debug_out[4]

Bit	Symbol	Access	Default	Description
[7:6]	pmur_mfs0[7:6]	RW	0x0	X_GPIO[3] pad out mux 2'b00 : GPIO[3] 2'b01 : SSP_2/rxd 2'b10 : debug_out[3]
[5:4]	pmur_mfs0[5:4]	RW	0x0	X_GPIO[2] pad out mux 2'b00 : GPIO[2] 2'b01 : SSP_2/txd 2'b10 : debug_out[2]
[3:2]	pmur_mfs0[3:2]	RW	0x0	X_GPIO[1] pad out mux 2'b00 : GPIO[1] 2'b01 : SSP_2/fs 2'b10 : debug_out[1]
[1:0]	pmur_mfs0[1:0]	RW	0x0	X_GPIO[0] pad out mux 2'b00 : GPIO[0] 2'b01 : SSP_2/sclk 2'b10 : debug_out[0]

Table 0-88. Pad pin mux "pmur_mfs1" (0x0C10_089C)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs1[31:30]	RW	0x0	X_I2C3_SDA pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : IIC_3/sdain 2'b01 : GPIO[31]
[29:28]	pmur_mfs1[29:28]	RW	0x0	X_I2C3_SCL pad out mux 2'b00 : IIC_3/sclin 2'b01 : GPIO[30]
[27:26]	pmur_mfs1[27:26]	RW	0x0	X_I2C2_SDA pad out mux 2'b00 : IIC_2/sdain 2'b01 : GPIO[29]
[25:24]	pmur_mfs1[25:24]	RW	0x0	X_I2C2_SCL pad out mux 2'b00 : IIC_2/sclin 2'b01 : GPIO[28]
[23:22]	pmur_mfs1[23:22]	RW	0x0	X_UART2_RTSN pad out mux 2'b00 : UART2/irda_nrts 2'b01 : GPIO[27]
[21:20]	pmur_mfs1[21:20]	RW	0x0	X_UART2_CTSN pad out mux 2'b00 : UART2/irda_ncts 2'b01 : GPIO[26]
[19:18]	pmur_mfs1[19:18]	RW	0x0	X_UART2_RX pad out mux 2'b00 : UART2/io_irda_sin 2'b01 : GPIO[25]

Bit	Symbol	Access	Default	Description
[17:16]	pmur_mfs1[17:16]	RW	0x0	X_UART2_TX pad out mux 2'b00 : UART2/io_irda_sout 2'b01 : GPIO[24]
[15:14]	pmur_mfs1[15:14]	RW	0x0	X_P1_LED1 pad out mux 2'b00 : p1_led1 2'b01 : GPIO[23]
[13:12]	pmur_mfs1[13:12]	RW	0x0	X_P1_LED0 pad out mux 2'b00 : p1_led0 2'b01 : GPIO[22]
[11:10]	pmur_mfs1[11:10]	RW	0x0	X_P0_LED1 pad out mux 2'b00 : p0_led1 2'b01 : GPIO[21]
[9:8]	pmur_mfs1[9:8]	RW	0x0	X_P0_LED0 pad out mux 2'b00 : p0_led0 2'b01 : GPIO[20]
[7:6]	pmur_mfs1[7:6]	RW	0x0	X_BOOT_FAIL pad out mux 2'b00 : boot_fail 2'b01 : GPIO[19] 2'b10 : debug_out[19]
[5:4]	pmur_mfs1[5:4]	RW	0x0	X_GPIO[18] pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : GPIO[18] 2'b01 : CANFD_2/canfd_rxd 2'b10 : debug_out[18]
[3:2]	pmur_mfs1[3:2]	RW	0x0	X_GPIO[17] pad out mux 2'b00 : GPIO[17] 2'b01 : CANFD_2/canfd_txd 2'b10 : debug_out[17]
[1:0]	pmur_mfs1[1:0]	RW	0x0	X_GPIO[16] pad out mux 2'b00 : GPIO[16] 2'b01 : DAC0/dac_ext_trig 2'b10 : debug_out[16]

Table 0-89. Pad pin mux "pmur_mfs2" (0x0C10_08A0)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs2[31:30]	RW	0x0	X_SEC_CPU_TMS pad out mux 2'b00 : Soteria/tms 2'b01 : UART_3/irda_ncts
[29:28]	pmur_mfs2[29:28]	RW	0x0	X_SEC_CPU_TRSTn pad out mux 2'b00 : Soteria/trstn 2'b01 : UART_3/irda_nrts

Bit	Symbol	Access	Default	Description
[27:26]	pmur_mfs2[27:26]	RW	0x0	X_SD0_WP pad out mux 2'b00 : SDC/io_sd_wp 2'b01 : Motor/ePWM9B
[25:24]	pmur_mfs2[25:24]	RW	0x0	X_SD0_DAT3 pad out mux 2'b00 : SDC/io_sd_dat3 2'b01 : Motor/ePWM9A
[23:22]	pmur_mfs2[23:22]	RW	0x0	X_SD0_DAT2 pad out mux 2'b00 : SDC/io_sd_dat2 2'b01 : Motor/ePWM8B
[21:20]	pmur_mfs2[21:20]	RW	0x0	X_SD0_DAT1 pad out mux 2'b00 : SDC/io_sd_dat1 2'b01 : Motor/ePWM8A
[19:18]	pmur_mfs2[19:18]	RW	0x0	X_SD0_DAT0 pad out mux 2'b00 : SDC/io_sd_dat0 2'b01 : Motor/ePWM7B
[17:16]	pmur_mfs2[17:16]	RW	0x0	X_SD0_CMD_RSP pad out mux 2'b00 : SDC/io_sd_rsp 2'b01 : Motor/ePWM7A
[15:14]	pmur_mfs2[15:14]	RW	0x0	X_SD1_RSTN pad out mux 2'b00 : SDC_1/io_sd_rstn

Bit	Symbol	Access	Default	Description
				2'b01 : GPIO[7]
[13:12]	pmur_mfs2[13:12]	RW	0x0	X_SD1_CD pad out mux 2'b00 : SDC_1/io_sd_cd 2'b01 : GPIO[6]
[11:10]	pmur_mfs2[11:10]	RW	0x0	X_SD1_WP pad out mux 2'b00 : SDC_1/io_sd_wp 2'b01 : GPIO[5]
[9:8]	pmur_mfs2[9:8]	RW	0x0	X_SD1_DAT3 pad out mux 2'b00 : SDC_1/io_sd_dat3 2'b01 : GPIO[4]
[7:6]	pmur_mfs2[7:6]	RW	0x0	X_SD1_DAT2 pad out mux 2'b00 : SDC_1/io_sd_dat2 2'b01 : GPIO[3]
[5:4]	pmur_mfs2[5:4]	RW	0x0	X_SD1_DAT1 pad out mux 2'b00 : SDC_1/io_sd_dat1 2'b01 : GPIO[2]
[3:2]	pmur_mfs2[3:2]	RW	0x0	X_SD1_DAT0 pad out mux 2'b00 : SDC_1/io_sd_dat0 2'b01 : GPIO[1]
[1:0]	pmur_mfs2[1:0]	RW	0x0	X_SD1_CMD_RSP pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : SDC_1/io_sd_rsp 2'b01 : GPIO[0]

Table 0-90. Pad pin mux "pmur_mfs3" (0x0C10_08A4)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs3[31:30]	RW	0x0	X_EQEP_S[1] pad out mux 2'b00 : Motor/eQep_sin[1] 2'b01 : UART_3/io_irda_sin
[29:28]	pmur_mfs3[29:28]	RW	0x0	X_EQEP_S[0] pad out mux 2'b00 : Motor/eQep_sin[0] 2'b01 : UART_3/io_irda_sout
[27:26]	pmur_mfs3[27:26]	RW	0x0	X_EQEP_I[3] pad out mux 2'b00 : Motor/eQep_iin[3] 2'b01 : UART_2/irda_nrts
[25:24]	pmur_mfs3[25:24]	RW	0x0	X_EQEP_I[2] pad out mux 2'b00 : Motor/eQep_iin[2] 2'b01 : UART_2/irda_ncts
[23:22]	pmur_mfs3[23:22]	RW	0x0	X_EQEP_I[1] pad out mux 2'b00 : Motor/eQep_iin[1] 2'b01 : UART_2/io_irda_sin

Bit	Symbol	Access	Default	Description
[21:20]	pmur_mfs3[21:20]	RW	0x0	X_EQEP_I[0] pad out mux 2'b00 : Motor/eQep_iin[0] 2'b01 : UART_2/io_irda_sout
[19:18]	pmur_mfs3[19:18]	RW	0x0	X_EQEP_BIN[3] pad out mux 2'b00 : Motor/eQep_bin[3] 2'b01 : SSP_2/rxd
[17:16]	pmur_mfs3[17:16]	RW	0x0	X_EQEP_BIN[2] pad out mux 2'b00 : Motor/eQep_bin[2] 2'b01 : SSP_2/txd
[15:14]	pmur_mfs3[15:14]	RW	0x0	X_EQEP_BIN[1] pad out mux 2'b00 : Motor/eQep_bin[1] 2'b01 : SSP_2/fs
[13:12]	pmur_mfs3[13:12]	RW	0x0	X_EQEP_BIN[0] pad out mux 2'b00 : Motor/eQep_bin[0] 2'b01 : SSP_2/sclk
[11:10]	pmur_mfs3[11:10]	RW	0x0	X_EQEP_AIN[3] pad out mux 2'b00 : Motor/eQep_ain[3] 2'b01 : SSP_3/rxd
[9:8]	pmur_mfs3[9:8]	RW	0x0	X_EQEP_AIN[2] pad out mux 2'b00 : Motor/eQep_ain[2]

Bit	Symbol	Access	Default	Description
				2'b01 : SSP_3/txd
[7:6]	pmur_mfs3[7:6]	RW	0x0	X_EQEP_AIN[1] pad out mux 2'b00 : Motor/eQep_ain[1] 2'b01 : SSP_3/fs
[5:4]	pmur_mfs3[5:4]	RW	0x0	X_EQEP_AIN[0] pad out mux 2'b00 : Motor/eQep_ain[0] 2'b01 : SSP_3/sclk
[3:2]	pmur_mfs3[3:2]	RW	0x0	X_SEC_CPU_TDO pad out mux 2'b00 : Soteria/tdi 2'b01 : UART_3/io_irda_sout
[1:0]	pmur_mfs3[1:0]	RW	0x0	X_SEC_CPU_TDI pad out mux 2'b00 : Soteria/tdi 2'b01 : UART_3/io_irda_sin

Table 0-91. Pad pin mux "pmur_mfs4" (0x0C10_08A8)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs4[31:30]	RW	0x0	X_EXADC3_A[1] pad out mux 2'b00 : Motor/exADC3_a[1] 2'b01 : FTPWMTMR/tmr5_out
[29:28]	pmur_mfs4[29:28]	RW	0x0	X_EXADC3_A[0] pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : Motor/exADC3_a[0] 2'b01 : FTPWMTMR/tmr4_out_n
[27:26]	pmur_mfs4[27:26]	RW	0x0	X_EXADC3_CSN pad out mux 2'b00 : Motor/exADC3_scsn 2'b01 : FTPWMTMR/tmr4_out
[25:24]	pmur_mfs4[25:24]	RW	0x0	X_EXADC3_SCLK pad out mux 2'b00 : Motor/exADC3_sclk 2'b01 : FTPWMTMR/tmr3_out_n
[23:22]	pmur_mfs4[23:22]	RW	0x0	X_ENDAT_DOUT[2] pad out mux 2'b00 : Motor/EnDat_dout[2] 2'b01 : FTPWMTMR/tmr3_out
[21:20]	pmur_mfs4[21:20]	RW	0x0	X_ENDAT_TCLK[2] pad out mux 2'b00 : Motor/EnDat_tclk[2] 2'b01 : FTPWMTMR/tmr2_out_n
[19:18]	pmur_mfs4[19:18]	RW	0x0	X_EQEP_BOUT[3] pad out mux 2'b00 : Motor/eQep_bout[3] 2'b01 : FTPWMTMR/tmr2_out
[17:16]	pmur_mfs4[17:16]	RW	0x0	X_EQEP_BOUT[2] pad out mux 2'b00 : Motor/eQep_bout[2] 2'b01 : FTPWMTMR/tmr1_out_n

Bit	Symbol	Access	Default	Description
[15:14]	pmur_mfs4[15:14]	RW	0x0	X_EQEP_BOUT[1] pad out mux 2'b00 : Motor/eQep_bout[1] 2'b01 : FTPWMTMR/tmr1_out
[13:12]	pmur_mfs4[13:12]	RW	0x0	X_EQEP_BOUT[0] pad out mux 2'b00 : Motor/eQep_bout[0] 2'b01 : USART_1/rts
[11:10]	pmur_mfs4[11:10]	RW	0x0	X_EQEP_AOUT[3] pad out mux 2'b00 : Motor/eQep_aout[3] 2'b01 : USART_1/cts
[9:8]	pmur_mfs4[9:8]	RW	0x0	X_EQEP_AOUT[2] pad out mux 2'b00 : Motor/eQep_aout[2] 2'b01 : USART_1/usart_rx
[7:6]	pmur_mfs4[7:6]	RW	0x0	X_EQEP_AOUT[1] pad out mux 2'b00 : Motor/eQep_aout[1] 2'b01 : USART_1/usart_tx
[5:4]	pmur_mfs4[5:4]	RW	0x0	X_EQEP_AOUT[0] pad out mux 2'b00 : Motor/eQep_aout[0] 2'b01 : USART_1/clk_out
[3:2]	pmur_mfs4[3:2]	RW	0x0	X_EQEP_S[3] pad out mux 2'b00 : Motor/eQep_sin[3]

Bit	Symbol	Access	Default	Description
				2'b01 : UART_3/irda_nrts
[1:0]	pmur_mfs4[1:0]	RW	0x0	X_EQEP_S[2] pad out mux 2'b00 : Motor/eQep_sin[2] 2'b01 : UART_3/irda_ncts

Table 0-92. Pad pin mux "pmur_mfs5" (0x0C10_08AC)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs5[31:30]	RW	0x0	X_XMC_ADDR[12] pad out mux 2'b00 : Motor/ADDR[12] 2'b01 : FTPWMTMR/tmr5_out
[29:28]	pmur_mfs5[29:28]	RW	0x0	X_XMC_ADDR[11] pad out mux 2'b00 : Motor/ADDR[11] 2'b01 : FTPWMTMR/tmr4_out_n
[27:26]	pmur_mfs5[27:26]	RW	0x0	X_XMC_ADDR[10] pad out mux 2'b00 : Motor/ADDR[10] 2'b01 : FTPWMTMR/tmr4_out
[25:24]	pmur_mfs5[25:24]	RW	0x0	X_XMC_ADDR[9] pad out mux 2'b00 : Motor/ADDR[9] 2'b01 : FTPWMTMR/tmr3_out_n
[23:22]	pmur_mfs5[23:22]	RW	0x0	X_XMC_ADDR[8] pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : Motor/ADDR[8] 2'b01 : FTPWMTMR/tmr3_out
[21:20]	pmur_mfs5[21:20]	RW	0x0	X_XMC_ADDR[7] pad out mux 2'b00 : Motor/ADDR[7] 2'b01 : FTPWMTMR/tmr2_out_n
[19:18]	pmur_mfs5[19:18]	RW	0x0	X_XMC_ADDR[6] pad out mux 2'b00 : Motor/ADDR[6] 2'b01 : FTPWMTMR/tmr2_out
[17:16]	pmur_mfs5[17:16]	RW	0x0	X_XMC_ADDR[5] pad out mux 2'b00 : Motor/ADDR[5] 2'b01 : FTPWMTMR/tmr1_out_n
[15:14]	pmur_mfs5[15:14]	RW	0x0	X_XMC_ADDR[4] pad out mux 2'b00 : Motor/ADDR[4] 2'b01 : FTPWMTMR/tmr1_out
[13:12]	pmur_mfs5[13:12]	RW	0x0	X_XMC_ADDR[0] pad out mux 2'b00 : Motor/ADDR[0] 2'b01 : SSP_2/txd
[11:10]	pmur_mfs5[11:10]	RW	0x0	X_XMC_WEN pad out mux 2'b00 : Motor/WEn 2'b01 : SSP_2/rxd

Bit	Symbol	Access	Default	Description
[9:8]	pmur_mfs5[9:8]	RW	0x0	X_XMC_OEN pad out mux 2'b00 : Motor/OEn 2'b01 : SSP_2/sclk
[7:6]	pmur_mfs5[7:6]	RW	0x0	X_XMC_CSN[0] pad out mux 2'b00 : Motor/CSn[0] 2'b01 : SSP_2/fs
[5:4]	pmur_mfs5[5:4]	RW	0x0	X_EXADC3_DOUTB pad out mux 2'b00 : Motor/exADC3_doutb 2'b01 : FTPWMTMR/tmr6_out_n
[3:2]	pmur_mfs5[3:2]	RW	0x0	X_EXADC3_DOUTA pad out mux 2'b00 : Motor/exADC3_douta 2'b01 : FTPWMTMR/tmr6_out
[1:0]	pmur_mfs5[1:0]	RW	0x0	X_EXADC3_A[2] pad out mux 2'b00 : Motor/exADC3_a[2] 2'b01 : FTPWMTMR/tmr5_out_n

Table 0-93. Pad pin mux "pmur_mfs6" (0x0C10_08B0)

Bit	Symbol	Access	Default	Description
[31:30]	pmur_mfs6[31:30]	RW	0x0	X_XMC_DATA[12] pad out mux 2'b00 : Motor/DATA_in[12]

Bit	Symbol	Access	Default	Description
				2'b01 : GPIO[28]
[29:28]	pmur_mfs6[29:28]	RW	0x0	X_XMC_DATA[11] pad out mux 2'b00 : Motor/DATA_in[11] 2'b01 : GPIO[27]
[27:26]	pmur_mfs6[27:26]	RW	0x0	X_XMC_DATA[10] pad out mux 2'b00 : Motor/DATA_in[10] 2'b01 : GPIO[26]
[25:24]	pmur_mfs6[25:24]	RW	0x0	X_XMC_DATA[9] pad out mux 2'b00 : Motor/DATA_in[9] 2'b01 : GPIO[25]
[23:22]	pmur_mfs6[23:22]	RW	0x0	X_XMC_DATA[8] pad out mux 2'b00 : Motor/DATA_in[8] 2'b01 : GPIO[24]
[21:20]	pmur_mfs6[21:20]	RW	0x0	X_XMC_DATA[7] pad out mux 2'b00 : Motor/DATA_in[7] 2'b01 : GPIO[23]
[19:18]	pmur_mfs6[19:18]	RW	0x0	X_XMC_DATA[6] pad out mux 2'b00 : Motor/DATA_in[6] 2'b01 : GPIO[22]
[17:16]	pmur_mfs6[17:16]	RW	0x0	X_XMC_DATA[5] pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : Motor/DATA_in[5] 2'b01 : GPIO[21]
[15:14]	pmur_mfs6[15:14]	RW	0x0	X_XMC_DATA[4] pad out mux 2'b00 : Motor/DATA_in[4] 2'b01 : GPIO[20]
[13:12]	pmur_mfs6[13:12]	RW	0x0	X_XMC_DATA[3] pad out mux 2'b00 : Motor/DATA_in[3] 2'b01 : GPIO[19]
[11:10]	pmur_mfs6[11:10]	RW	0x0	X_XMC_DATA[2] pad out mux 2'b00 : Motor/DATA_in[2] 2'b01 : GPIO[18]
[9:8]	pmur_mfs6[9:8]	RW	0x0	X_XMC_DATA[1] pad out mux 2'b00 : Motor/DATA_in[1] 2'b01 : GPIO[17]
[7:6]	pmur_mfs6[7:6]	RW	0x0	X_XMC_DATA[0] pad out mux 2'b00 : Motor/DATA_in[0] 2'b01 : GPIO[16]
[5:4]	pmur_mfs6[5:4]	RW	0x0	X_XMC_ADDR[15] pad out mux 2'b00 : Motor/ADDR[15] 2'b01 : FTPWMTMR/tmr6_out_n

Bit	Symbol	Access	Default	Description
[3:2]	pmur_mfs6[3:2]	RW	0x0	X_XMC_ADDR[14] pad out mux 2'b00 : Motor/ADDR[14] 2'b01 : FTPWMTMR/tmr6_out
[1:0]	pmur_mfs6[1:0]	RW	0x0	X_XMC_ADDR[13] pad out mux 2'b00 : Motor/ADDR[13] 2'b01 : FTPWMTMR/tmr5_out_n

Table 0-94. Pad pin mux "pmur_mfs7" (0x0C10_08B4)

Bit	Symbol	Access	Default	Description
[31:19]	-	RW	0x0	Reserved
[18]	pmur_mfs7[18]	RW	0x0	X_CA53_TRACE* pad out mux 1'b0 : CA53 trace function 1'b1 : Motor_Top ds_mclk[5:0] output
[17:16]	pmur_mfs7[17:16]	RW	0x0	X_USART0_RTSN pad out mux 2'b00 : USART/usart_rtsn 2'b01 : DAC_3/dac_ext_trig
[15:14]	pmur_mfs7[15:14]	RW	0x0	X_USART0_CTSN pad out mux 2'b00 : USART/usart_ctsn 2'b01 : DAC_2/dac_ext_trig
[13:12]	pmur_mfs7[13:12]	RW	0x0	X_USART0_RX pad out mux

Bit	Symbol	Access	Default	Description
				2'b00 : USART/usart_rx 2'b01 : DAC_1/dac_ext_trig
[11:10]	pmur_mfs7[11:10]	RW	0x0	X_USART0_TX pad out mux 2'b00 : USART/usart_tx 2'b01 : DAC/dac_ext_trig
[9:8]	pmur_mfs7[9:8]	RW	0x0	X_UART1_CTSN pad out mux 2'b00 : UART_1/irda_ncts 2'b01 : DAC_2/dac_ext_trig
[7:6]	pmur_mfs7[7:6]	RW	0x0	X_UART0_CTSN pad out mux 2'b00 : UART_0/irda_ncts 2'b01 : DAC_1/dac_ext_trig
[5:4]	pmur_mfs7[5:4]	RW	0x0	X_XMC_DATA[15] pad out mux 2'b00 : Motor/DATA_in[15] 2'b01 : GPIO[31]
[3:2]	pmur_mfs7[3:2]	RW	0x0	X_XMC_DATA[14] pad out mux 2'b00 : Motor/DATA_in[14] 2'b01 : GPIO[30]
[1:0]	pmur_mfs7[1:0]	RW	0x0	X_XMC_DATA[13] pad out mux 2'b00 : Motor/DATA_in[13] 2'b01 : GPIO[29]

24.16 Pad PU control (0x0C10_08B8 ~ 0x0C10_08D8)

Table 0-95. Pad PU control (0x0C10_08B8)

Bit	Symbol	Access	Default	Description
[31:30]	-	RW	0x0	Reserved
[29]	xmc_wen_pu	RW	0x0	X_XMC_WEN PU control
[28]	xmc_oen_pu	RW	0x0	X_XMC_OEN PU control
[27:24]	xmc_csn_pu	RW	0x0	X_XMC_CSN PU control
[23:21]	endat_tclk_pu	RW	0x0	X_ENDAT_TCLK PU control
[20:18]	endat_dout_pu	RW	0x0	X_ENDAT_DOUT PU control
[17:15]	endat_din_pu	RW	0x0	X_ENDAT_DIN PU control
[14:12]	endat_oe_pu	RW	0x0	X_ENDAT_OE PU control
[11]	ca53_trstn_pu	RW	0x1	X_CA53_TRSTN PU control
[10:7]	ca53_traced_pu	RW	0x0	X_CA53_TRACED PU control
[6]	ca53_tracectl_pu	RW	0x0	X_CA53_TRACECTL PU control
[5]	ca53_traceclk_pu	RW	0x0	X_CA53_TRACECLK PU control
[4]	ca53_tms_pu	RW	0x1	X_CA53_TMS PU control
[3]	ca53_tdo_pu	RW	0x0	X_CA53_TDO PU control
[2]	ca53_tdi_pu	RW	0x1	X_CA53_TDI PU control
[1]	ca53_tck_pu	RW	0x1	X_CA53_TCK PU control
[0]	boot_fail_pu	RW	0x0	X_BOOT_FAIL PU control

Table 0-96. Pad PU control (0x0C10_08BC)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	exadc1_sclk_pu	RW	0x0	X_EXADC1_SCLK PU control
[29]	exadc1_doutb_pu	RW	0x0	X_EXADC1_DOUTB PU control
[28]	exadc1_douta_pu	RW	0x0	X_EXADC1_DOUTA PU control
[27]	exadc1_csn_pu	RW	0x0	X_EXADC1_CSN PU control
[26:24]	exadc1_a_pu	RW	0x0	X_EXADC1_A PU control
[23:12]	eqep_aout_pu	RW	0x0	X_EQEP_AOUT PU control
[11:0]	eqep_ain_pu	RW	0x0	X_EQEP_AIN PU control

Table 0-97. Pad PU control (0x0C10_08C0)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	exadc2_sclk_pu	RW	0x0	X_EXADC2_SCLK PU control
[29]	exadc2_doutb_pu	RW	0x0	X_EXADC2_DOUTB PU control
[28]	exadc2_douta_pu	RW	0x0	X_EXADC2_DOUTA PU control
[27]	exadc2_csn_pu	RW	0x0	X_EXADC2_CSN PU control
[26:24]	exadc2_a_pu	RW	0x0	X_EXADC2_A PU control
[23:12]	eqep_bout_pu	RW	0x0	X_EQEP_BOUT PU control

Bit	Symbol	Access	Default	Description
[11:0]	eqep_bin_pu	RW	0x0	X_EQEP_BIN PU control

Table 0-98. Pad PU control (0x0C10_08C4)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	exadc3_sclk_pu	RW	0x0	X_EXADC3_SCLK PU control
[29]	exadc3_doutb_pu	RW	0x0	X_EXADC3_DOUTB PU control
[28]	exadc3_douta_pu	RW	0x0	X_EXADC3_DOUTA PU control
[27]	exadc3_csn_pu	RW	0x0	X_EXADC3_CSN PU control
[26:24]	exadc3_a_pu	RW	0x0	X_EXADC3_A PU control
[23:12]	eqep_s_pu	RW	0x0	X_EQEP_S PU control
[11:0]	eqep_i_pu	RW	0x0	X_EQEP_I PU control

Table 0-99. Pad PU control (0x0C10_08C8)

Bit	Symbol	Access	Default	Description
[31:16]	xmc_data_pu	RW	0x0	X_XMC_DATA PU control
[15:0]	xmc_addr_pu	RW	0x0	X_XMC_ADDR PU control

Table 0-100. Pad PU control (0x0C10_08CC)

Bit	Symbol	Access	Default	Description
[31:13]	gpio_pu	RW	0x0	X_GPIO[18:0] PU control
[12]	pwm_oe_pu	RW	0x0	X_PWM_OE PU control
[11]	pwm6b_pu	RW	0x0	X_PWM6B PU control
[10]	pwm6a_pu	RW	0x0	X_PWM6A PU control
[9]	pwm5b_pu	RW	0x0	X_PWM5B PU control
[8]	pwm5a_pu	RW	0x0	X_PWM5A PU control
[7]	pwm4b_pu	RW	0x0	X_PWM4B PU control
[6]	pwm4a_pu	RW	0x0	X_PWM4A PU control
[5]	pwm3b_pu	RW	0x0	X_PWM3B PU control
[4]	pwm3a_pu	RW	0x0	X_PWM3A PU control
[3]	pwm2b_pu	RW	0x0	X_PWM2B PU control
[2]	pwm2a_pu	RW	0x0	X_PWM2A PU control
[1]	pwm1b_pu	RW	0x0	X_PWM1B PU control
[0]	pwm1a_pu	RW	0x0	X_PWM1A PU control

Table 0-101. Pad PU control (0x0C10_08D0)

Bit	Symbol	Access	Default	Description
[31:30]	-	RW	0x0	Reserved
[29]	sd1_wp_pu	RW	0x1	X_SD1_WP PU control

Bit	Symbol	Access	Default	Description
[28]	sd1_rstn_pu	RW	0x0	X_SD1_RSTN PU control
[27]	sd1_dat3_pu	RW	0x0	X_SD1_DAT3 PU control
[26]	sd1_dat2_pu	RW	0x0	X_SD1_DAT2 PU control
[25]	sd1_dat1_pu	RW	0x0	X_SD1_DAT1 PU control
[24]	sd1_dat0_pu	RW	0x0	X_SD1_DAT0 PU control
[23]	sd1_cmd_rsp_pu	RW	0x0	X_SD1_CMD_RSP PU control
[22]	sd1_clk_pu	RW	0x0	X_SD1_CLK PU control
[21]	sd1_cd_pu	RW	0x1	X_SD1_CD PU control
[20]	sd0_wp_pu	RW	0x1	X_SD0_WP PU control
[19]	sd0_rstn_pu	RW	0x0	X_SD0_RSTN PU control
[18]	sd0_dat3_pu	RW	0x0	X_SD0_DAT3 PU control
[17]	sd0_dat2_pu	RW	0x0	X_SD0_DAT2 PU control
[16]	sd0_dat1_pu	RW	0x0	X_SD0_DAT1 PU control
[15]	sd0_dat0_pu	RW	0x0	X_SD0_DAT0 PU control
[14]	sd0_cmd_rsp_pu	RW	0x0	X_SD0_CMD_RSP PU control
[13]	sd0_clk_pu	RW	0x0	X_SD0_CLK PU control
[12]	sd0_cd_pu	RW	0x1	X_SD0_CD PU control
[11]	p1_led1_pu	RW	0x0	X_P1_LED1 PU control
[10]	p1_led0_pu	RW	0x0	X_P1_LED0 PU control

Bit	Symbol	Access	Default	Description
[9]	p0_led1_pu	RW	0x0	X_P0_LED1 PU control
[8]	p0_led0_pu	RW	0x0	X_P0_LED0 PU control
[7]	i2c3_sda_pu	RW	0x0	X_I2C3_SDA PU control
[6]	i2c3_scl_pu	RW	0x0	X_I2C3_SCL PU control
[5]	i2c2_sda_pu	RW	0x0	X_I2C2_SDA PU control
[4]	i2c2_scl_pu	RW	0x0	X_I2C2_SCL PU control
[3]	i2c1_sda_pu	RW	0x0	X_I2C1_SDA PU control
[2]	i2c1_scl_pu	RW	0x0	X_I2C1_SCL PU control
[1]	i2c0_sda_pu	RW	0x0	X_I2C0_SDA PU control
[0]	i2c0_scl_pu	RW	0x0	X_I2C0_SCL PU control

Table 0-102. Pad PU control (0x0C10_08D4)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	uart0_tx_pu	RW	0x0	X_UART0_TX PU control
[29]	uart0_rx_pu	RW	0x1	X_UART0_RX PU control
[28]	uart0_rtsn_pu	RW	0x0	X_UART0_RTSN PU control
[27]	uart0_ctsn_pu	RW	0x1	X_UART0_CTSN PU control
[26]	u3_vbus_pu	RW	0x0	U3_VBUS PU control

Bit	Symbol	Access	Default	Description
[25]	u3_port_pwr_en_pu	RW	0x0	U3_PORT_PWR_EN PU control
[24]	u21_vbus_pu	RW	0x0	U21_VBUS PU control
[23]	u21_drvbus_pu	RW	0x0	U21_DRVBUS PU control
[22]	u20_vbus_pu	RW	0x0	U20_VBUS PU control
[21]	u20_drvbus_pu	RW	0x0	U20_DRVBUS PU control
[20]	spif_wp_n_pu	RW	0x0	X_SPIF_WP_N PU control
[19]	spif_tx_pu	RW	0x0	X_SPIF_TX PU control
[18]	spif_sck_pu	RW	0x0	X_SPIF_SCK PU control
[17]	spif_rx_pu	RW	0x0	X_SPIF_RX PU control
[16]	spif_hold_n_pu	RW	0x0	X_SPIF_HOLD_N PU control
[15]	spif_cs_n_pu	RW	0x0	X_SPIF_CS_N PU control
[14]	spi1_tx_pu	RW	0x0	X_SPI1_TX PU control
[13]	spi1_sclk_pu	RW	0x0	X_SPI1_SCLK PU control
[12]	spi1_rx_pu	RW	0x0	X_SPI1_RX PU control
[11]	spi1_fs_pu	RW	0x0	X_SPI1_FS PU control
[10]	spi1_fs1_pu	RW	0x0	X_SPI1_FS1 PU control
[9]	spi0_tx_pu	RW	0x0	X_SPI0_TX PU control
[8]	spi0_sclk_pu	RW	0x0	X_SPI0_SCLK PU control
[7]	spi0_rx_pu	RW	0x0	X_SPI0_RX PU control

Bit	Symbol	Access	Default	Description
[6]	spi0_fs_pu	RW	0x0	X_SPI0_FS PU control
[5]	spi0_fs1_pu	RW	0x0	X_SPI0_FS1 PU control
[4]	sec_cpu_trstn_pu	RW	0x1	X_SEC_CPU_TRSTN PU control
[3]	sec_cpu_tms_pu	RW	0x1	X_SEC_CPU_TMS PU control
[2]	sec_cpu_tdo_pu	RW	0x0	X_SEC_CPU_TDO PU control
[1]	sec_cpu_tdi_pu	RW	0x1	X_SEC_CPU_TDI PU control
[0]	sec_cpu_tck_pu	RW	0x1	X_SEC_CPU_TCK PU control

Table 0-103. Pad PU control (0x0C10_08D8)

Bit	Symbol	Access	Default	Description
[31:20]	-	RW	0x0	Reserved
[19]	pcie_wake_pu	RW	0x1	X_PCIE_WAKE PU control
[18]	pcie_perstn_pu	RW	0x0	X_PCIE_PERSTN PU control
[17]	pcie_clkreqn_pu	RW	0x1	X_PCIE_CLKREQN PU control
[16]	canfd1_tx_pu	RW	0x0	X_CANFD1_TX PU control
[15]	canfd1_rx_pu	RW	0x0	X_CANFD1_RX PU control
[14]	canfd0_tx_pu	RW	0x0	X_CANFD0_TX PU control
[13]	canfd0_rx_pu	RW	0x0	X_CANFD0_RX PU control
[12]	usart0_tx_pu	RW	0x0	X_USART0_TX PU control

Bit	Symbol	Access	Default	Description
[11]	usart0_rx_pu	RW	0x1	X_USART0_RX PU control
[10]	usart0_rtsn_pu	RW	0x0	X_USART0_RTSN PU control
[9]	usart0_ctsn_pu	RW	0x1	X_USART0_CTSN PU control
[8]	usart0_clk_pu	RW	0x0	X_USART0_CLK PU control
[7]	uart2_tx_pu	RW	0x0	X_UART2_TX PU control
[6]	uart2_rx_pu	RW	0x1	X_UART2_RX PU control
[5]	uart2_rtsn_pu	RW	0x0	X_UART2_RTSN PU control
[4]	uart2_ctsn_pu	RW	0x1	X_UART2_CTSN PU control
[3]	uart1_tx_pu	RW	0x0	X_UART1_TX PU control
[2]	uart1_rx_pu	RW	0x1	X_UART1_RX PU control
[1]	uart1_rtsn_pu	RW	0x0	X_UART1_RTSN PU control
[0]	uart1_ctsn_pu	RW	0x1	X_UART1_CTSN PU control

24.17 Pad PD control (0x0C10_08DC ~ 0x0C10_08FC)

Table 0-104. Pad PD control (0x0C10_08DC)

Bit	Symbol	Access	Default	Description
[31:30]	-	RW	0x0	Reserved
[29]	xmc_wen_pd	RW	0x0	X_XMC_WEN PD control
[28]	xmc_oen_pd	RW	0x0	X_XMC_OEN PD control

Bit	Symbol	Access	Default	Description
[27:24]	xmc_csn_pd	RW	0x0	X_XMC_CSN PD control
[23:21]	endat_tclk_pd	RW	0x0	X_ENDAT_TCLK PD control
[20:18]	endat_dout_pd	RW	0x0	X_ENDAT_DOUT PD control
[17:15]	endat_din_pd	RW	0x0	X_ENDAT_DIN PD control
[14:12]	endat_oe_pd	RW	0x0	X_ENDAT_OE PD control
[11]	ca53_trstn_pd	RW	0x0	X_CA53_TRSTN PD control
[10:7]	ca53_traced_pd	RW	0x0	X_CA53_TRACED PD control
[6]	ca53_tracectl_pd	RW	0x0	X_CA53_TRACECTL PD control
[5]	ca53_traceclk_pd	RW	0x0	X_CA53_TRACECLK PD control
[4]	ca53_tms_pd	RW	0x0	X_CA53_TMS PD control
[3]	ca53_tdo_pd	RW	0x0	X_CA53_TDO PD control
[2]	ca53_tdi_pd	RW	0x0	X_CA53_TDI PD control
[1]	ca53_tck_pd	RW	0x0	X_CA53_TCK PD control
[0]	boot_fail_pd	RW	0x0	X_BOOT_FAIL PD control

Table 0-105. Pad PD control (0x0C10_08E0)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	exadc1_sclk_pd	RW	0x0	X_EXADC1_SCLK PD control

Bit	Symbol	Access	Default	Description
[29]	exadc1_doutb_pd	RW	0x0	X_EXADC1_DOUTB PD control
[28]	exadc1_douta_pd	RW	0x0	X_EXADC1_DOUTA PD control
[27]	exadc1_csn_pd	RW	0x0	X_EXADC1_CSN PD control
[26:24]	exadc1_a_pd	RW	0x0	X_EXADC1_A PD control
[23:12]	eqep_aout_pd	RW	0x0	X_EQEP_AOUT PD control
[11:0]	eqep_ain_pd	RW	0x0	X_EQEP_AIN PD control

Table 0-106. Pad PD control (0x0C10_08E4)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	exadc2_sclk_pd	RW	0x0	X_EXADC2_SCLK PD control
[29]	exadc2_doutb_pd	RW	0x0	X_EXADC2_DOUTB PD control
[28]	exadc2_douta_pd	RW	0x0	X_EXADC2_DOUTA PD control
[27]	exadc2_csn_pd	RW	0x0	X_EXADC2_CSN PD control
[26:24]	exadc2_a_pd	RW	0x0	X_EXADC2_A PD control
[23:12]	eqep_bout_pd	RW	0x0	X_EQEP_BOUT PD control
[11:0]	eqep_bin_pd	RW	0x0	X_EQEP_BIN PD control

Table 0-107. Pad PD control (0x0C10_08E8)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	exadc3_sclk_pd	RW	0x0	X_EXADC3_SCLK PD control
[29]	exadc3_doutb_pd	RW	0x0	X_EXADC3_DOUTB PD control
[28]	exadc3_douta_pd	RW	0x0	X_EXADC3_DOUTA PD control
[27]	exadc3_csn_pd	RW	0x0	X_EXADC3_CSN PD control
[26:24]	exadc3_a_pd	RW	0x0	X_EXADC3_A PD control
[23:12]	eqep_s_pd	RW	0x0	X_EQEP_S PD control
[11:0]	eqep_i_pd	RW	0x0	X_EQEP_I PD control

Table 0-108. Pad PD control (0x0C10_08EC)

Bit	Symbol	Access	Default	Description
[31:16]	xmc_data_pd	RW	0x0	X_XMC_DATA PD control
[15:0]	xmc_addr_pd	RW	0x0	X_XMC_ADDR PD control

Table 0-109. Pad PD control (0x0C10_08F0)

Bit	Symbol	Access	Default	Description
[31:13]	gpio_pd	RW	0x0	X_GPIO[18:0] PD control
[12]	pwm_oe_pd	RW	0x0	X_PWM_OE PD control
[11]	pwm6b_pd	RW	0x0	X_PWM6B PD control

Bit	Symbol	Access	Default	Description
[10]	pwm6a_pd	RW	0x0	X_PWM6A PD control
[9]	pwm5b_pd	RW	0x0	X_PWM5B PD control
[8]	pwm5a_pd	RW	0x0	X_PWM5A PD control
[7]	pwm4b_pd	RW	0x0	X_PWM4B PD control
[6]	pwm4a_pd	RW	0x0	X_PWM4A PD control
[5]	pwm3b_pd	RW	0x0	X_PWM3B PD control
[4]	pwm3a_pd	RW	0x0	X_PWM3A PD control
[3]	pwm2b_pd	RW	0x0	X_PWM2B PD control
[2]	pwm2a_pd	RW	0x0	X_PWM2A PD control
[1]	pwm1b_pd	RW	0x0	X_PWM1B PD control
[0]	pwm1a_pd	RW	0x0	X_PWM1A PD control

Table 0-110. Pad PD control (0x0C10_08F4)

Bit	Symbol	Access	Default	Description
[31:30]	-	RW	0x0	Reserved
[29]	sd1_wp_pd	RW	0x0	X_SD1_WP PD control
[28]	sd1_rstn_pd	RW	0x0	X_SD1_RSTN PD control
[27]	sd1_dat3_pd	RW	0x0	X_SD1_DAT3 PD control
[26]	sd1_dat2_pd	RW	0x0	X_SD1_DAT2 PD control
[25]	sd1_dat1_pd	RW	0x0	X_SD1_DAT1 PD control

Bit	Symbol	Access	Default	Description
[24]	sd1_dat0_pd	RW	0x0	X_SD1_DAT0 PD control
[23]	sd1_cmd_rsp_pd	RW	0x0	X_SD1_CMD_RSP PD control
[22]	sd1_clk_pd	RW	0x0	X_SD1_CLK PD control
[21]	sd1_cd_pd	RW	0x0	X_SD1_CD PD control
[20]	sd0_wp_pd	RW	0x0	X_SD0_WP PD control
[19]	sd0_rstn_pd	RW	0x0	X_SD0_RSTN PD control
[18]	sd0_dat3_pd	RW	0x0	X_SD0_DAT3 PD control
[17]	sd0_dat2_pd	RW	0x0	X_SD0_DAT2 PD control
[16]	sd0_dat1_pd	RW	0x0	X_SD0_DAT1 PD control
[15]	sd0_dat0_pd	RW	0x0	X_SD0_DAT0 PD control
[14]	sd0_cmd_rsp_pd	RW	0x0	X_SD0_CMD_RSP PD control
[13]	sd0_clk_pd	RW	0x0	X_SD0_CLK PD control
[12]	sd0_cd_pd	RW	0x0	X_SD0_CD PD control
[11]	p1_led1_pd	RW	0x0	X_P1_LED1 PD control
[10]	p1_led0_pd	RW	0x0	X_P1_LED0 PD control
[9]	p0_led1_pd	RW	0x0	X_P0_LED1 PD control
[8]	p0_led0_pd	RW	0x0	X_P0_LED0 PD control
[7]	i2c3_sda_pd	RW	0x0	X_I2C3_SDA PD control
[6]	i2c3_scl_pd	RW	0x0	X_I2C3_SCL PD control

Bit	Symbol	Access	Default	Description
[5]	i2c2_sda_pd	RW	0x0	X_I2C2_SDA PD control
[4]	i2c2_scl_pd	RW	0x0	X_I2C2_SCL PD control
[3]	i2c1_sda_pd	RW	0x0	X_I2C1_SDA PD control
[2]	i2c1_scl_pd	RW	0x0	X_I2C1_SCL PD control
[1]	i2c0_sda_pd	RW	0x0	X_I2C0_SDA PD control
[0]	i2c0_scl_pd	RW	0x0	X_I2C0_SCL PD control

Table 0-111. Pad PD control (0x0C10_08F8)

Bit	Symbol	Access	Default	Description
[31]	-	RW	0x0	Reserved
[30]	uart0_tx_pd	RW	0x0	X_UART0_TX PD control
[29]	uart0_rx_pd	RW	0x0	X_UART0_RX PD control
[28]	uart0_rtsn_pd	RW	0x0	X_UART0_RTSN PD control
[27]	uart0_ctsn_pd	RW	0x0	X_UART0_CTSN PD control
[26]	u3_vbus_pd	RW	0x1	U3_VBUS PD control
[25]	u3_port_pwr_en_pd	RW	0x0	U3_PORT_PWR_EN PD control
[24]	u21_vbus_pd	RW	0x1	U21_VBUS PD control
[23]	u21_drvbus_pd	RW	0x0	U21_DRVBUS PD control
[22]	u20_vbus_pd	RW	0x1	U20_VBUS PD control

Bit	Symbol	Access	Default	Description
[21]	u20_drvbus_pd	RW	0x0	U20_DRVBUS PD control
[20]	spif_wp_n_pd	RW	0x0	X_SPIF_WP_N PD control
[19]	spif_tx_pd	RW	0x0	X_SPIF_TX PD control
[18]	spif_sck_pd	RW	0x0	X_SPIF_SCK PD control
[17]	spif_rx_pd	RW	0x0	X_SPIF_RX PD control
[16]	spif_hold_n_pd	RW	0x0	X_SPIF_HOLD_N PD control
[15]	spif_cs_n_pd	RW	0x0	X_SPIF_CS_N PD control
[14]	spi1_tx_pd	RW	0x0	X_SPI1_TX PD control
[13]	spi1_sclk_pd	RW	0x0	X_SPI1_SCLK PD control
[12]	spi1_rx_pd	RW	0x0	X_SPI1_RX PD control
[11]	spi1_fs_pd	RW	0x0	X_SPI1_FS PD control
[10]	spi1_fs1_pd	RW	0x0	X_SPI1_FS1 PD control
[9]	spi0_tx_pd	RW	0x0	X_SPI0_TX PD control
[8]	spi0_sclk_pd	RW	0x0	X_SPI0_SCLK PD control
[7]	spi0_rx_pd	RW	0x0	X_SPI0_RX PD control
[6]	spi0_fs_pd	RW	0x0	X_SPI0_FS PD control
[5]	spi0_fs1_pd	RW	0x0	X_SPI0_FS1 PD control
[4]	sec_cpu_trstn_pd	RW	0x0	X_SEC_CPU_TRSTN PD control
[3]	sec_cpu_tms_pd	RW	0x0	X_SEC_CPU_TMS PD control

Bit	Symbol	Access	Default	Description
[2]	sec_cpu_tdo_pd	RW	0x0	X_SEC_CPU_TDO PD control
[1]	sec_cpu_tdi_pd	RW	0x0	X_SEC_CPU_TDI PD control
[0]	sec_cpu_tck_pd	RW	0x0	X_SEC_CPU_TCK PD control

Table 0-112. Pad PD control (0x0C10_08FC)

Bit	Symbol	Access	Default	Description
[31:20]	-	RW	0x0	Reserved
[19]	pcie_wake_pd	RW	0x0	X_PCIE_WAKE PD control
[18]	pcie_perstn_pd	RW	0x0	X_PCIE_PERSTN PD control
[17]	pcie_clkreqn_pd	RW	0x0	X_PCIE_CLKREQN PD control
[16]	canfd1_tx_pd	RW	0x0	X_CANFD1_TX PD control
[15]	canfd1_rx_pd	RW	0x0	X_CANFD1_RX PD control
[14]	canfd0_tx_pd	RW	0x0	X_CANFD0_TX PD control
[13]	canfd0_rx_pd	RW	0x0	X_CANFD0_RX PD control
[12]	usart0_tx_pd	RW	0x0	X_USART0_TX PD control
[11]	usart0_rx_pd	RW	0x0	X_USART0_RX PD control
[10]	usart0_rtsn_pd	RW	0x0	X_USART0_RTSN PD control
[9]	usart0_ctsn_pd	RW	0x0	X_USART0_CTSN PD control
[8]	usart0_clk_pd	RW	0x0	X_USART0_CLK PD control

Bit	Symbol	Access	Default	Description
[7]	uart2_tx_pd	RW	0x0	X_UART2_TX PD control
[6]	uart2_rx_pd	RW	0x0	X_UART2_RX PD control
[5]	uart2_rtsn_pd	RW	0x0	X_UART2_RTSN PD control
[4]	uart2_ctsn_pd	RW	0x0	X_UART2_CTSN PD control
[3]	uart1_tx_pd	RW	0x0	X_UART1_TX PD control
[2]	uart1_rx_pd	RW	0x0	X_UART1_RX PD control
[1]	uart1_rtsn_pd	RW	0x0	X_UART1_RTSN PD control
[0]	uart1_ctsn_pd	RW	0x0	X_UART1_CTSN PD control

24.18 FTSCU100 Software reset mask control (0x0C10_0920 ~ 0x0C10_092C)

Table 0-113. FTSCU100 Software reset mask control (0x0C10_0920)

Bit	Symbol	Access	Default	Description
[31]	swrst_msk[31:0]	RW	0x0	u_SBS_GMAC_1 aresetn software reset mask in axi-clk domain 1'b1 : mask scu_swrst request
[30]		RW	0x0	u_SBS_GMAC aresetn software reset mask in axi-clk domain
[29]		RW	0x0	u_PCIE_SBS dbi_aresetn software reset mask in axi-clk domain
[28]		RW	0x0	u_PCIE_SBS slv_aresetn software reset mask in axi-clk domain
[27]		RW	0x0	u_PCIE_SBS mstr_aresetn software reset mask in axi-clk domain
[26]		RW	0x0	u_GIC400 nRESET software reset mask in axi-clk domain

Bit	Symbol	Access	Default	Description
[25]		RW	0x0	u_FTX2X030 AResetnM software reset mask in axi-clk domain
[24]		RW	0x0	u_FTX2X030 AResetnS software reset mask in axi-clk domain
[23]		RW	0x0	u_FTX2P030_1 aresetn software reset mask in axi-clk domain
[22]		RW	0x0	u_FTX2P030 aresetn software reset mask in axi-clk domain
[21]		RW	0x0	u_FTSPI020 aresetn software reset mask in axi-clk domain
[20]		RW	0x0	u_FTEMC030_1 AResetn software reset mask in axi-clk domain
[19]		RW	0x0	u_FTEMC030 AResetn software reset mask in axi-clk domain
[18]		RW	0x0	u_FTDMAC030_1 AResetn software reset mask in axi-clk domain
[17]		RW	0x0	u_FTDMAC030 AResetn software reset mask in axi-clk domain
[16]		RW	0x0	DDR sub-system aresetn_3 software reset mask
[15]		RW	0x0	DDR sub-system aresetn_2 software reset mask
[14]		RW	0x0	DDR sub-system aresetn_1 software reset mask
[13]		RW	0x0	DDR sub-system aresetn_0 software reset mask
[12]		RW	0x0	DDR sub-system presetn software reset mask
[11]		RW	0x0	DDR sub-system core_ddrc_rstn software reset mask
[10]		RW	0x0	DDR sub-system Reset software reset mask
[9]		RW	0x0	DDR sub-system top__PwrOkIn software reset mask
[8]		RW	0x0	CA53 sub-system nL2RESET software reset mask
[7]		RW	0x0	CA53 sub-system presetn software reset mask

Bit	Symbol	Access	Default	Description
[6]		RW	0x0	CA53 sub-system nPRESETDBG software reset mask
[5]		RW	0x0	CA53 sub-system presetdbgn software reset mask
[4]		RW	0x0	CA53 sub-system tresetn software reset mask
[3]		RW	0x0	CA53 sub-system nCORERESET[3] software reset mask
[2]		RW	0x0	CA53 sub-system nCORERESET[2] software reset mask
[1]		RW	0x0	CA53 sub-system nCORERESET[1] software reset mask
[0]		RW	0x0	CA53 sub-system nCORERESET[0] software reset mask

Table 0-114. FTSCU100 Software reset mask control (0x0C10_0924)

Bit	Symbol	Access	Default	Description
[31]	swrst_msk[63:32]	RW	0x0	u_FTSSP010_1 presetn software reset mask in apb-clk domain 1'b1 : mask scu_swrst request
[30]		RW	0x0	u_FTSSP010 presetn software reset mask in apb-clk domain
[29]		RW	0x0	u_FTPWMTMR010_1 presetn software reset mask in apb-clk domain
[28]		RW	0x0	u_FTPWMTMR010 presetn software reset mask in apb-clk domain
[27]		RW	0x0	u_FTIIC010_3 PRESETn software reset mask in apb-clk domain
[26]		RW	0x0	u_FTIIC010_2 PRESETn software reset mask in apb-clk domain
[25]		RW	0x0	u_FTIIC010_1 PRESETn software reset mask in apb-clk domain

Bit	Symbol	Access	Default	Description
[24]		RW	0x0	u_FTIIC010 PRESETn software reset mask in apb-clk domain
[23]		RW	0x0	u_FTGPIO010 PRSTn software reset mask in apb-clk domain
[22]		RW	0x0	u_FTEMC030_1 PResetn software reset mask in apb-clk domain
[21]		RW	0x0	u_FTEMC030 PResetn software reset mask in apb-clk domain
[20]		RW	0x0	u_FTDMAC030_1 PResetn software reset mask in apb-clk domain
[19]		RW	0x0	u_FTDMAC030 PResetn software reset mask in apb-clk domain
[18]		RW	0x0	u_DAC_TOP_3 RESET_N software reset mask in apb-clk domain
[17]		RW	0x0	u_DAC_TOP_2 RESET_N software reset mask in apb-clk domain
[16]		RW	0x0	u_DAC_TOP_1 RESET_N software reset mask in apb-clk domain
[15]		RW	0x0	u_DAC_TOP RESET_N software reset mask in apb-clk domain
[14]		RW	0x0	u_ADC_TOP preset_n software reset mask in apb-clk domain
[13]		RW	0x0	u_nic400_UniMap hresetn software reset mask in ahb-clk domain
[12]		RW	0x0	u_VIVANTE_GPU HRESETn software reset mask in ahb-clk domain
[11]		RW	0x0	u_SEC_SUBSYS hrstn software reset mask in ahb-clk domain
[10]		RW	0x0	u_SBS_CAN_FD_2 hresetn software reset mask in ahb-clk domain
[9]		RW	0x0	u_SBS_CAN_FD_1 hresetn software reset mask in ahb-clk domain
[8]		RW	0x0	u_SBS_CAN_FD hresetn software reset mask in ahb-clk domain
[7]		RW	0x0	u_FTSDC021_1 hresetn software reset mask in ahb-clk domain
[6]		RW	0x0	u_FTSDC021 hresetn software reset mask in ahb-clk domain

Bit	Symbol	Access	Default	Description
[5]		RW	0x0	u_nic400_UniMap aresetn software reset mask in axi-clk domain
[4]		RW	0x0	u_VIVANTE_GPU ARESETn software reset mask in axi-clk domain
[3]		RW	0x0	u_SBS_USB3_TOP hwrst_n software reset mask in axi-clk domain
[2]		RW	0x0	u_SBS_OTG2_TOP_1 rst_n software reset mask in axi-clk domain
[1]		RW	0x0	u_SBS_OTG2_TOP rst_n software reset mask in axi-clk domain
[0]		RW	0x0	u_SBS_LVDS_TX aresetn software reset mask in axi-clk domain

Table 0-115. FTSCU100 Software reset mask control (0x0C10_0928)

Bit	Symbol	Access	Default	Description
[31]	swrst_msk[95:64]	RW	0x0	u_SBS_LVDS_TX LC_SCALER_RSTn software reset mask 1'b1 : mask scu_swrst request
[30]		RW	0x0	u_SBS_LVDS_TX LC_RSTn software reset mask
[29]		RW	0x0	u_FTUSART010_1 usart_rst_n software reset mask
[28]		RW	0x0	u_FTUSART010 usart_rst_n software reset mask
[27]		RW	0x0	u_ADC_TOP adc_rst_n software reset mask
[26]		RW	0x0	u_TDC_TOP presetn software reset mask in apb-clk domain
[25]		RW	0x0	u_SBS_USB3_TOP APB_RST_N software reset mask in apb-clk domain
[24]		RW	0x0	u_SBS_OTG2_TOP_1 presetn software reset mask in apb-clk

Bit	Symbol	Access	Default	Description
				domain
[23]		RW	0x0	u_SBS_OTG2_TOP presetn software reset mask in apb-clk domain
[22]		RW	0x0	u_SBS_LVDS_TX presetn software reset mask in apb-clk domain
[21]		RW	0x0	u_SBS_GMAC_1 presetn_phy software reset mask in apb-clk domain
[20]		RW	0x0	u_SBS_GMAC presetn_phy software reset mask in apb-clk domain
[19]		RW	0x0	u_SBS_GMAC_1 presetn software reset mask in apb-clk domain
[18]		RW	0x0	u_SBS_GMAC presetn software reset mask in apb-clk domain
[17]		RW	0x0	u_PCIE_SBS I_APB_RSTN software reset mask in apb-clk domain
[16]		RW	0x0	u_PCIE_SBS perst_n software reset mask in apb-clk domain
[15]		RW	0x0	u_PCIE_EXTREG presetn software reset mask in apb-clk domain
[14]		RW	0x0	reserved
[13]		RW	0x0	u_FTX2X030 PResetn software reset mask in apb-clk domain
[12]		RW	0x0	u_FTX2P030_1 presetn software reset mask in apb-clk domain
[11]		RW	0x0	u_FTX2P030 presetn software reset mask in apb-clk domain
[10]		RW	0x0	u_FTWDT011_1 PRESET_N software reset mask in apb-clk domain
[9]		RW	0x0	u_FTWDT011 PRESET_N software reset mask in apb-clk domain
[8]		RW	0x0	u_FTUSART010_1 preset_n software reset mask in apb-clk domain
[7]		RW	0x0	u_FTUSART010 preset_n software reset mask in apb-clk domain

Bit	Symbol	Access	Default	Description
[6]		RW	0x0	u_FTUART010_3 PRSTn software reset mask in apb-clk domain
[5]		RW	0x0	u_FTUART010_2 PRSTn software reset mask in apb-clk domain
[4]		RW	0x0	u_FTUART010_1 PRSTn software reset mask in apb-clk domain
[3]		RW	0x0	u_FTUART010 PRSTn software reset mask in apb-clk domain
[2]		RW	0x0	u_FTTMR010 presetn software reset mask in apb-clk domain
[1]		RW	0x0	u_FTSSP010_3 presetn software reset mask in apb-clk domain
[0]		RW	0x0	u_FTSSP010_2 presetn software reset mask in apb-clk domain

Table 0-116. FTSCU100 Software reset mask control (0x0C10_092C)

Bit	Symbol	Access	Default	Description
[31:14]		RW	0x0	Reserved
[13]		RW	0x0	u_SBS_USB3_TOP PONRST software reset mask 1'b1 : mask scu_swrst request
[12]		RW	0x0	u_SBS_OTG2_TOP_1 PONRST software reset mask
[11]	swrst_msk[127:96]	RW	0x0	u_SBS_OTG2_TOP PONRST software reset mask
[10]		RW	0x0	u_Motor_Top EnDat_rstn software reset mask
[9]		RW	0x0	u_Motor_Top ePWM_rstn software reset mask
[8]		RW	0x0	u_Motor_Top exADC_presetn software reset mask
[7]		RW	0x0	u_Motor_Top eQep_PRESETn software reset mask

Bit	Symbol	Access	Default	Description
[6]		RW	0x0	u_Motor_Top ePWM_prstn software reset mask
[5]		RW	0x0	u_Motor_Top XMC_Ctl_PRESETn software reset mask
[4]		RW	0x0	u_nic400_UniMap usb3_aresetn software reset mask
[3]		RW	0x0	u_nic400_UniMap cpu_aresetn software reset mask
[2]		RW	0x0	u_VIVANTE_GPU resetPin_ software reset mask
[1]		RW	0x0	u_SBS_LVDS_TX p2lvds_rstn software reset mask
[0]		RW	0x0	u_SBS_LVDS_TX TV_RSTn software reset mask

24.19 FTSCU100 reboot mask control (0x0C10_0930 ~ 0x0C10_093C)

Table 0-117. FTSCU100 Reboot mask control (0x0C10_0930)

Bit	Symbol	Access	Default	Description
[31]	reboot_msk [31:0]	RW	0x0	u_SBS_GMAC_1 aresetn reboot mask in axi-clk domain 1'b1 : mask scu_reboot request
[30]		RW	0x0	u_SBS_GMAC aresetn reboot mask in axi-clk domain
[29]		RW	0x0	u_PCIE_SBS dbi_aresetn reboot mask in axi-clk domain
[28]		RW	0x0	u_PCIE_SBS slv_aresetn reboot mask in axi-clk domain
[27]		RW	0x0	u_PCIE_SBS mstr_aresetn reboot mask in axi-clk domain
[26]		RW	0x0	u_GIC400 nRESET reboot mask in axi-clk domain
[25]		RW	0x0	u_FTX2X030 AResetnM reboot mask in axi-clk domain

Bit	Symbol	Access	Default	Description
[24]		RW	0x0	u_FTX2X030 AResetnS reboot mask in axi-clk domain
[23]		RW	0x0	u_FTX2P030_1 aresetn reboot mask in axi-clk domain
[22]		RW	0x0	u_FTX2P030 aresetn reboot mask in axi-clk domain
[21]		RW	0x0	u_FTSPI020 aresetn reboot mask in axi-clk domain
[20]		RW	0x0	u_FTEMC030_1 AResetn reboot mask in axi-clk domain
[19]		RW	0x0	u_FTEMC030 AResetn reboot mask in axi-clk domain
[18]		RW	0x0	u_FTDMAC030_1 AResetn reboot mask in axi-clk domain
[17]		RW	0x0	u_FTDMAC030 AResetn reboot mask in axi-clk domain
[16]		RW	0x0	DDR sub-system aresetn_3 reboot mask
[15]		RW	0x0	DDR sub-system aresetn_2 reboot mask
[14]		RW	0x0	DDR sub-system aresetn_1 reboot mask
[13]		RW	0x0	DDR sub-system aresetn_0 reboot mask
[12]		RW	0x0	DDR sub-system presetn reboot mask
[11]		RW	0x0	DDR sub-system core_ddrc_rstn reboot mask
[10]		RW	0x0	DDR sub-system Reset reboot mask
[9]		RW	0x0	DDR sub-system top__PwrOkIn reboot mask
[8]		RW	0x0	CA53 sub-system nL2RESET reboot mask
[7]		RW	0x0	CA53 sub-system presetn reboot mask
[6]		RW	0x0	CA53 sub-system nPRESETDBG reboot mask

Bit	Symbol	Access	Default	Description
[5]		RW	0x0	CA53 sub-system presetdbgn reboot mask
[4]		RW	0x0	CA53 sub-system tresetn reboot mask
[3]		RW	0x0	CA53 sub-system nCORERESET[3] reboot mask
[2]		RW	0x0	CA53 sub-system nCORERESET[2] reboot mask
[1]		RW	0x0	CA53 sub-system nCORERESET[1] reboot mask
[0]		RW	0x0	CA53 sub-system nCORERESET[0] reboot mask

Table 0-118. FTSCU100 Reboot mask control (0x0C10_0934)

Bit	Symbol	Access	Default	Description
[31]	reboot_msk [63:32]	RW	0x0	u_FTSSP010_1 presetn reboot mask in apb-clk domain 1'b1 : mask scu_reboot request
[30]		RW	0x0	u_FTSSP010 presetn reboot mask in apb-clk domain
[29]		RW	0x0	u_FTPWMTMR010_1 presetn reboot mask in apb-clk domain
[28]		RW	0x0	u_FTPWMTMR010 presetn reboot mask in apb-clk domain
[27]		RW	0x0	u_FTIIC010_3 PRESETn reboot mask in apb-clk domain
[26]		RW	0x0	u_FTIIC010_2 PRESETn reboot mask in apb-clk domain
[25]		RW	0x0	u_FTIIC010_1 PRESETn reboot mask in apb-clk domain
[24]		RW	0x0	u_FTIIC010 PRESETn reboot mask in apb-clk domain
[23]		RW	0x0	u_FTGPIO010 PRSTn reboot mask in apb-clk domain

Bit	Symbol	Access	Default	Description
[22]		RW	0x0	u_FTEMC030_1 PResetn reboot mask in apb-clk domain
[21]		RW	0x0	u_FTEMC030 PResetn reboot mask in apb-clk domain
[20]		RW	0x0	u_FTDMAC030_1 PResetn reboot mask in apb-clk domain
[19]		RW	0x0	u_FTDMAC030 PResetn reboot mask in apb-clk domain
[18]		RW	0x0	u_DAC_TOP_3 RESET_N reboot mask in apb-clk domain
[17]		RW	0x0	u_DAC_TOP_2 RESET_N reboot mask in apb-clk domain
[16]		RW	0x0	u_DAC_TOP_1 RESET_N reboot mask in apb-clk domain
[15]		RW	0x0	u_DAC_TOP RESET_N reboot mask in apb-clk domain
[14]		RW	0x0	u_ADC_TOP preset_n reboot mask in apb-clk domain
[13]		RW	0x0	u_nic400_UniMap hresetn reboot mask in ahb-clk domain
[12]		RW	0x0	u_VIVANTE_GPU HRESETn reboot mask in ahb-clk domain
[11]		RW	0x0	u_SEC_SUBSYS hrstn reboot mask in ahb-clk domain
[10]		RW	0x0	u_SBS_CAN_FD_2 hresetn reboot mask in ahb-clk domain
[9]		RW	0x0	u_SBS_CAN_FD_1 hresetn reboot mask in ahb-clk domain
[8]		RW	0x0	u_SBS_CAN_FD hresetn reboot mask in ahb-clk domain
[7]		RW	0x0	u_FTSDC021_1 hresetn reboot mask in ahb-clk domain
[6]		RW	0x0	u_FTSDC021 hresetn reboot mask in ahb-clk domain
[5]		RW	0x0	u_nic400_UniMap aresetn reboot mask in axi-clk domain
[4]		RW	0x0	u_VIVANTE_GPU ARESETn reboot mask in axi-clk domain

Bit	Symbol	Access	Default	Description
[3]		RW	0x0	u_SBS_USB3_TOP hwrst_n reboot mask in axi-clk domain
[2]		RW	0x0	u_SBS_OTG2_TOP_1 rst_n reboot mask in axi-clk domain
[1]		RW	0x0	u_SBS_OTG2_TOP rst_n reboot mask in axi-clk domain
[0]		RW	0x0	u_SBS_LVDS_TX aresetn reboot mask in axi-clk domain

Table 0-119. FTSCU100 Reboot mask control (0x0C10_0938)

Bit	Symbol	Access	Default	Description
[31]	reboot_msk [95:64]	RW	0x0	u_SBS_LVDS_TX LC_SCALER_RSTn reboot mask 1'b1 : mask scu_reboot request
[30]		RW	0x0	u_SBS_LVDS_TX LC_RSTn reboot mask
[29]		RW	0x0	u_FTUSART010_1 usart_rst_n reboot mask
[28]		RW	0x0	u_FTUSART010 usart_rst_n reboot mask
[27]		RW	0x0	u_ADC_TOP adc_rst_n reboot mask
[26]		RW	0x0	u_TDC_TOP presetn reboot mask in apb-clk domain
[25]		RW	0x0	u_SBS_USB3_TOP APB_RST_N reboot mask in apb-clk domain
[24]		RW	0x0	u_SBS_OTG2_TOP_1 presetn reboot mask in apb-clk domain
[23]		RW	0x0	u_SBS_OTG2_TOP presetn reboot mask in apb-clk domain
[22]		RW	0x0	u_SBS_LVDS_TX presetn reboot mask in apb-clk domain
[21]		RW	0x0	u_SBS_GMAC_1 presetn_phy reboot mask in apb-clk domain

Bit	Symbol	Access	Default	Description
[20]		RW	0x0	u_SBS_GMAC presetn_phy reboot mask in apb-clk domain
[19]		RW	0x0	u_SBS_GMAC_1 presetn reboot mask in apb-clk domain
[18]		RW	0x0	u_SBS_GMAC presetn reboot mask in apb-clk domain
[17]		RW	0x0	u_PCIE_SBS I_APB_RSTN reboot mask in apb-clk domain
[16]		RW	0x0	u_PCIE_SBS perst_n reboot mask in apb-clk domain
[15]		RW	0x0	u_PCIE_EXTREG presetn reboot mask in apb-clk domain
[14]		RW	0x0	reserved
[13]		RW	0x0	u_FTX2X030 PResetn reboot mask in apb-clk domain
[12]		RW	0x0	u_FTX2P030_1 presetn reboot mask in apb-clk domain
[11]		RW	0x0	u_FTX2P030 presetn reboot mask in apb-clk domain
[10]		RW	0x0	u_FTWDT011_1 PRESET_N reboot mask in apb-clk domain
[9]		RW	0x0	u_FTWDT011 PRESET_N reboot mask in apb-clk domain
[8]		RW	0x0	u_FTUSART010_1 preset_n reboot mask in apb-clk domain
[7]		RW	0x0	u_FTUSART010 preset_n reboot mask in apb-clk domain
[6]		RW	0x0	u_FTUART010_3 PRSTn reboot mask in apb-clk domain
[5]		RW	0x0	u_FTUART010_2 PRSTn reboot mask in apb-clk domain
[4]		RW	0x0	u_FTUART010_1 PRSTn reboot mask in apb-clk domain
[3]		RW	0x0	u_FTUART010 PRSTn reboot mask in apb-clk domain
[2]		RW	0x0	u_FTTMR010 presetn reboot mask in apb-clk domain

Bit	Symbol	Access	Default	Description
[1]		RW	0x0	u_FTSSP010_3 presetn reboot mask in apb-clk domain
[0]		RW	0x0	u_FTSSP010_2 presetn reboot mask in apb-clk domain

Table 0-120. FTSCU100 Reboot mask control (0x0C10_093C)

Bit	Symbol	Access	Default	Description
[31:14]	reboot_msk [127:96]	RW	0x0	Reserved
[13]		RW	0x0	u_SBS_USB3_TOP PONRST reboot mask 1'b1 : mask scu_reboot request
[12]		RW	0x0	u_SBS_OTG2_TOP_1 PONRST reboot mask
[11]		RW	0x0	u_SBS_OTG2_TOP PONRST reboot mask
[10]		RW	0x0	u_Motor_Top EnDat_rstn reboot mask
[9]		RW	0x0	u_Motor_Top ePWM_rstn reboot mask
[8]		RW	0x0	u_Motor_Top exADC_presetn reboot mask
[7]		RW	0x0	u_Motor_Top eQep_PRESETn reboot mask
[6]		RW	0x0	u_Motor_Top ePWM_prstn reboot mask
[5]		RW	0x0	u_Motor_Top XMC_Ctl_PRESETn reboot mask
[4]		RW	0x0	u_nic400_UniMap usb3_aresetn reboot mask
[3]		RW	0x0	u_nic400_UniMap cpu_aresetn reboot mask
[2]		RW	0x0	u_VIVANTE_GPU resetPin_ reboot mask

Bit	Symbol	Access	Default	Description
[1]		RW	0x0	u_SBS_LVDS_TX p2lvds_rstn reboot mask
[0]		RW	0x0	u_SBS_LVDS_TX TV_RSTn reboot mask

24.20 DDR reset control (0x0C10_0940)

Table 0-121. DDR reset control (0x0C10_0940)

Bit	Symbol	Access	Default	Description
[7]	ddr_rstn_ctrl[7:0]	RW	0x0	DDR sub-system aresetn_3 control 1'b1 : release control of aresetn_3, and it would be controlled by SCU reboot or SW reset function 1'b0 : set aresetn_3 == 1'b0
[6]		RW	0x0	DDR sub-system aresetn_2 control 1'b1 : release control of aresetn_2, and it would be controlled by SCU reboot or SW reset function 1'b0 : set aresetn_2 == 1'b0
[5]		RW	0x0	DDR sub-system aresetn_1 control 1'b1 : release control of aresetn_1, and it would be controlled by SCU reboot or SW reset function 1'b0 : set aresetn_1 == 1'b0
[4]		RW	0x0	DDR sub-system aresetn_0 control 1'b1 : release control of aresetn_0, and it would be controlled by SCU reboot or SW reset function

Bit	Symbol	Access	Default	Description
				1'b0 : set aresetn_0 == 1'b0
[3]		RW	0x0	DDR sub-system presetn control 1'b1 : release control of presetn, and it would be controlled by SCU reboot or SW reset function 1'b0 : set presetn == 1'b0
[2]		RW	0x0	DDR sub-system core_ddrc_rstn control 1'b1 : release control of core_ddrc_rstn, and it would be controlled by SCU reboot or SW reset function 1'b0 : set core_ddrc_rstn == 1'b0
[1]		RW	0x0	DDR sub-system Reset control 1'b1 : release control of Reset, and it would be controlled by SCU reboot or SW reset function 1'b0 : set Reset == 1'b0
[0]		RW	0x0	DDR sub-system top__PwrOkIn control 1'b1 : release control of top__PwrOkIn, and it would be controlled by SCU reboot or SW reset function 1'b0 : set top__PwrOkIn == 1'b0

24.21 WFI control (0x0C10_0944)

Table 0-122. WFI control (0x0C10_0944)

Bit	Symbol	Access	Default	Description
[31:24]	wfi_edge_sync_delay	RW	0x18	edge sync circuit delay counter
[23:21]	-	RW	0x0	reserved
[20]	wfi_L2_mask	RW	0x0	1'b1 : force CA53 L2 WFI signal =1'b1 , 1'b0 : use CA53 L2 original WFI
[19:16]	wfi_mask	RW	0x0	User should use wfi_mask to mask WFI signal when CPU_SPEED command is asserted. [0] : 1'b1 : force CA53 core0 WFI signal =1'b1 , 1'b0 : use CA53 core0 original WFI [1] : 1'b1 : force CA53 core1 WFI signal =1'b1 , 1'b0 : use CA53 core1 original WFI [2] : 1'b1 : force CA53 core2 WFI signal =1'b1 , 1'b0 : use CA53 core2 original WFI [3] : 1'b1 : force CA53 core3 WFI signal =1'b1 , 1'b0 : use CA53 core3 original WFI
[15:10]	-	RW	0x0	reserved
[9]	ACINACTM_force_clr	RW	0x0	force ACINACTM = 1'b0
[8]	ACINACTM_force_set	RW	0x0	force ACINACTM = 1'b1
[7:0]	ACINACTM_delay_cnt	RW	0x20	CA53 ACINACTM delay counter control

24.22 IP MSB address extension (0x0C10_0948)

Table 0-123. IP MSB address extension (0x0C10_0948)

Bit	Symbol	Access	Default	Description
[31:22]	-	RW	0x0	Reserved
[21:20]	IP_addr_msb[21:0]	RW	0x0	ARAddrM_u_SBS_LVDX_TX[33:32] address extension, from LCDC to DDR
[19:18]		RW	0x0	ARADDR_u_VIVANTE_GPU[33:32] address extension, from GPU to DDR
[17:16]		RW	0x0	AWADDR_u_VIVANTE_GPU[33:32] address extension, from GPU to DDR
[15:14]		RW	0x0	haddr_m12_sdc_1[33:32] address extension, from SDC_1 master to NIC400.
[13:12]		RW	0x0	haddr_m11_sdc_0[33:32] address extension, from SDC_0 master to NIC400.
[11:10]		RW	0x0	araddr_m3_usb2_1[33:32] address extension, from USB2_1 master to NIC400.
[9:8]		RW	0x0	awaddr_m3_usb2_1[33:32] address extension, from USB2_1 master to NIC400.
[7:6]		RW	0x0	araddr_m2_usb2_0[33:32] address extension, from USB2_0 master to NIC400.
[5:4]		RW	0x0	awaddr_m2_usb2_0[33:32] address extension, from USB2_0 master to NIC400.

Bit	Symbol	Access	Default	Description
[3:2]		RW	0x0	araddr_m1_usb3[33:32] address extension, from USB3 master to NIC400.
[1:0]		RW	0x0	awaddr_m1_usb3[33:32] address extension, from USB3 master to NIC400.

24.23 Pad DCSR control (0x0C10_094C ~ 0x0C10_0958)

Table 0-124. Pad dcsr control (0x0C10_094C)

Bit	Symbol	Access	Default	Description
[31:28]	i2c2_dcsr	RW	0x0	dcsr[2:1] : output driving control (3.3V/1.8V) 2'b00 : 4 mA / 3 mA 2'b01 : 8 mA / 6 mA 2'b10 : 12 mA / 9 mA 2'b11 : 16 mA / 12 mA dcsr[3] : outptu slew rate control 1'b0 : fast 1'b1 : slow
[27:24]	i2c1_dcsr	RW	0x0	same as above
[23:20]	i2c0_dcsr	RW	0x0	same as above
[19:16]	gpio_dcsr	RW	0x0	same as above
[15:12]	gmac_dcsr	RW	0x6	same as above

Bit	Symbol	Access	Default	Description
[11:8]	gmac1_dcsr	RW	0x6	same as above
[7:4]	can_fd_dcsr	RW	0x0	same as above
[3:0]	can_fd1_dcsr	RW	0x0	same as above

Table 0-125. Pad dcsr control (0x0C10_0950)

Bit	Symbol	Access	Default	Description
[31:28]	spi0_dcsr	RW	0x0	dcsr[2:1] : output driving control (3.3V/1.8V) 2'b00 : 4 mA / 3 mA 2'b01 : 8 mA / 6 mA 2'b10 : 12 mA / 9 mA 2'b11 : 16 mA / 12 mA dcsr[3] : outptu slew rate control 1'b0 : fast 1'b1 : slow
[27:24]	soteria_dcsr	RW	0x0	same as above
[23:20]	sd1_dcsr	RW	0x0	EMMC output driving reference to 0x0c10_0858[26:24] dcsr[3] : outptu slew rate control 1'b0 : fast 1'b1 : slow
[19:16]	sd0_dcsr	RW	0x0	same as spi0_dcsr
[15:12]	pcie_dcsr	RW	0x0	same as spi0_dcsr

Bit	Symbol	Access	Default	Description
[11:8]	misc_dcsr	RW	0x6	same as spi0_dcsr
[7:4]	jtag_dcsr	RW	0x0	same as spi0_dcsr
[3:0]	i2c3_dcsr	RW	0x0	same as spi0_dcsr

Table 0-126. Pad dcsr control (0x0C10_0954)

Bit	Symbol	Access	Default	Description
[31:28]	uart2_dcsr	RW	0x0	dcsr[2:1] : output driving control (3.3V/1.8V) 2'b00 : 4 mA / 3 mA 2'b01 : 8 mA / 6 mA 2'b10 : 12 mA / 9 mA 2'b11 : 16 mA / 12 mA dcsr[3] : outptu slew rate control 1'b0 : fast 1'b1 : slow
[27:24]	uart1_dcsr	RW	0x0	same as above
[23:20]	uart0_dcsr	RW	0x0	same as above
[19:16]	u2_1_dcsr	RW	0x0	same as above
[15:12]	u2_0_dcsr	RW	0x0	same as above
[11:8]	trace_dcsr	RW	0x0	same as above
[7:4]	spi_dcsr	RW	0x0	same as above

Bit	Symbol	Access	Default	Description
[3:0]	spi1_dcsr	RW	0x0	same as above

Table 0-127. Pad dcsr control (0x0C10_0958)

Bit	Symbol	Access	Default	Description
[31:28]	-	RW	0x0	Reserved
[27:24]	motor_endat_dcsr	RW	0x0	dcsr[2:1] : output driving control (3.3V/1.8V) 2'b00 : 4 mA / 3 mA 2'b01 : 8 mA / 6 mA 2'b10 : 12 mA / 9 mA 2'b11 : 16 mA / 12 mA dcsr[3] : outptu slew rate control 1'b0 : fast 1'b1 : slow
[23:20]	motor_pwm_dcsr	RW	0x0	same as above
[19:16]	motor_exadc_dcsr	RW	0x0	same as above
[15:12]	motor_eqep_dcsr	RW	0x0	same as above
[11:8]	motor_xmc_dcsr	RW	0x0	same as above
[7:4]	usb3_dcsr	RW	0x0	same as above
[3:0]	usart0_dcsr	RW	0x0	same as above

24.24 DDR mode register read (0x0C10_095C ~ 0x0C10_0978)

Table 0-128. DDR mode register read (0x0C10_095C ~ 0x0C10_0978)

Bit	Symbol	Access	Default	Description
[31:0]	rd_mrr_data[31:0]	R	0x0	DDR mode register read data, (0x0C10_095C) valid when 0x82c[2] == 1'b1
[31:0]	rd_mrr_data[63:32]	R	0x0	DDR mode register read data, (0x0C10_0960)
[31:0]	rd_mrr_data[95:64]	R	0x0	DDR mode register read data, (0x0C10_0964)
[31:0]	rd_mrr_data[127:96]	R	0x0	DDR mode register read data, (0x0C10_0968)
[31:0]	rd_mrr_data[159:128]	R	0x0	DDR mode register read data, (0x0C10_096C)
[31:0]	rd_mrr_data[191:160]	R	0x0	DDR mode register read data, (0x0C10_0970)
[31:0]	rd_mrr_data[223:192]	R	0x0	DDR mode register read data, (0x0C10_0974)
[31:0]	rd_mrr_data[255:224]	R	0x0	DDR mode register read data, (0x0C10_0978)

25. PCIE External Control register

Synopsys PCIE controller and analog PHY have thousands of sideband signals. Most of these input signals are given fixed values, only a few of them are connected to external control registers. *Table 0-129* shows the PCIE sideband control register summary table.

Table 0-129. PCIE External Control Register Summary

Address	Description
0x05E0_0000	Clock and Reset Signals
0x05E0_0004	Core Control Signals
0x05E0_0008	Core Status Signals
0x05E0_0010	Spare Control Register 0
0x05E0_0014	Spare Control Register 1
0x05E0_0018	Spare Control Register 2
0x05E0_001C	Spare Control Register 3
0x05E0_0020	Spare Status Register 0
0x05E0_0024	Spare Status Register 0
0x05E0_0028	Spare Status Register 0
0x05E0_002C	Spare Status Register 0
0x05E0_0080	Interrupt Status
0x05E0_0084	Interrupt Enable
0x05E0_0100	PHY Status Signals

25.1 Clock and Reset Signals (0x05E0_0000)

Table 0-130. Clock and Reset Signals (0x05E0_0000)

Bit	Symbol	Access	Default	Description
[1]	app_init_rst	RW	0x0	Request from your application to send a hot reset to the upstream port.
[0]	app_clk_req_n	RW	0x0	Indicates that the application logic is ready to have reference clock removed.

25.2 Core Control Signals and Status Signals (0x05E0_0004 ~ 0x05E0_0008)

Table 0-131. Core Control Signals (0x05E0_0004)

Bit	Symbol	Access	Default	Description
[23:16]	app_bus_num	RW	0x0	Bus number
[12:8]	app_dev_num	RW	0x0	Device number
[5]	app_hold_phy_rst	RW	0x1	hold the PHY in reset
[4]	sys_aux_pwr_det	RW	0x0	Auxiliary Power Detected
[3]	apps_pm_xmt_turnoff	RW	0x0	Generate a PM_Turn_Off message
[2]	app_dbi_ro_wr_disable	RW	0x0	DBI Read-only Write Disable
[1]	app_ltssm_enable	RW	0x0	Enable LTSSM to strat Link Training
[0]	app_sris_mode	RW	0x0	SRIS operating mode

Table 0-132. Core Status Signals (0x05E0_0008)

Bit	Symbol	Access	Default	Description
[12:8]	frsq_int_msg_num	R	0x0	FRSQ Interrupt Message Number
[6]	pm_xtlh_block_tlp	R	0x0	Application must stop generating new outgoing request TLPs due to the current power management state
[5:0]	smlh_ltssm_state	R	0x0	Current state of the LTSSM

25.3 Spare Control Register and Status Register (0x05E0_0010 ~ 0x05E0_002C)

Table 0-133. Spare Control Register and Status Register (0x05E0_0010 ~ 0x05E0_002C)

Bit	Symbol	Access	Default	Description
[31:0]	spare_ctrl0	RW	0x0	Spare Control Register 0, (0x05E0_0010)
[31:0]	spare_ctrl1	RW	0x0	Spare Control Register 1, (0x05E0_0014)
[31:0]	spare_ctrl2	RW	0x0	Spare Control Register 2, (0x05E0_0018)
[31:0]	spare_ctrl3	RW	0x0	Spare Control Register 3, (0x05E0_001C)
[31:0]	spare_stat0	R	0x0	Spare Status Register 0, (0x05E0_0020)
[31:0]	spare_stat1	R	0x0	Spare Status Register 1, (0x05E0_0024)
[31:0]	spare_stat2	R	0x0	Spare Status Register 2, (0x05E0_0028)
[31:0]	spare_stat3	R	0x0	Spare Status Register 3, (0x05E0_002C)

25.4 Interrupt Status and Control (0x05E0_0080 ~ 0x05E0_0084)

Table 0-134. Interrupt Status (0x05E0_0080)

Bit	Symbol	Access	Default	Description
[23]	spare_int9	RW1C	0x0	Spare Interrupt 9
[22]	spare_int8	RW1C	0x0	Spare Interrupt 8
[21]	spare_int7	RW1C	0x0	Spare Interrupt 7
[20]	spare_int6	RW1C	0x0	Spare Interrupt 6
[19]	spare_int5	RW1C	0x0	Spare Interrupt 5
[18]	spare_int4	RW1C	0x0	Spare Interrupt 4
[17]	spare_int3	RW1C	0x0	Spare Interrupt 3
[16]	spare_int2	RW1C	0x0	Spare Interrupt 2
[15]	spare_int1	RW1C	0x0	Spare Interrupt 1
[14]	spare_int0	RW1C	0x0	Spare Interrupt 0
[13]	radm_pm_to_ack	RW1C	0x0	received a PME_TO_Ack message
[12]	link_down	RW1C	0x0	Link has gone down and that it should stop data transmission
[7]	intd	R	0x0	INTD
[6]	intc	R	0x0	INTC
[5]	intb	R	0x0	INTB
[4]	inta	R	0x0	INTA
[2]	frsq_msi	RW1C	0x0	FRSQ Interrupt
[1]	cfg_up_drs_to_frs	RW1C	0x0	DRS to FRSInterrupt
[0]	cfg_drs_msi	RW1C	0x0	DRS Message Received Interrupt

Table 0-135. Interrupt Enable (0x05E0_0084)

Bit	Symbol	Access	Default	Description
[23]	spare_int9_en	RW	0x0	Spare Interrupt 9 Enable
[22]	spare_int8_en	RW	0x0	Spare Interrupt 8 Enable
[21]	spare_int7_en	RW	0x0	Spare Interrupt 7 Enable
[20]	spare_int6_en	RW	0x0	Spare Interrupt 6 Enable
[19]	spare_int5_en	RW	0x0	Spare Interrupt 5 Enable
[18]	spare_int4_en	RW	0x0	Spare Interrupt 4 Enable
[17]	spare_int3_en	RW	0x0	Spare Interrupt 3 Enable
[16]	spare_int2_en	RW	0x0	Spare Interrupt 2 Enable
[15]	spare_int1_en	RW	0x0	Spare Interrupt 1 Enable
[14]	spare_int0_en	RW	0x0	Spare Interrupt 0 Enable
[13]	radm_pm_to_ack_en	RW	0x0	received a PME_TO_Ack message Enable
[12]	link_down_en	RW	0x0	Link has gone down and that it should stop data transmission Enable
[7]	intd_en	RW	0x0	INTD Interrupt Enable
[6]	intc_en	RW	0x0	INTC Interrupt Enable
[5]	intb_en	RW	0x0	INTB Interrupt Enable
[4]	inta_en	RW	0x0	INTA Interrupt Enable

Bit	Symbol	Access	Default	Description
[2]	frsq_msi_en	RW	0x0	FRSQ Interrupt Enable
[1]	cfg_up_drs_to_frs_en	RW	0x0	DRS to FR Interrupt Enable
[0]	cfg_drs_msi_en	RW	0x0	DRS Message Received Interrupt Enable

25.5 PHY Status Signals (0x05E0_0100)

Table 0-136. PHY Status Signals (0x05E0_0100)

Bit	Symbol	Access	Default	Description
[0]	sram_init_done	R	0x0	PHY SRAM init done

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